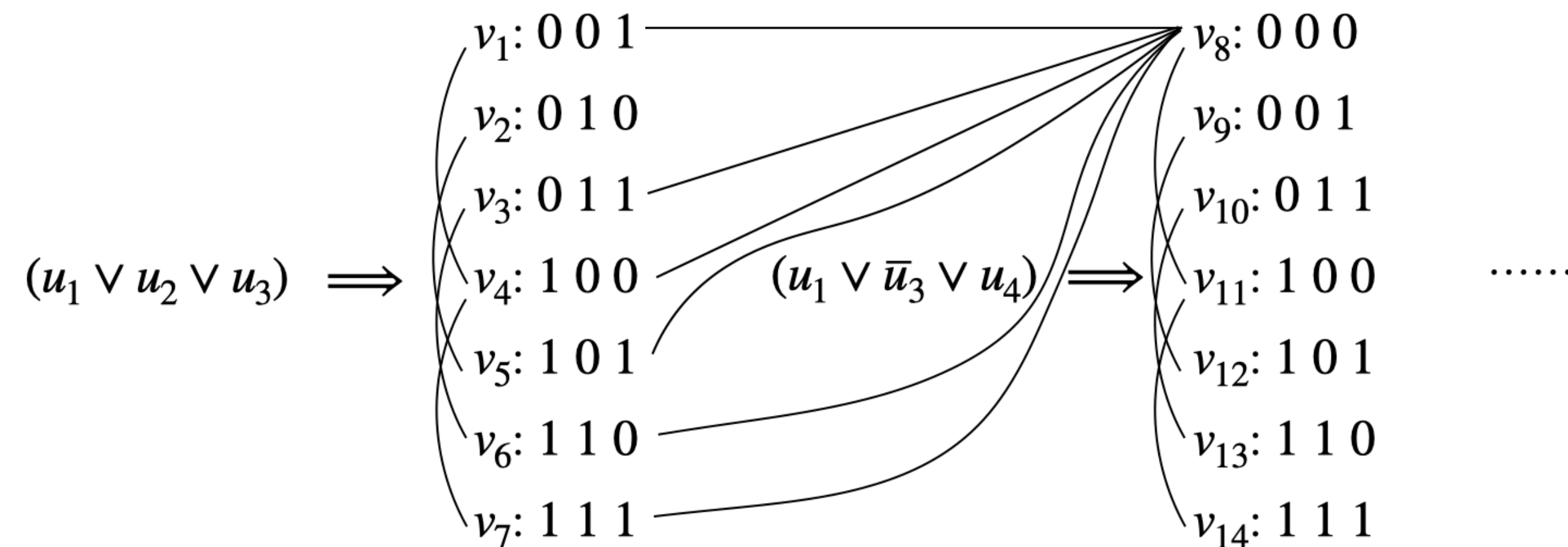


Lecture 36

Reductions: IndSet, VertexCover, DirHampath

$$3SAT \leq_p IndSet$$



Claim: ϕ is satisfiable iff G has an independent set of size $k = \#$ of clauses in ϕ

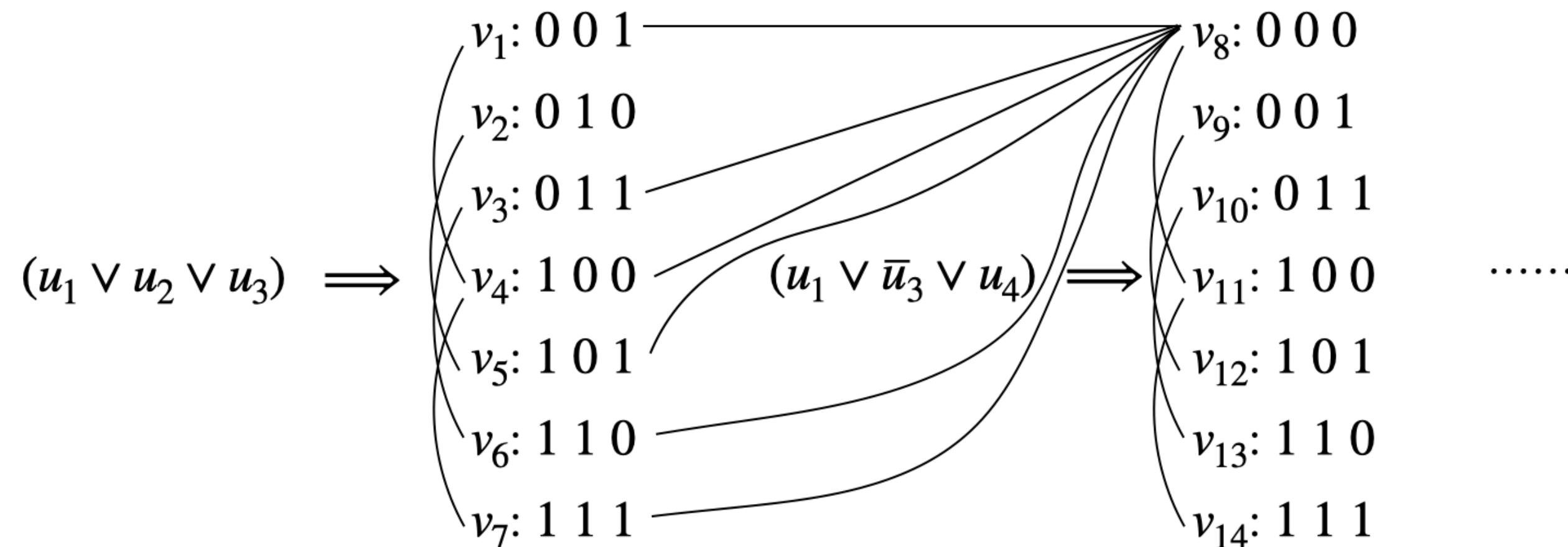
Proof: ($\Rightarrow$) Suppose ϕ has a satisfying assignment u .

Form an independent set S of size k for G :

By picking a vertex from every cluster whose values matches to that of u .

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Satisfying assignment of ϕ : $u_1 = 1, u_2 = 1, u_3 = 0, u_4 = 1, \dots$



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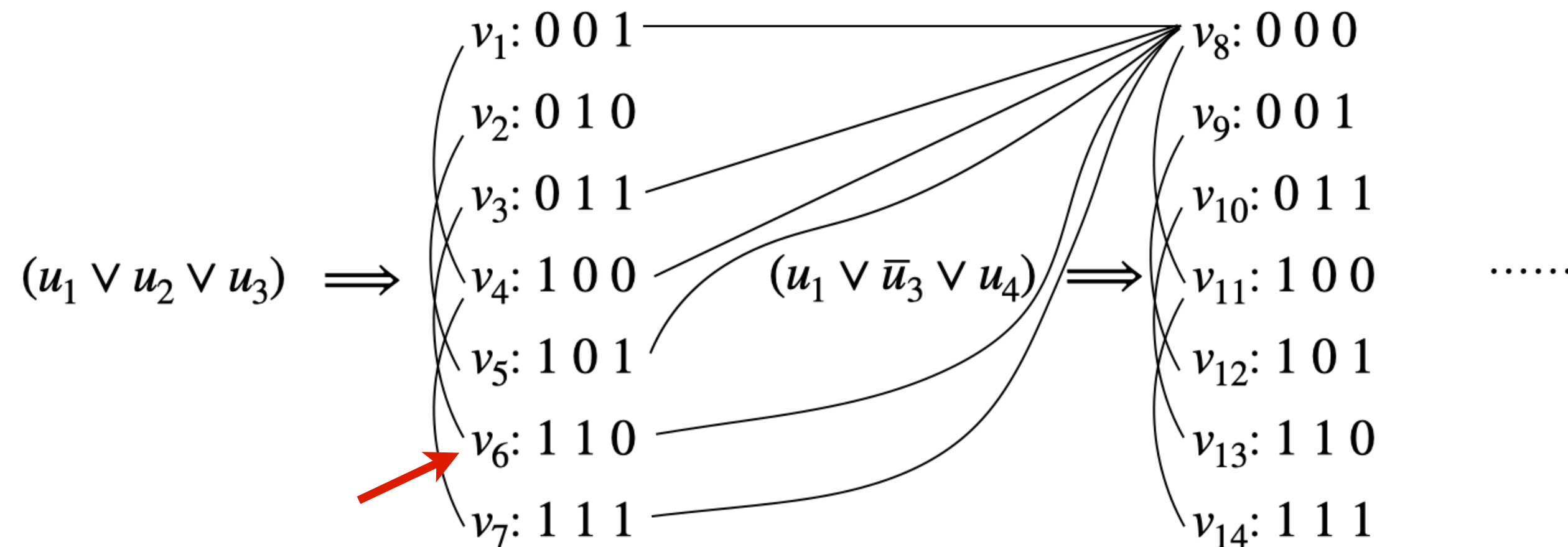
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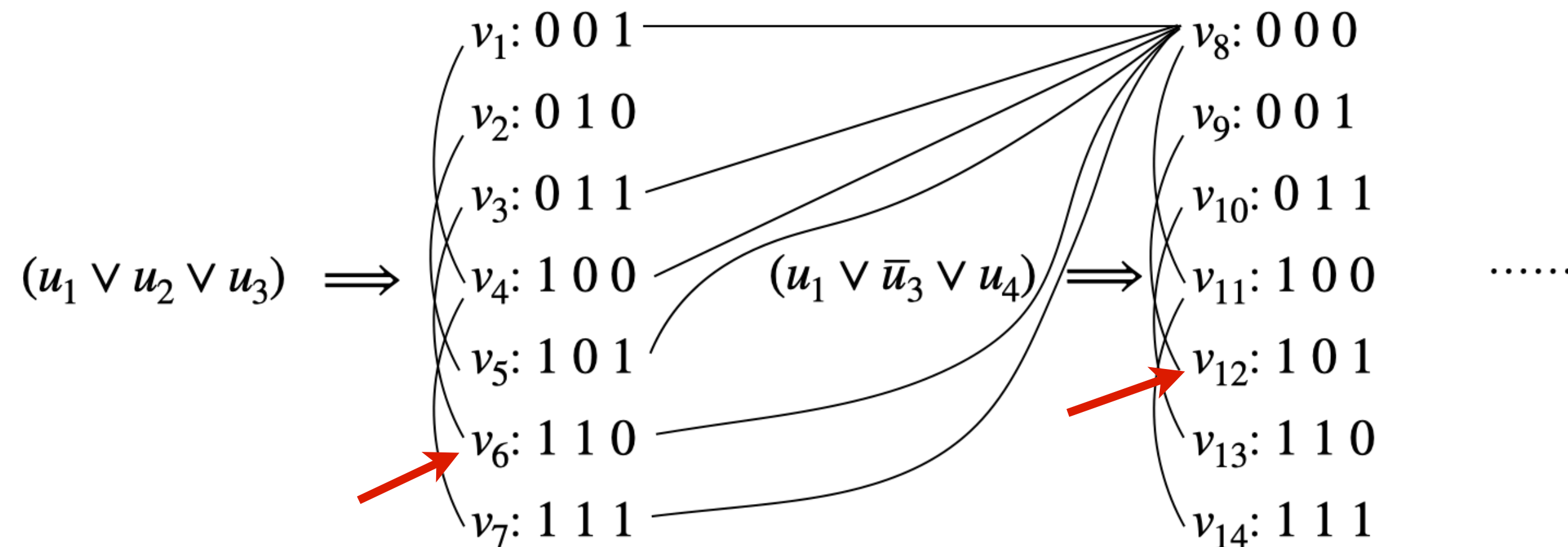
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$\phi \rightarrow \langle G, k \rangle$:

- A cluster of **7** vertices $\forall$ clause of ϕ corresponding to **satisfying partial assignments**.
- An edge between every pair of vertices in the **same cluster**.
- An edge between two vertices of different clusters, if they correspond to **inconsistent partial assignments**.
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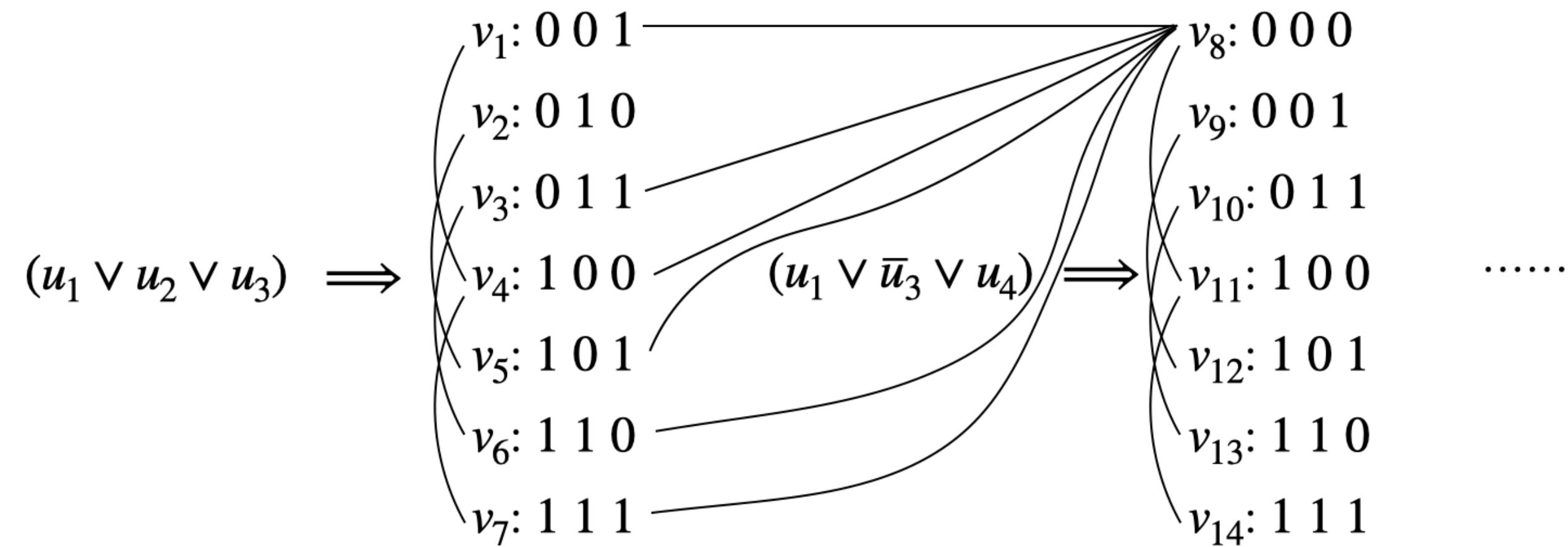
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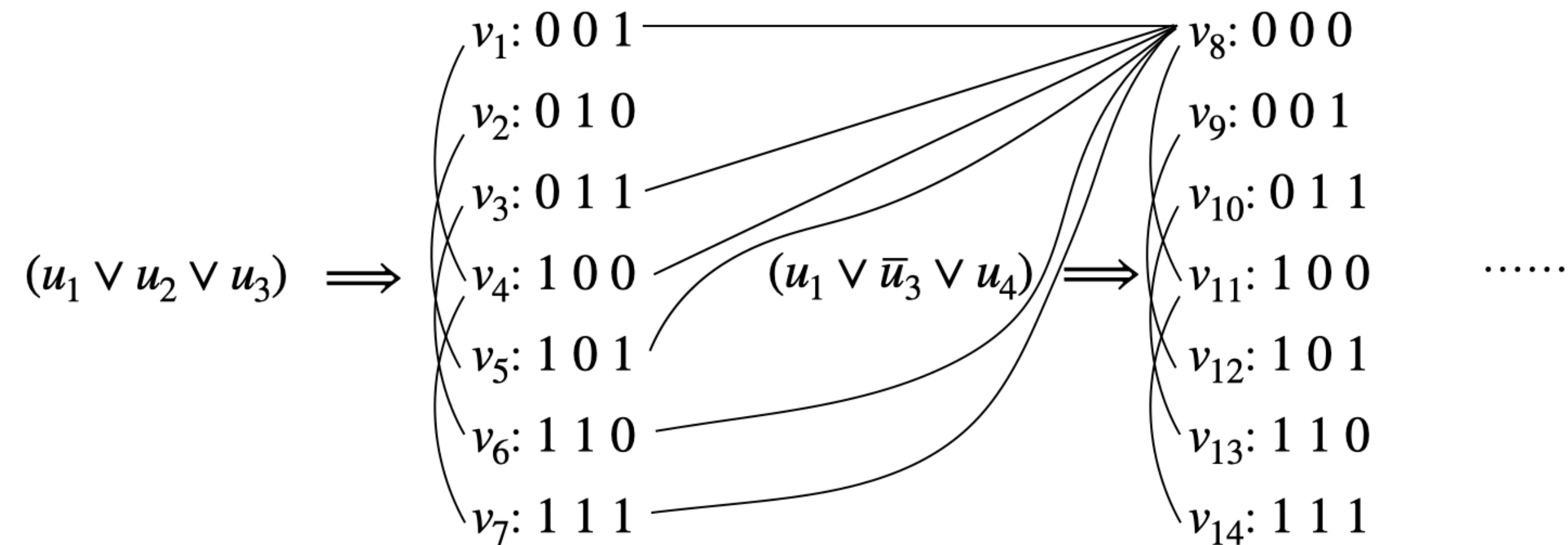
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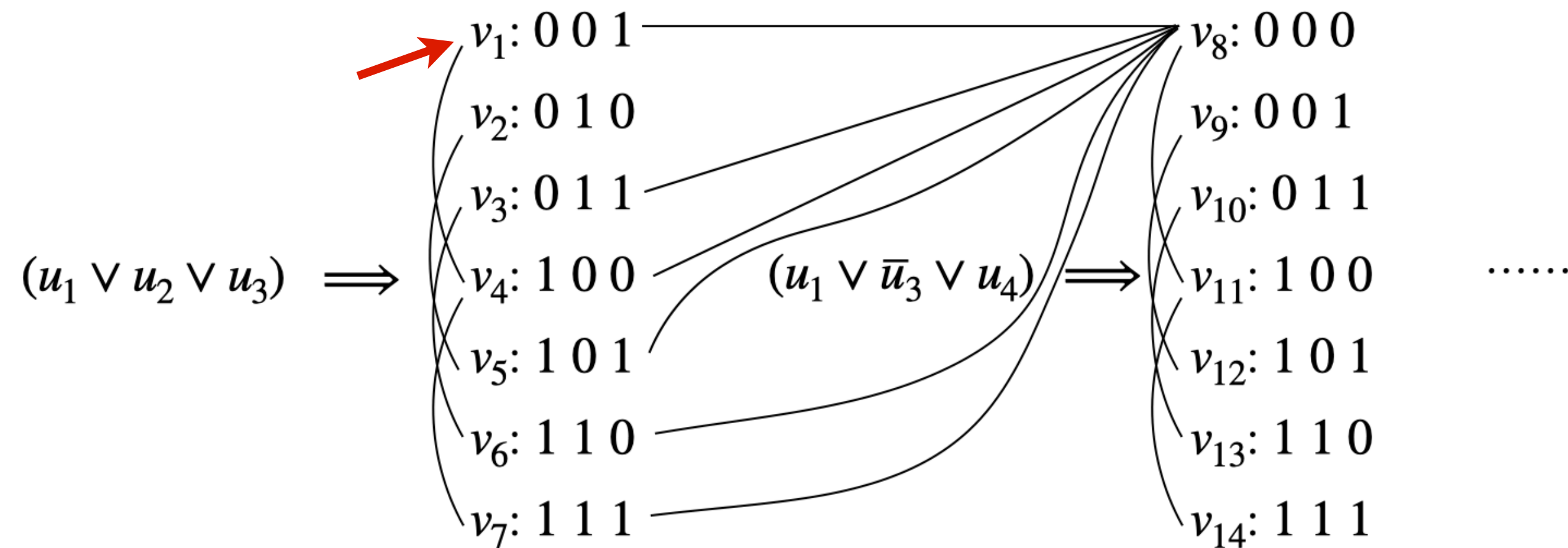


Claim: ϕ is satisfiable iff G has an independent set of size $k = \#$ of clauses in ϕ

Proof: ($\Leftarrow$) Suppose G has an independent set S of size k .

1) S has exactly one vertex from each cluster of 7 vertices.

$3SAT \leq_p IndSet$

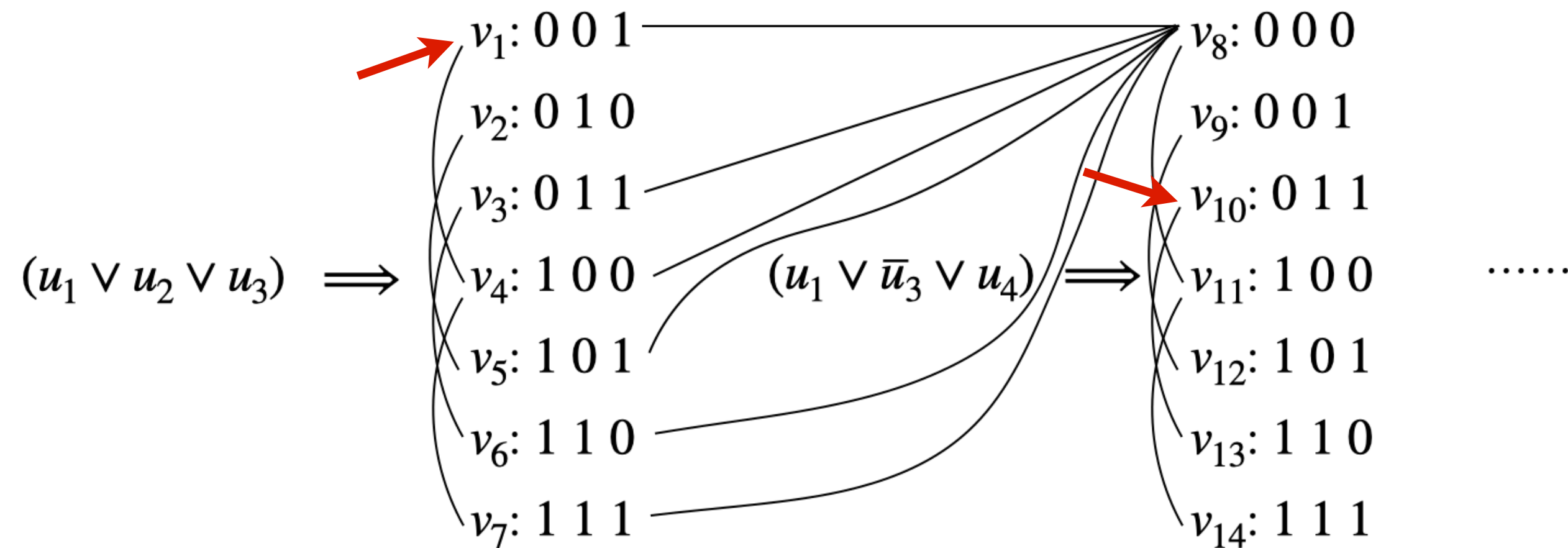


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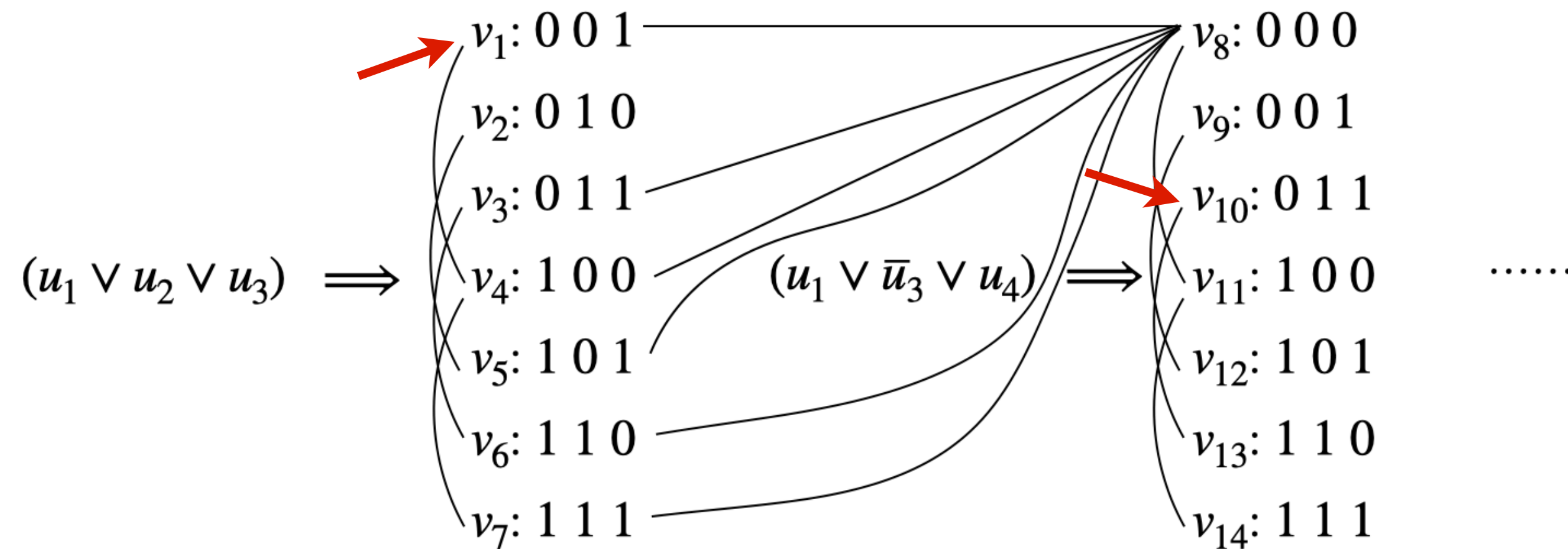


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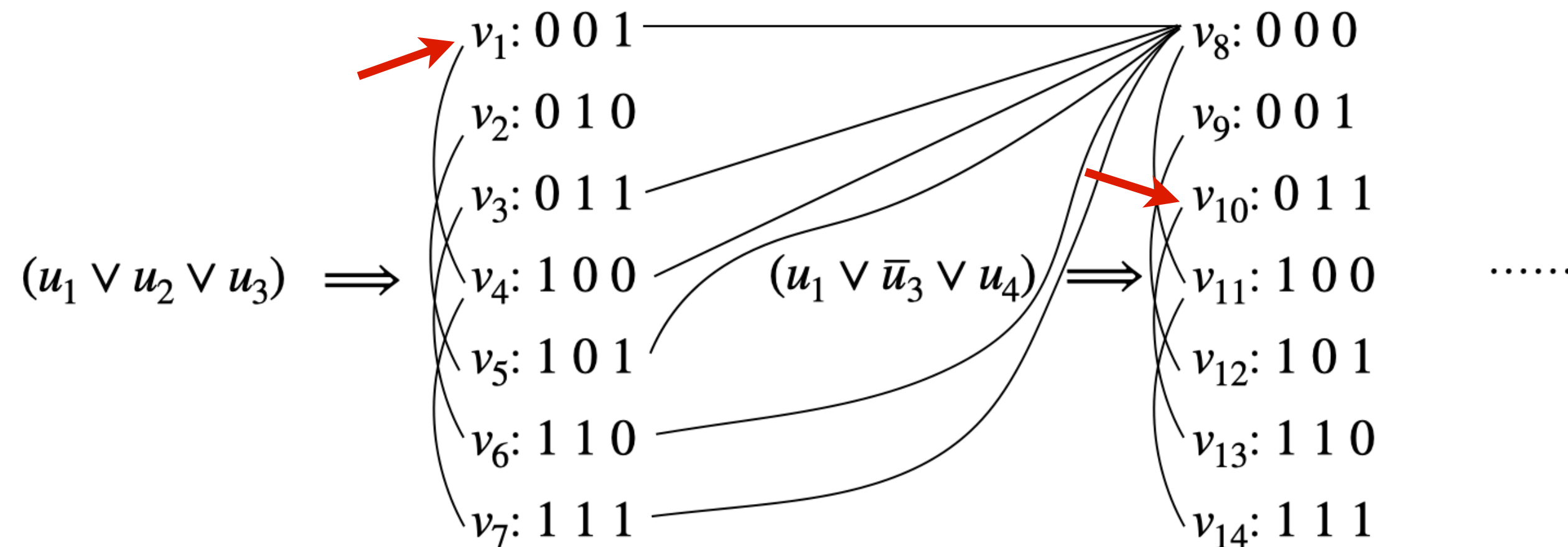
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- 2) Form a satisfying assignment u for ϕ using values corresponding to vertices of S .

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Satisfying assignment for ϕ : $u_1 = 0$



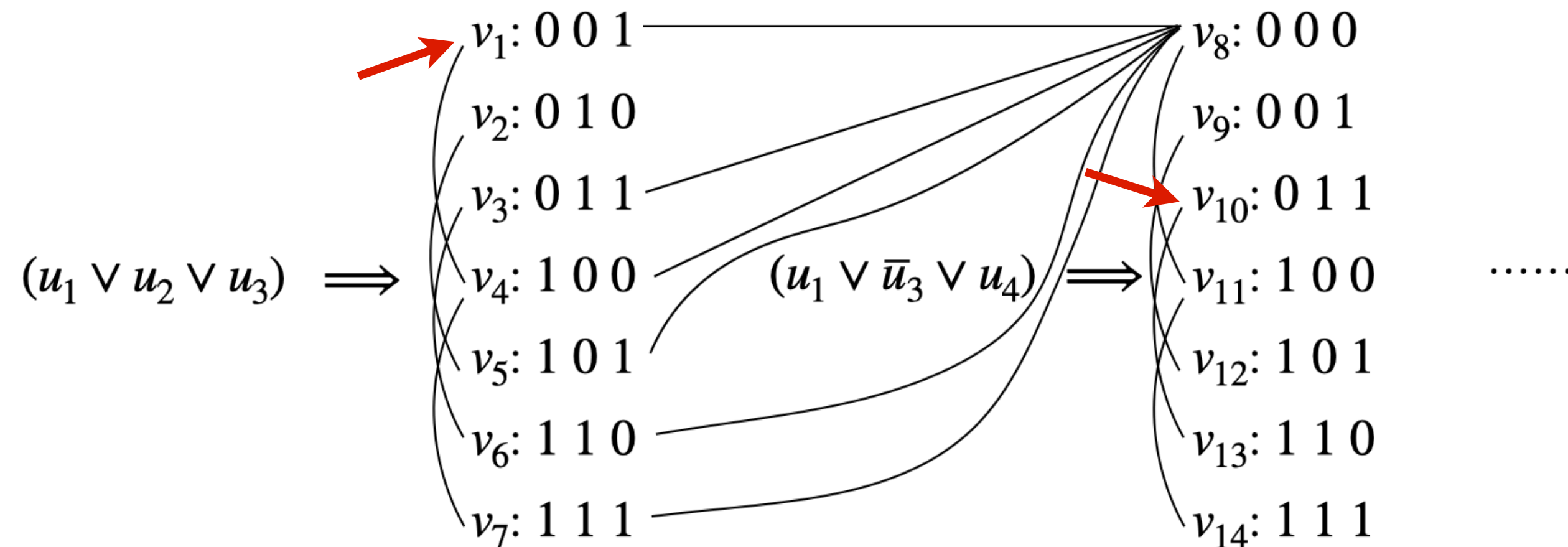
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Satisfying assignment for ϕ : $u_1 = 0, u_2 = 0$



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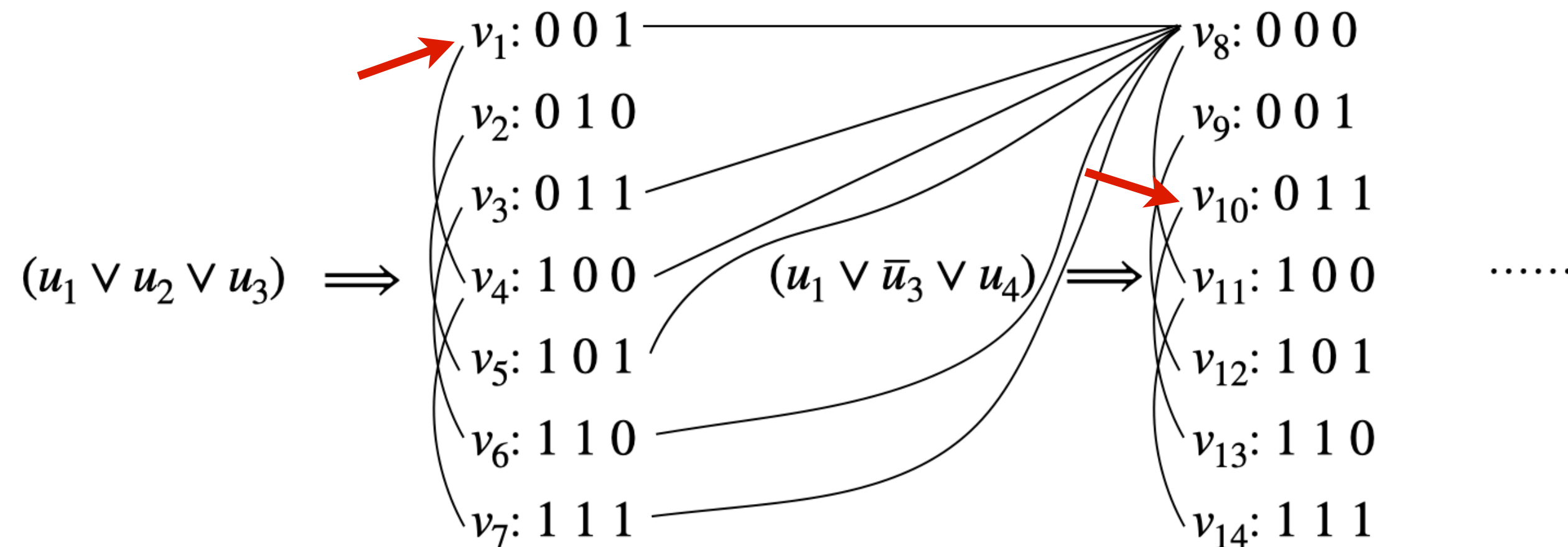
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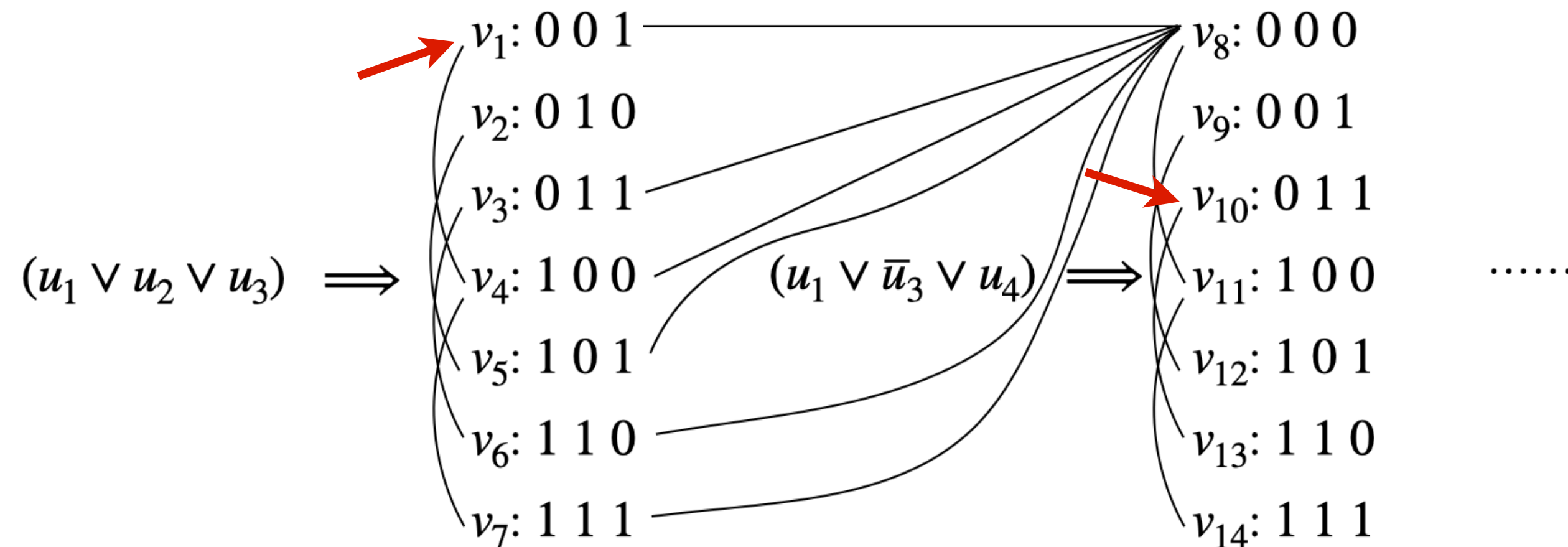
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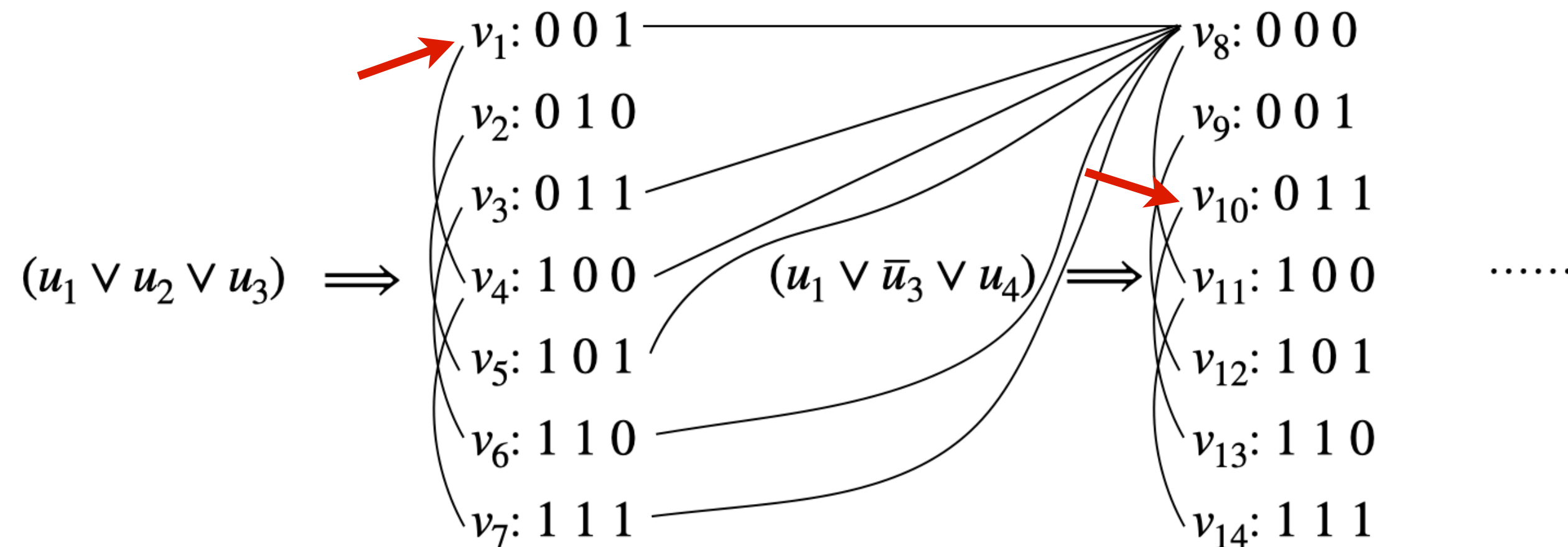
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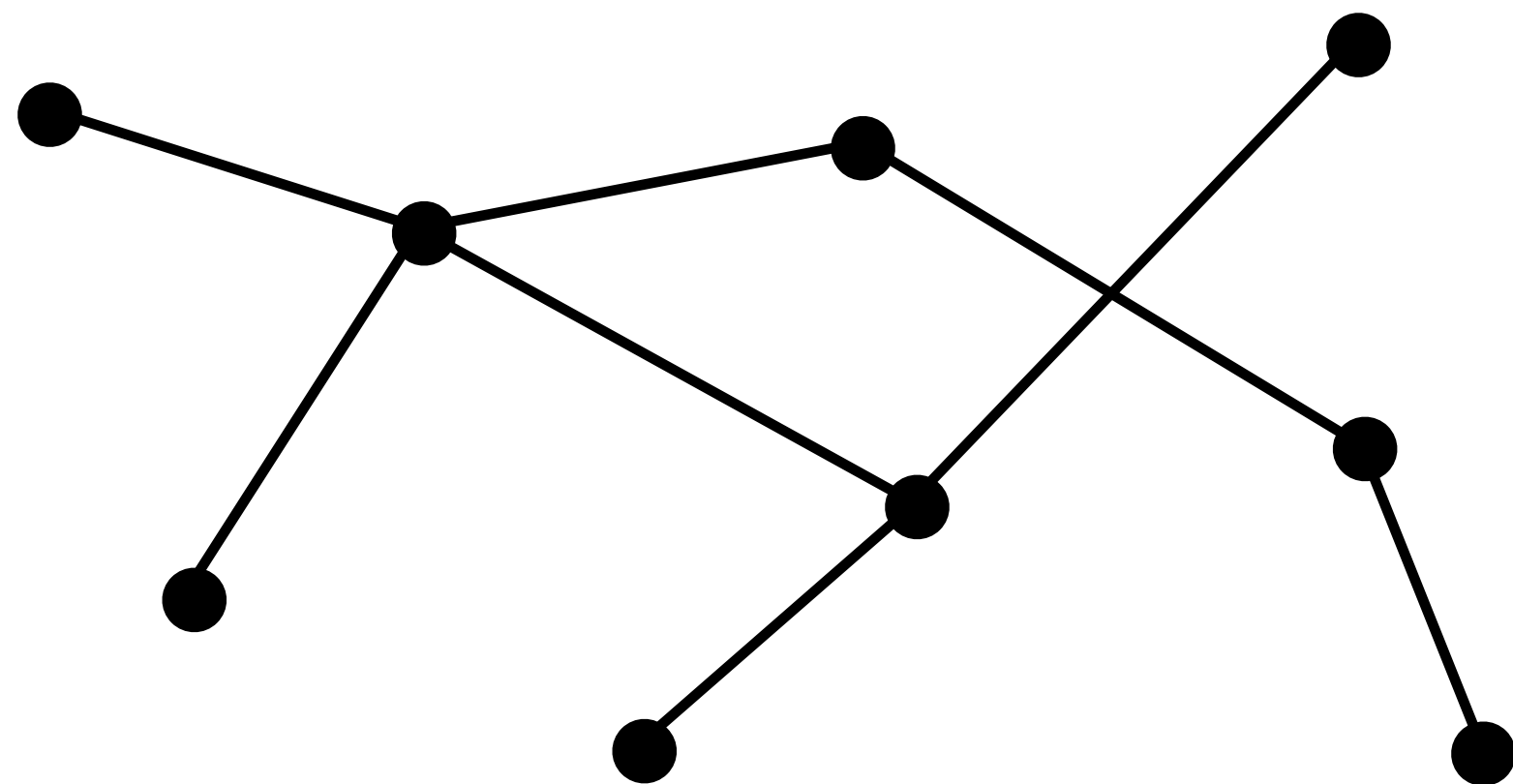
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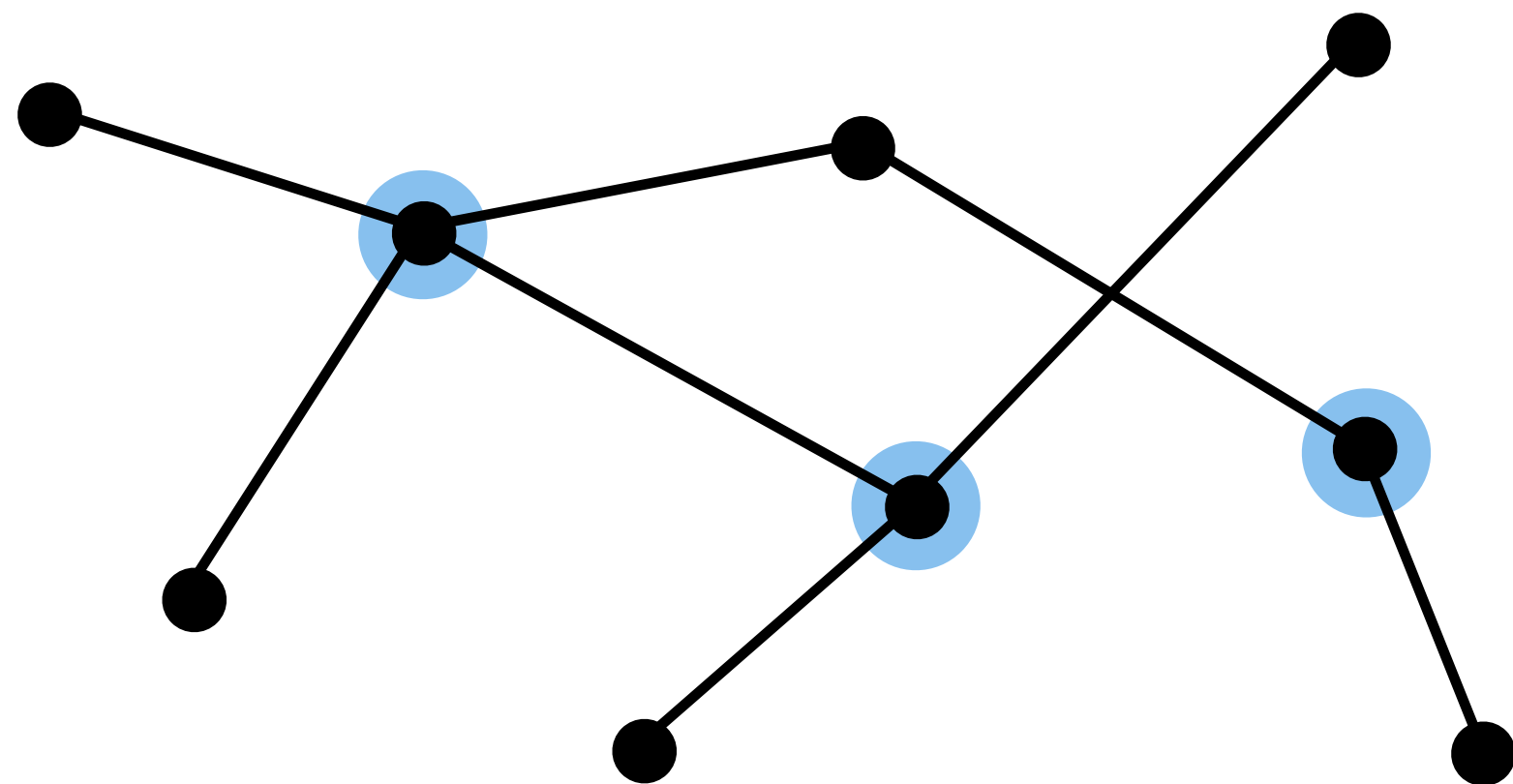


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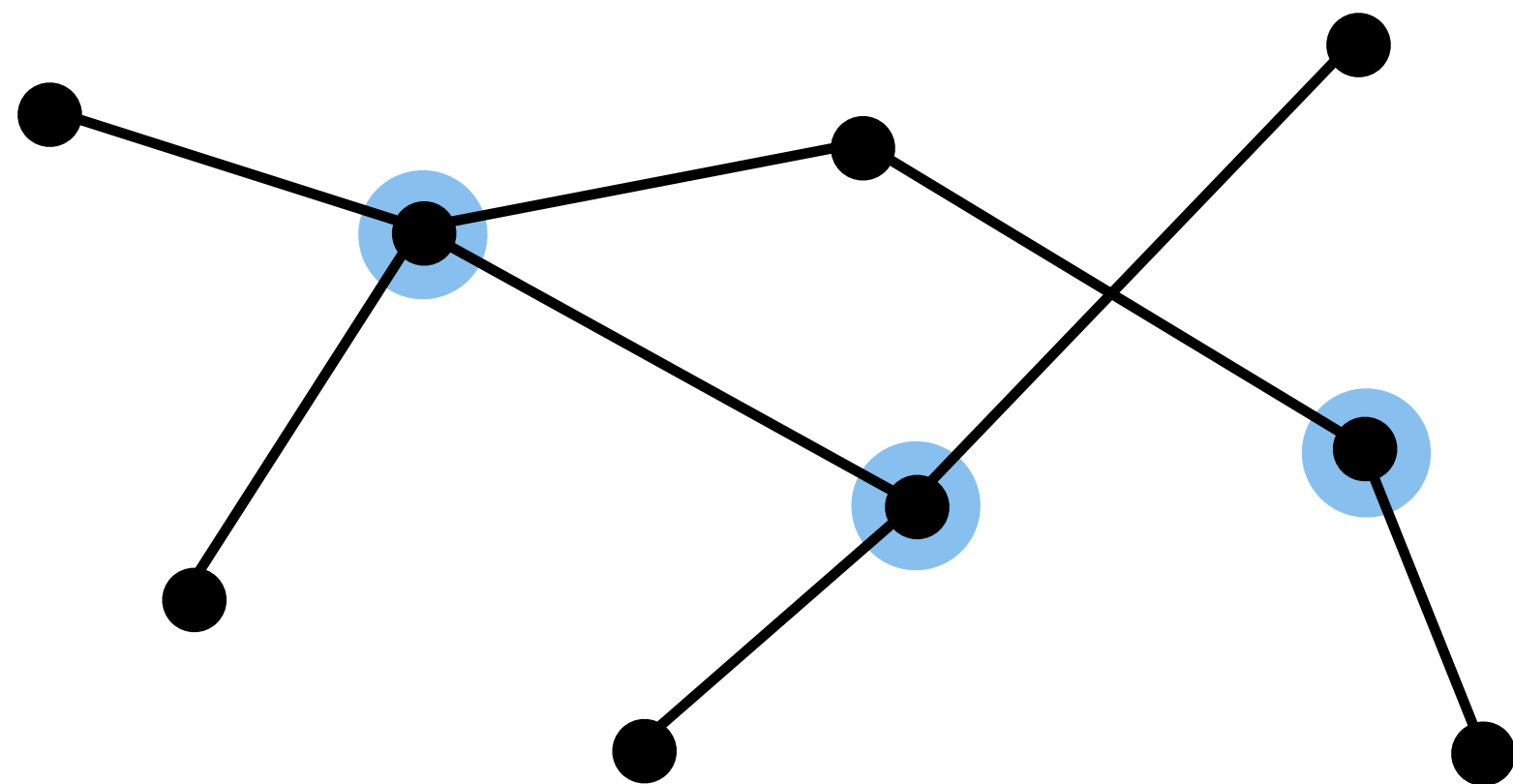
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Has a vertex cover of size 3.

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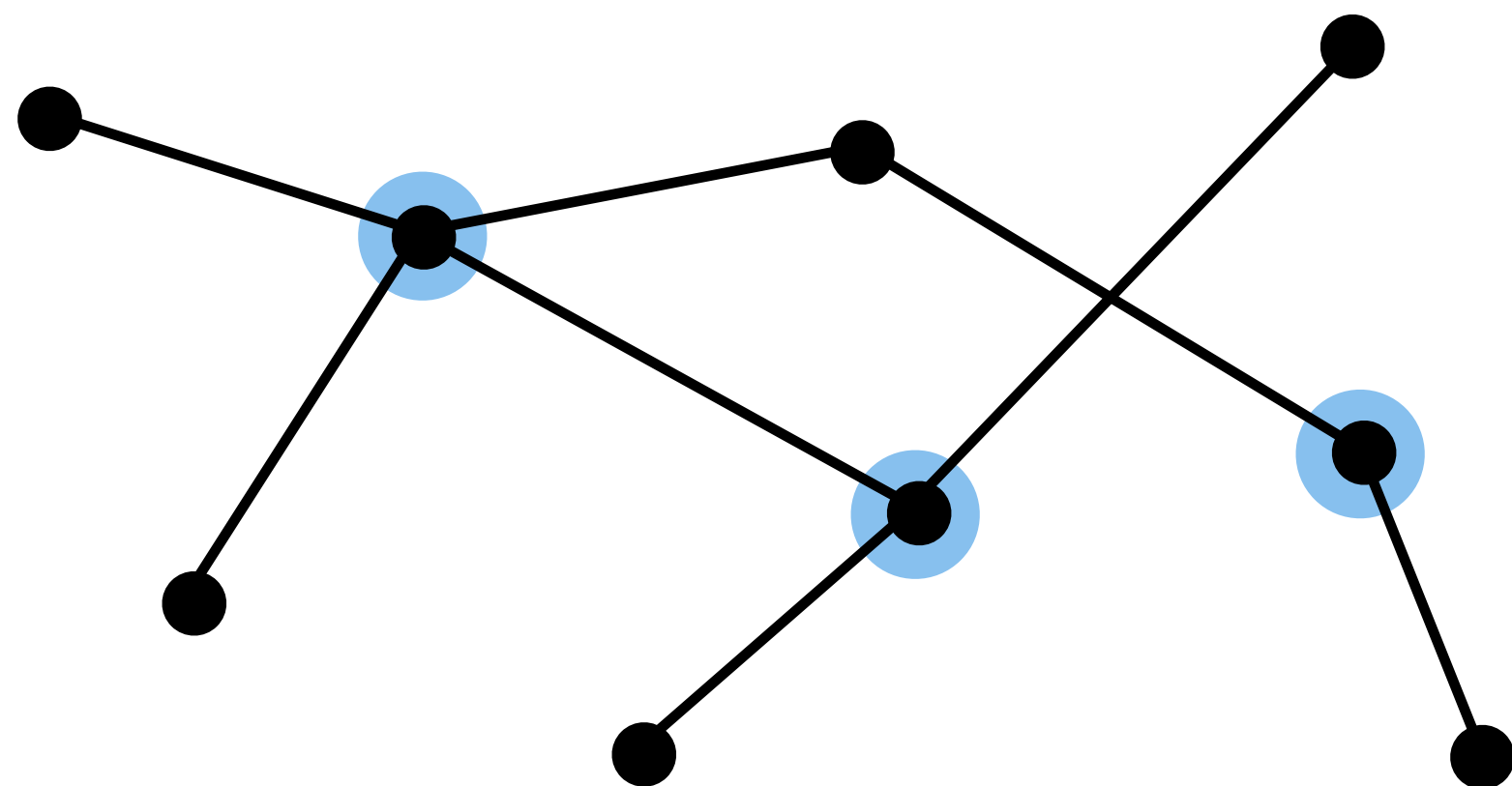
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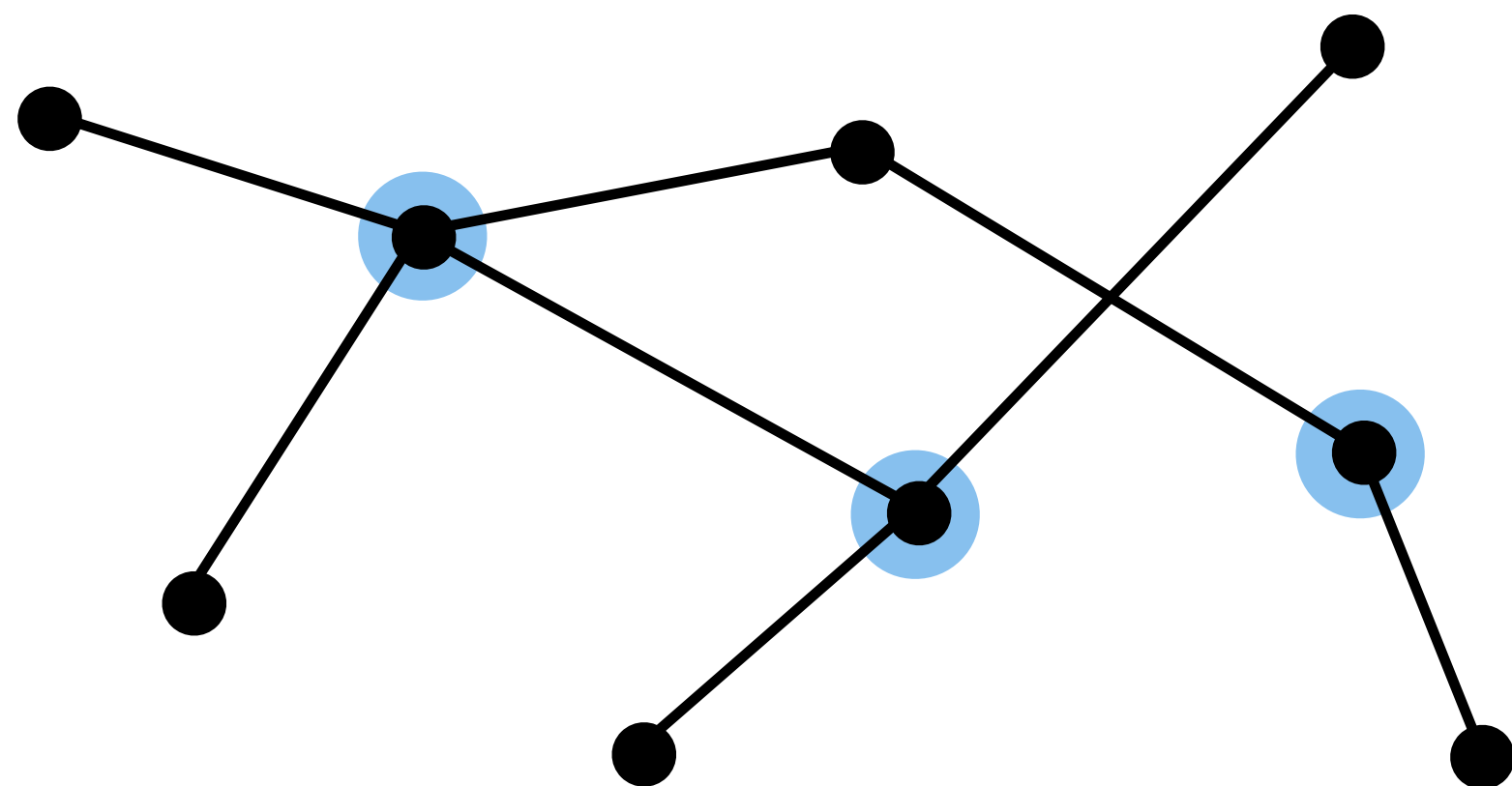
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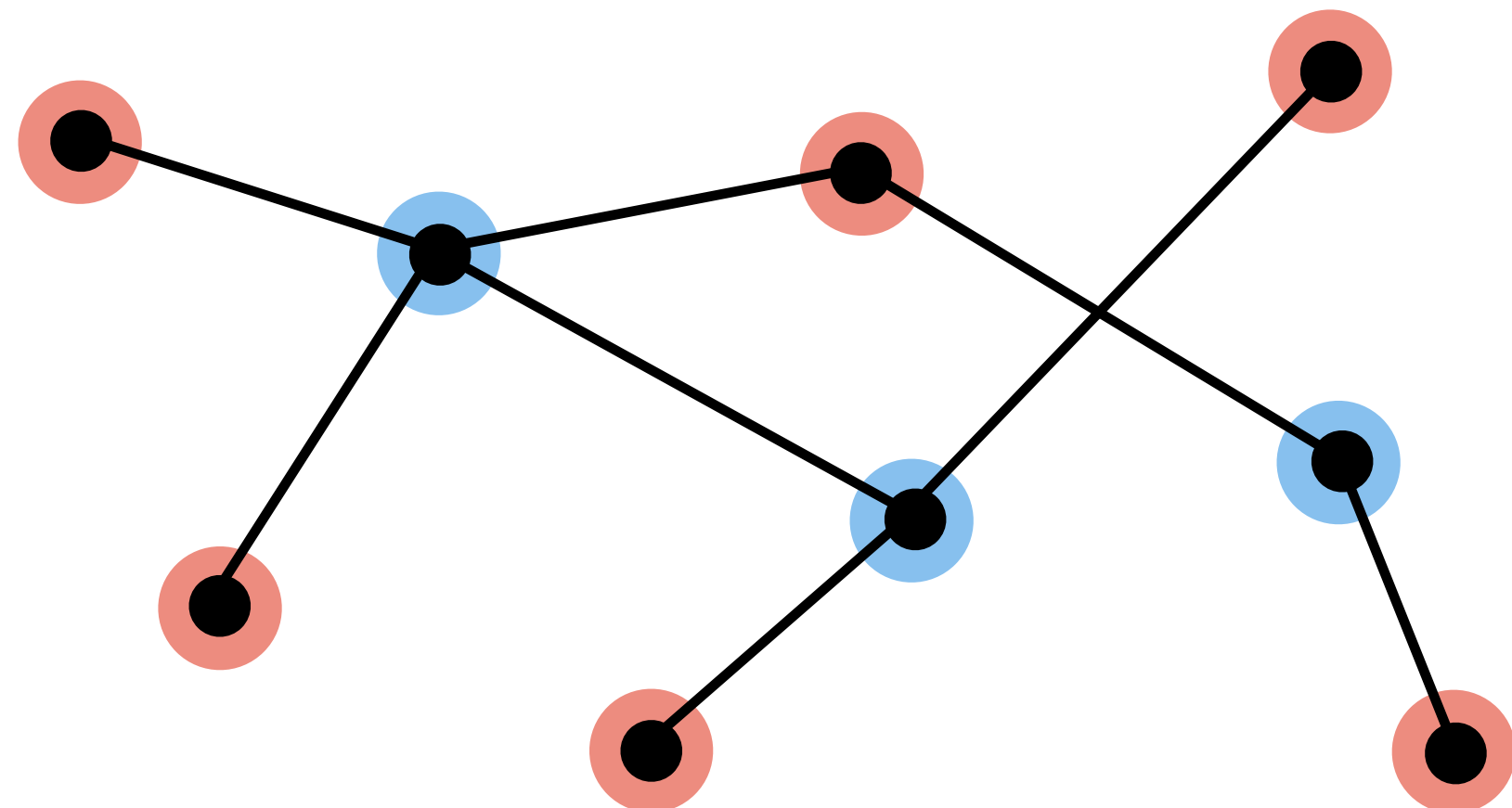


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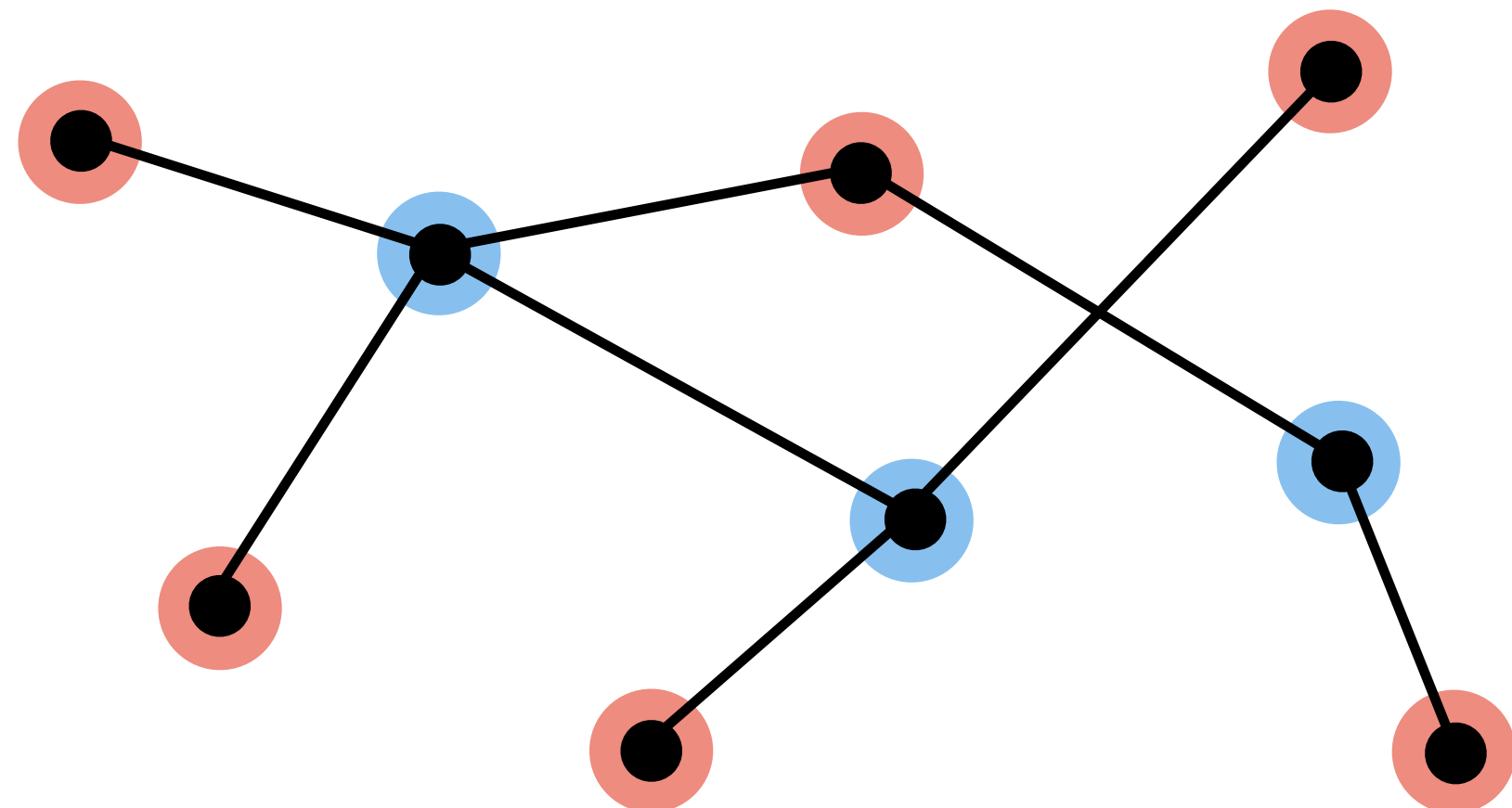


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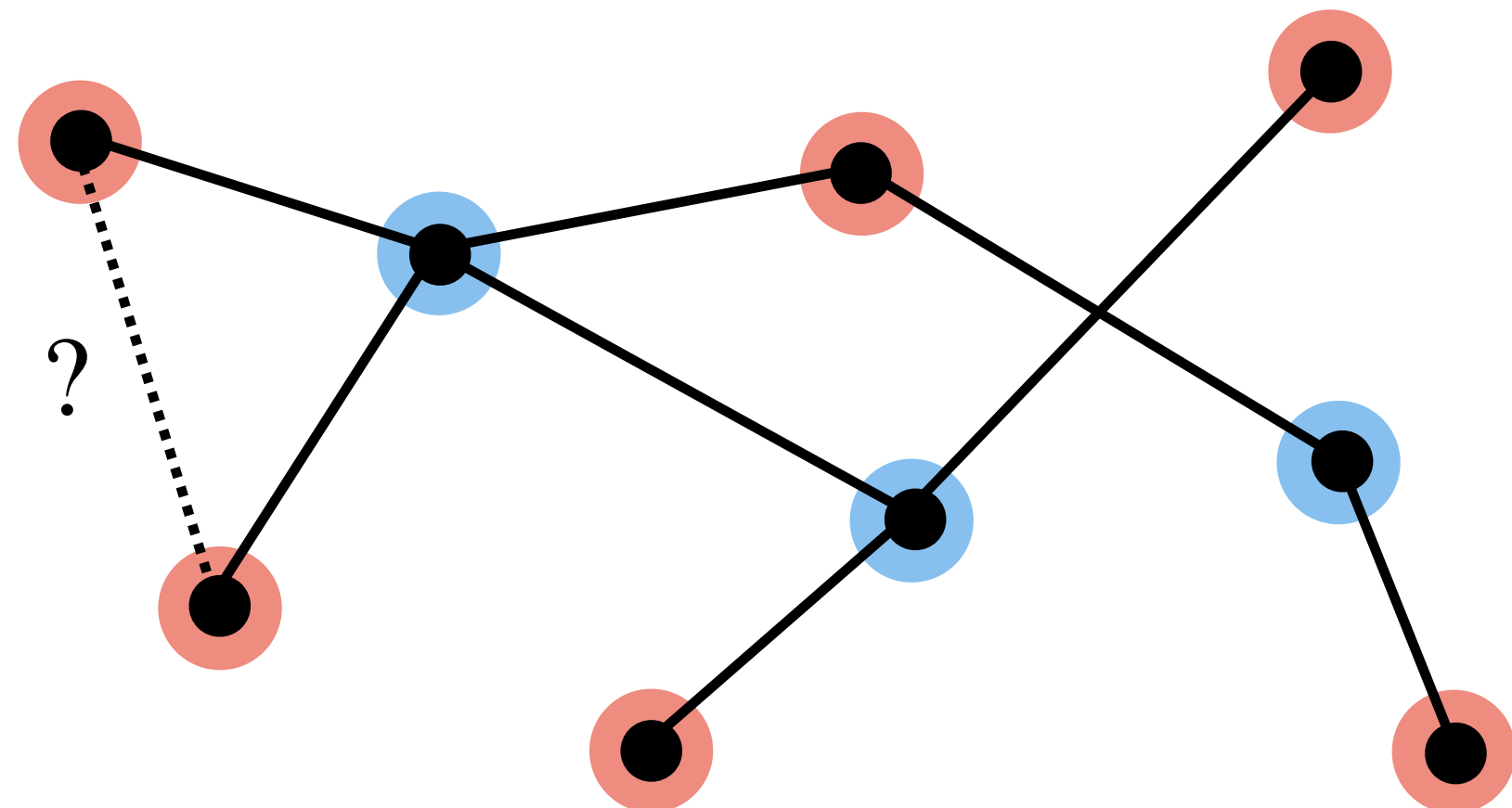


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$\langle G, k \rangle \rightarrow \langle G', k' \rangle$:

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- $k' = n - k$, where $n = |V(G)|$.

$$\textit{IndSet} \leq_p \textit{VertexCover}$$

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Claim: A graph $G(V, E)$ has an independent set of size k iff it has a vertex cover of size $(n - k)$.

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- $3SAT = \{\phi \mid \phi \text{ is a satisfiable 3CNF formula}\}$
- $DirHampath = \{\langle G, s, t \rangle \mid G \text{ is a directed graph with a hamiltonian path from } s \text{ to } t\}$

$$3SAT \leq_p DirHampath$$

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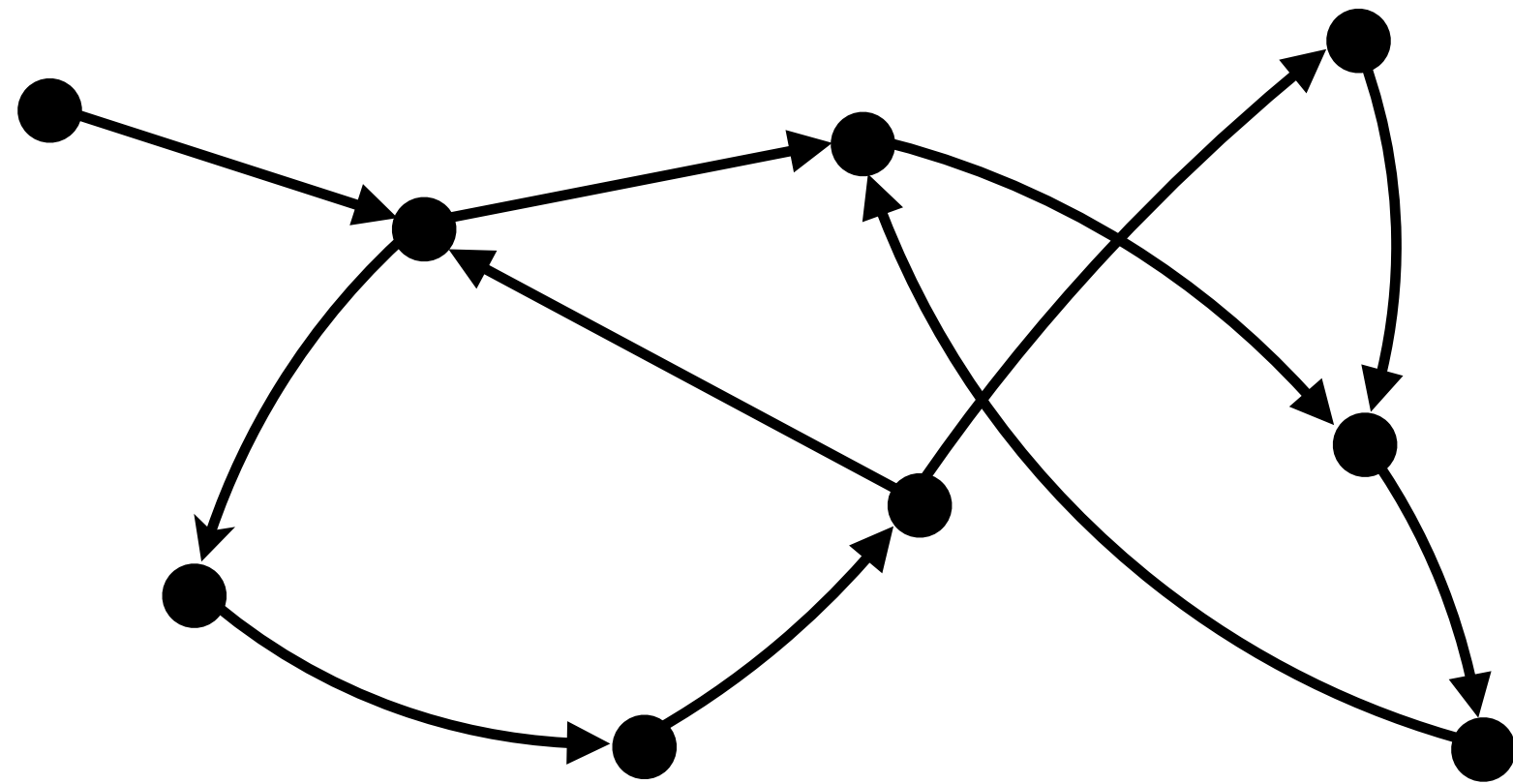


A path that goes via all the vertices

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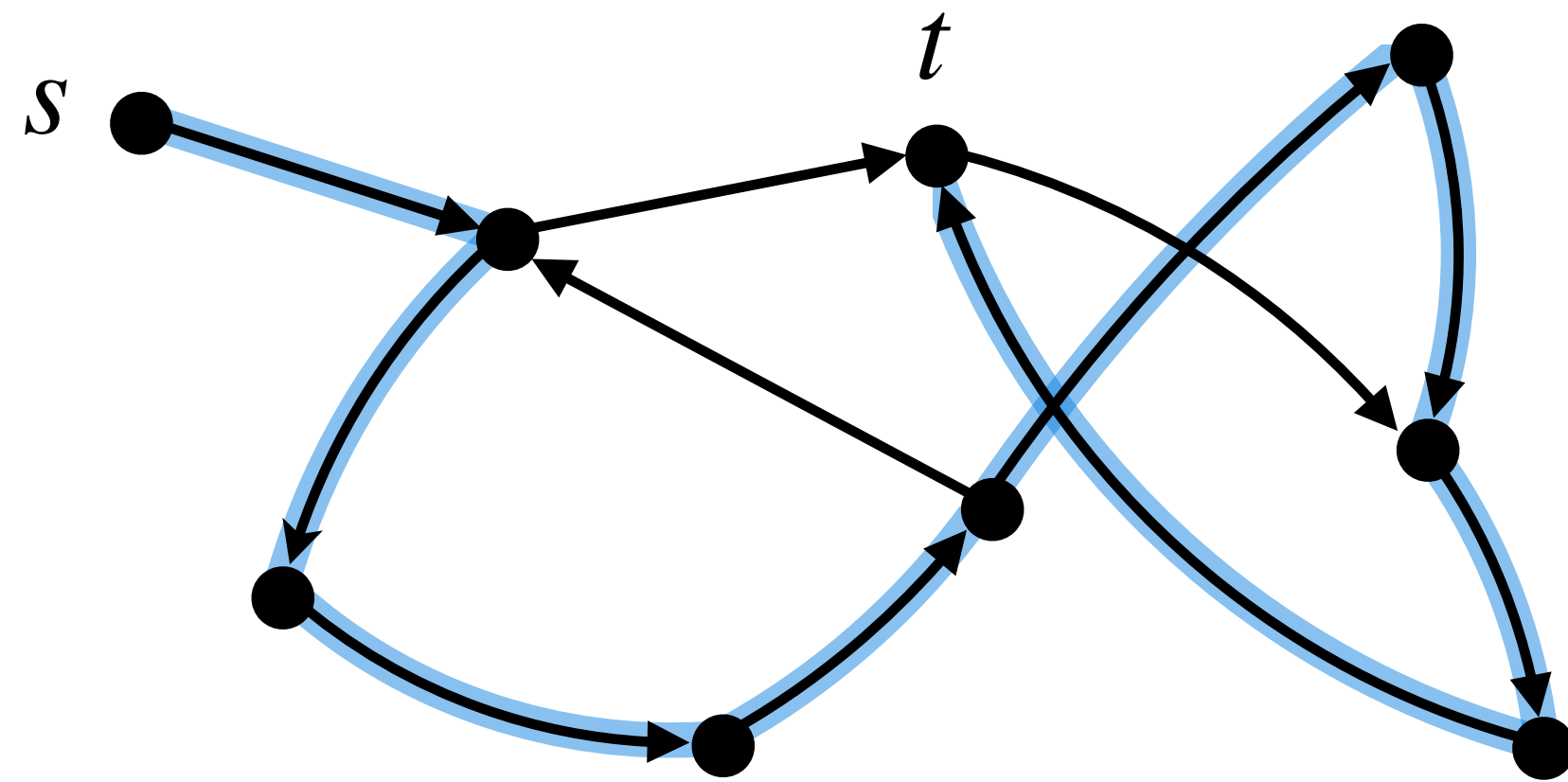
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A path that goes via all the vertices



Blue edges form a hamiltonian path from s to t

$$3SAT \leq_p DirHampath$$

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Reduction: Let's convert ϕ into $\langle G, s, t \rangle$. Suppose ϕ has n variables and m clauses.

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s ●

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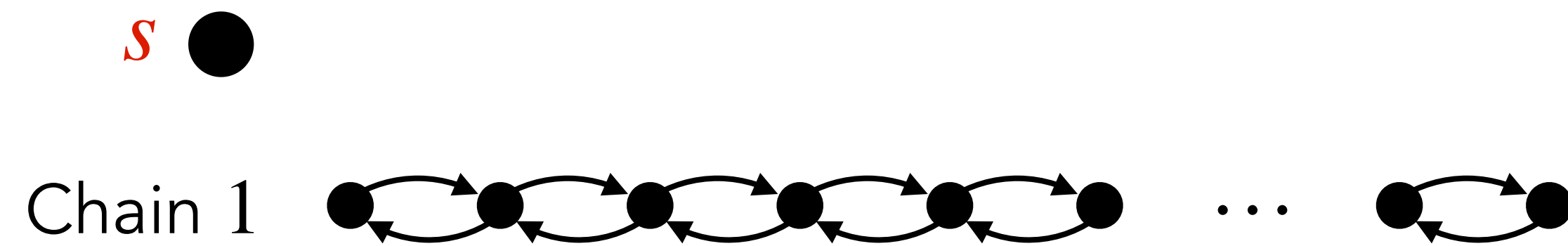
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s ●

● t

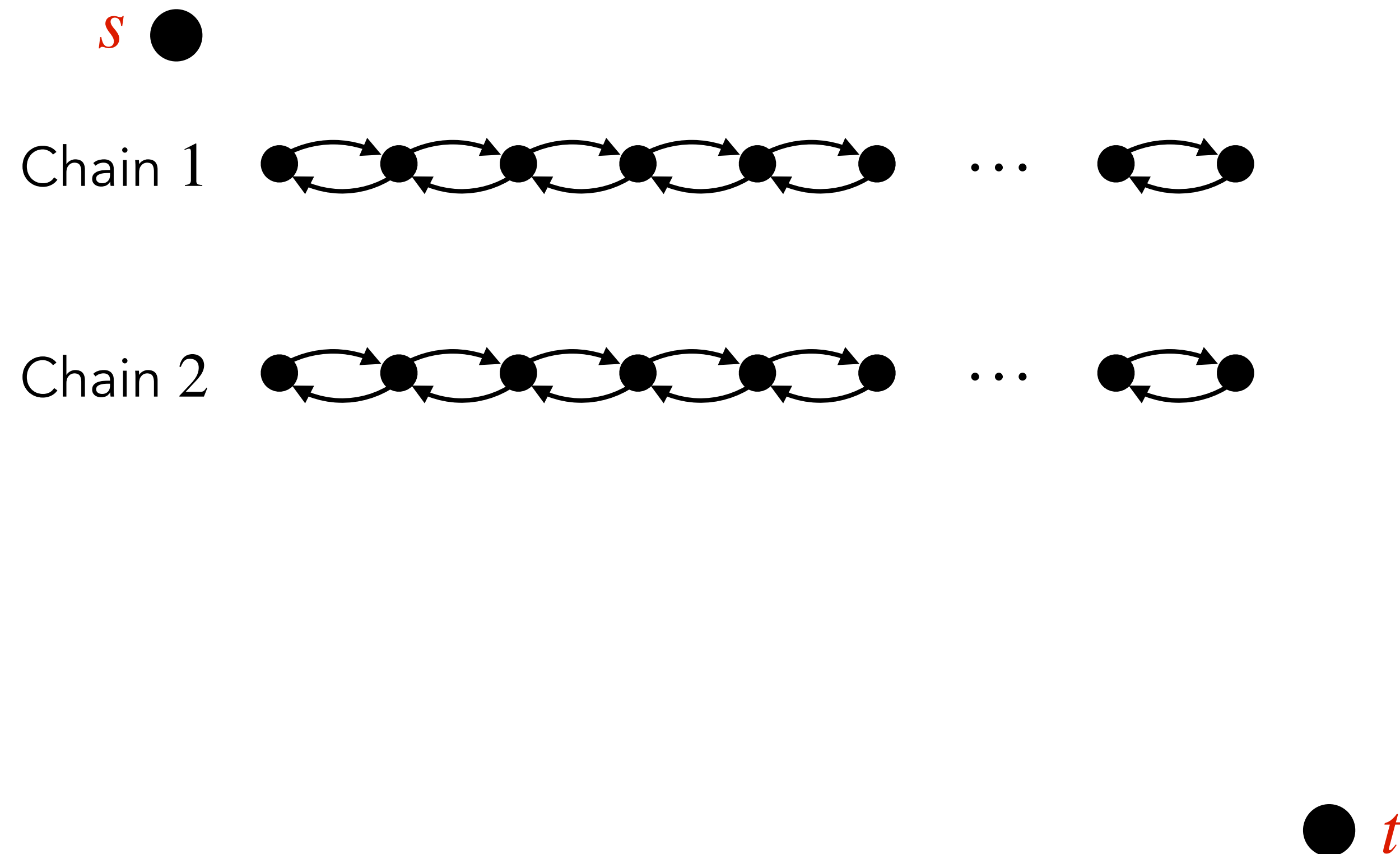
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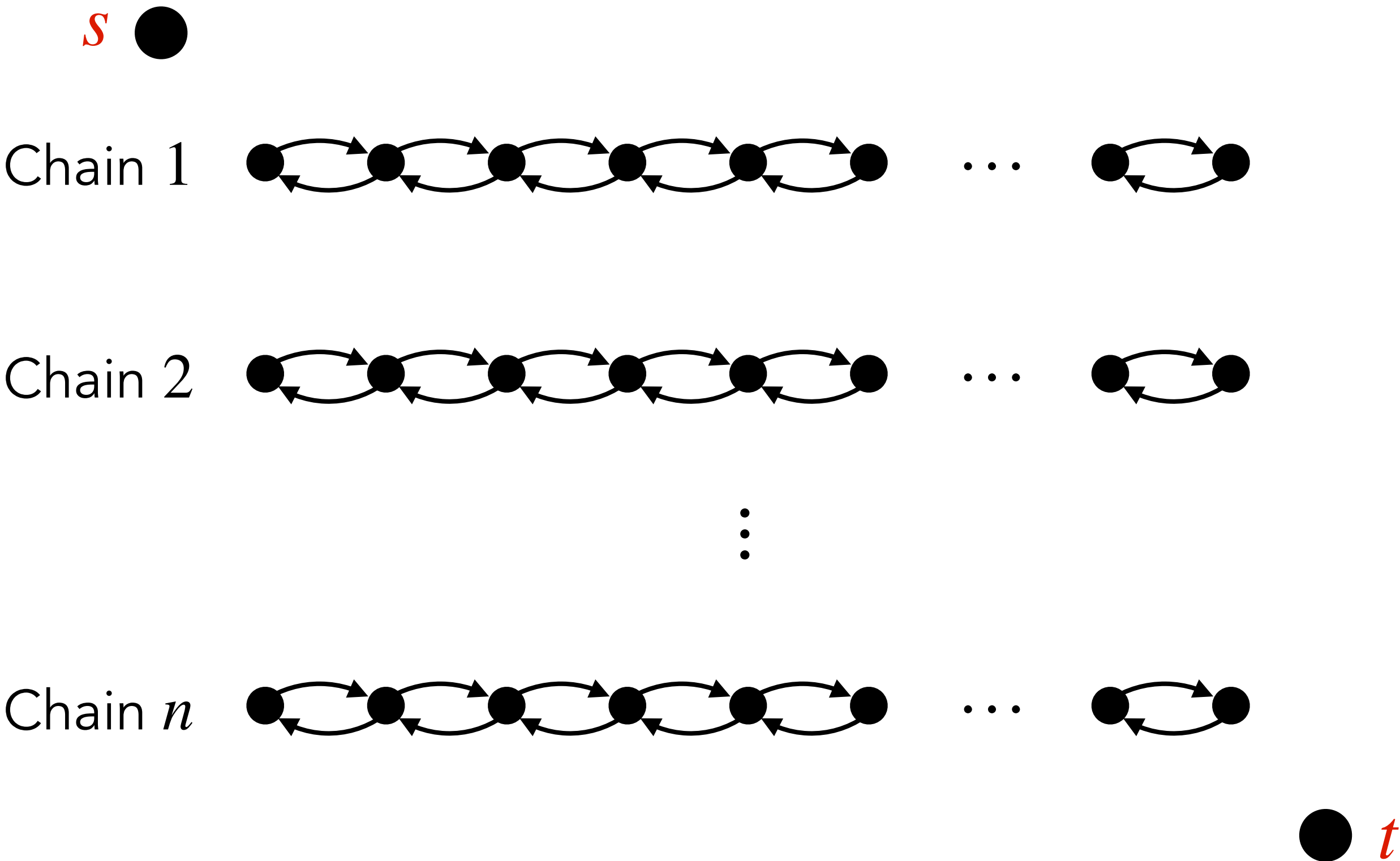
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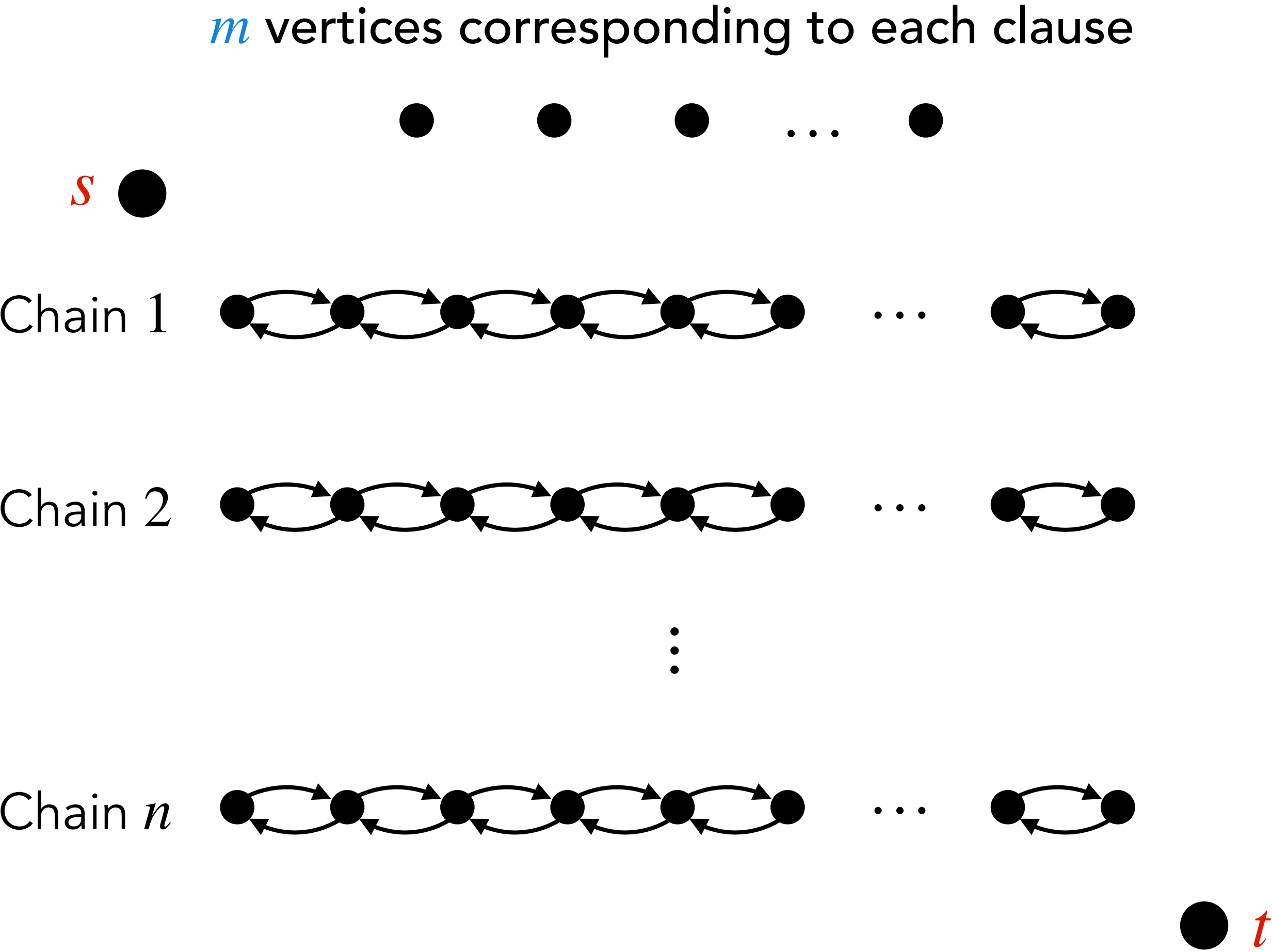
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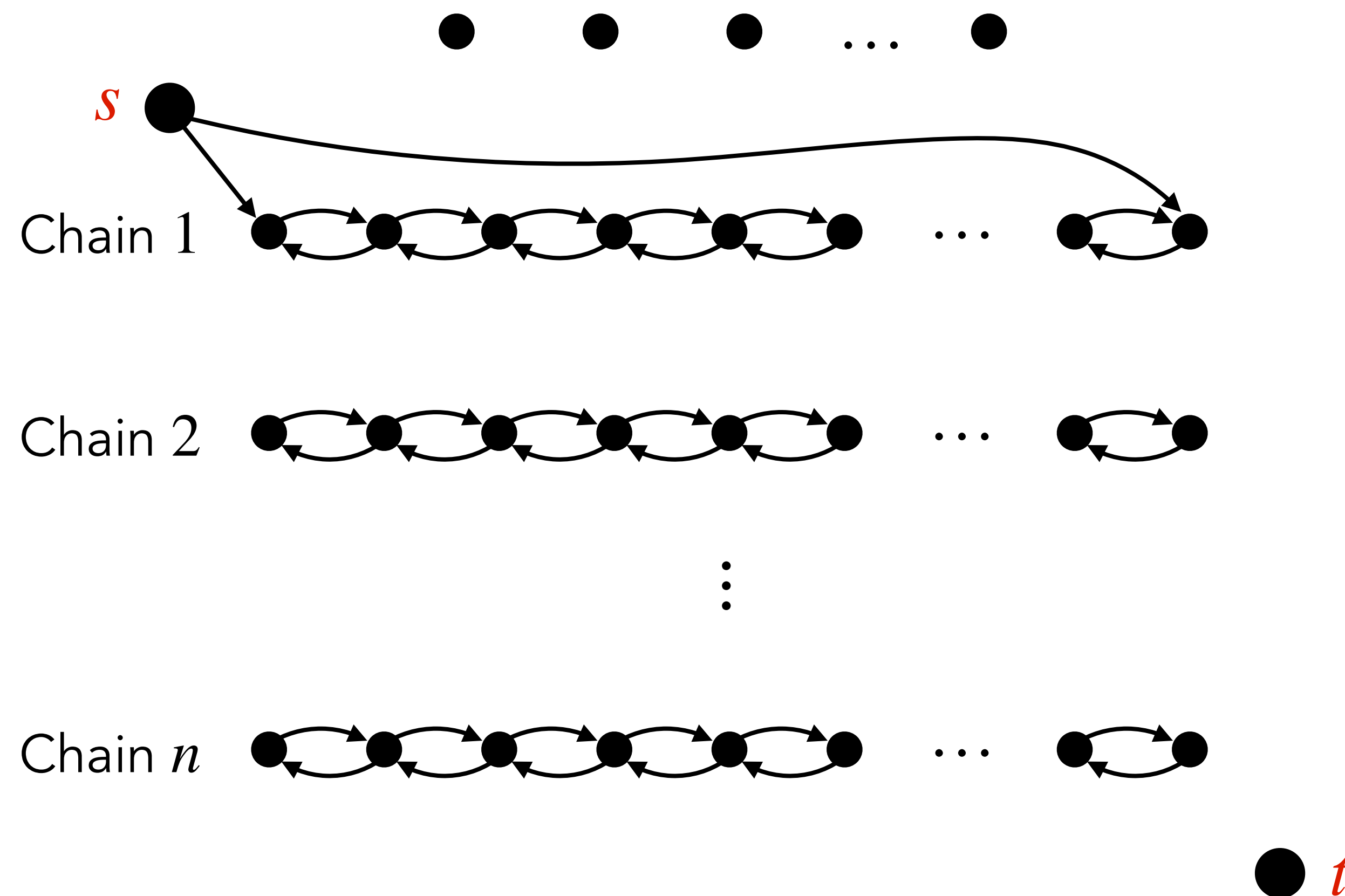
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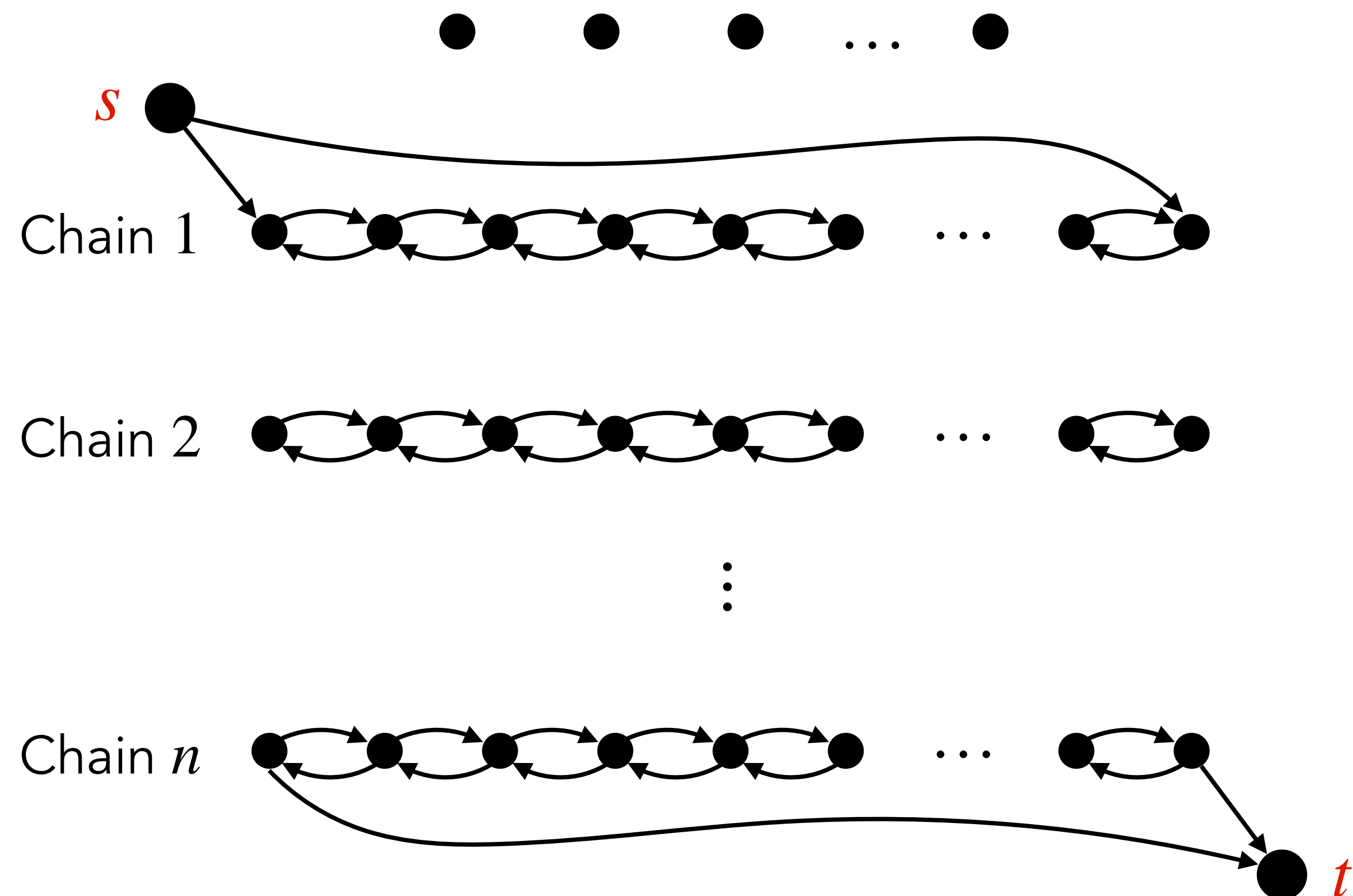
m vertices corresponding to each clause



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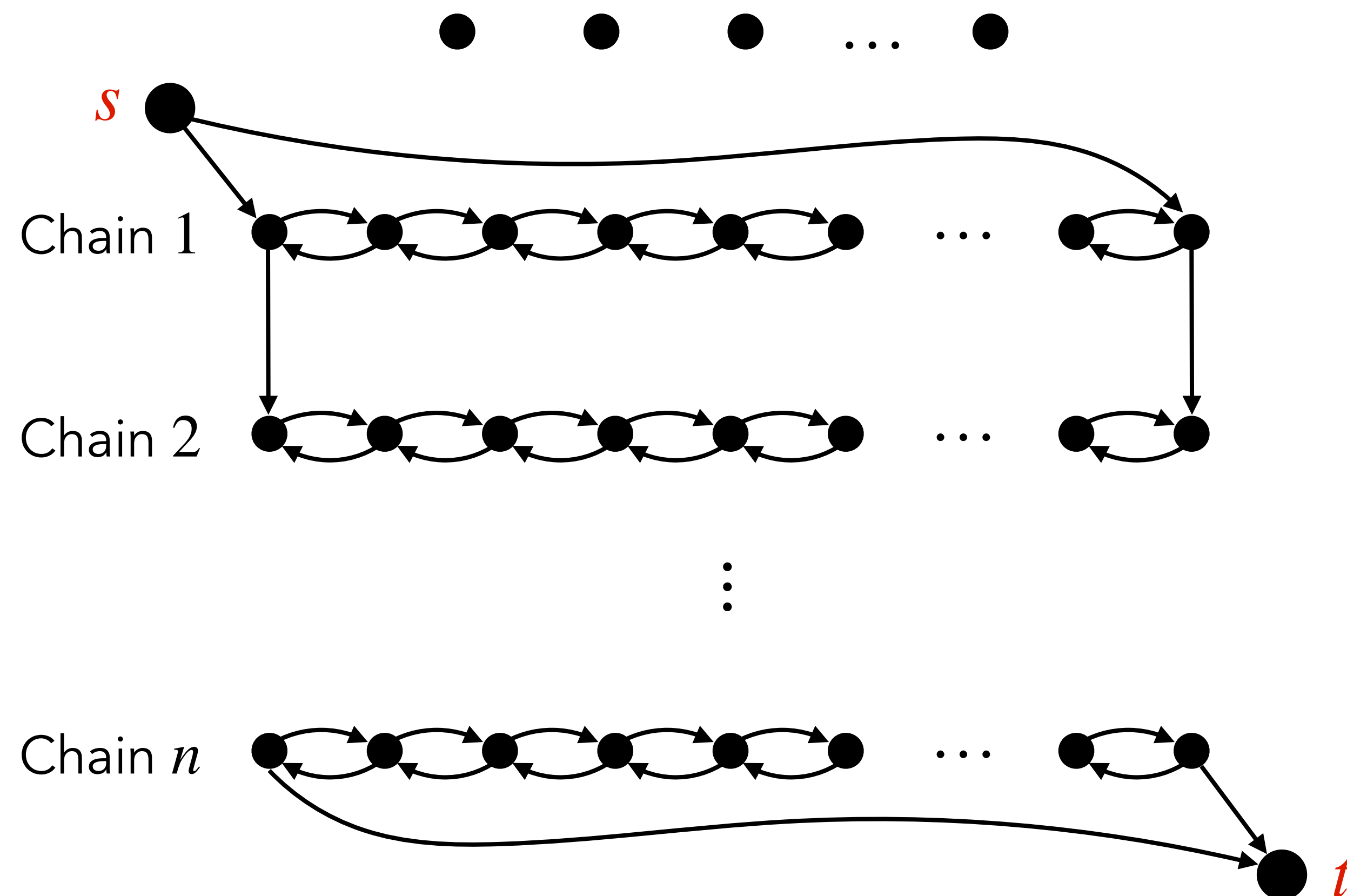
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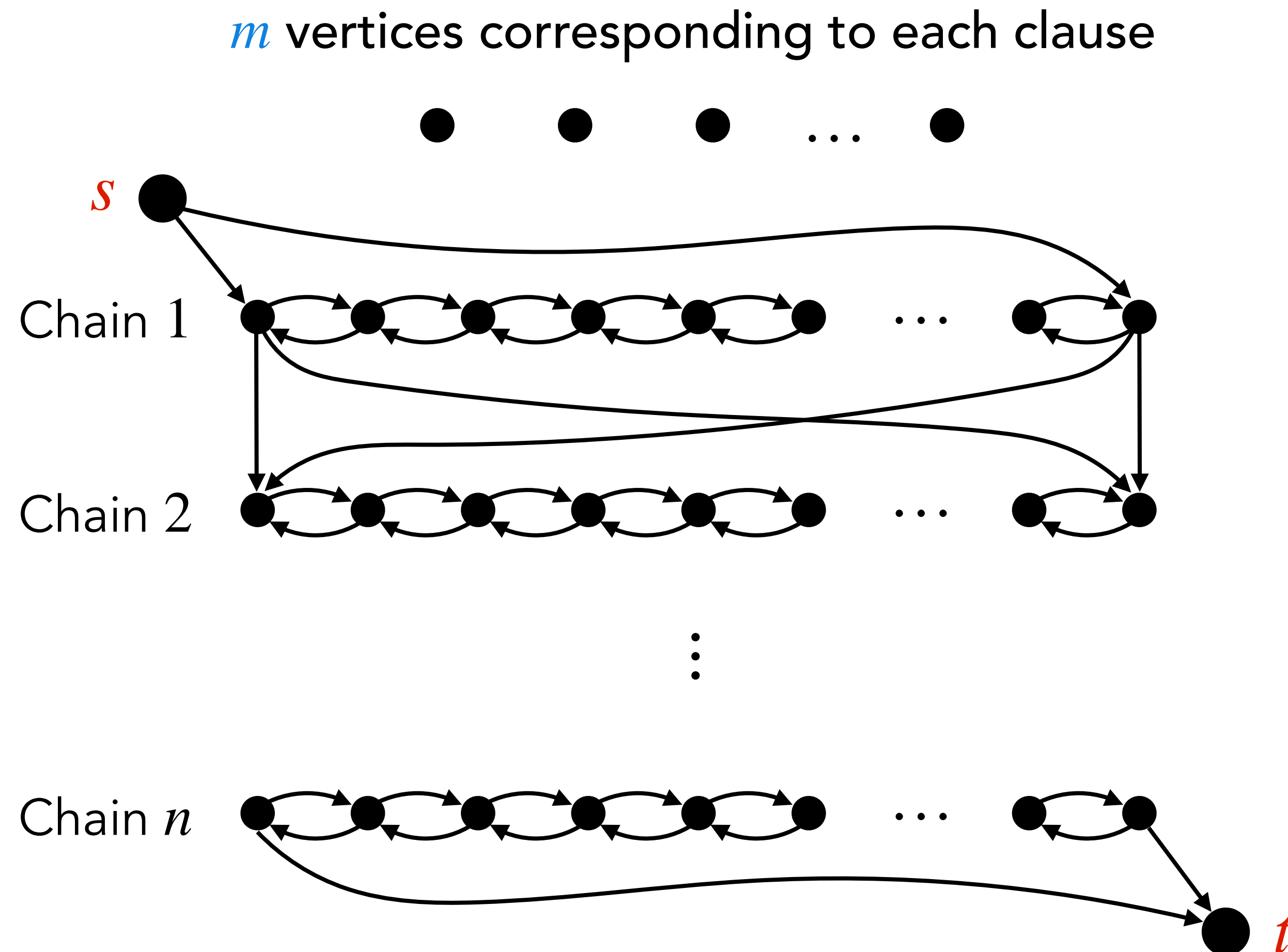
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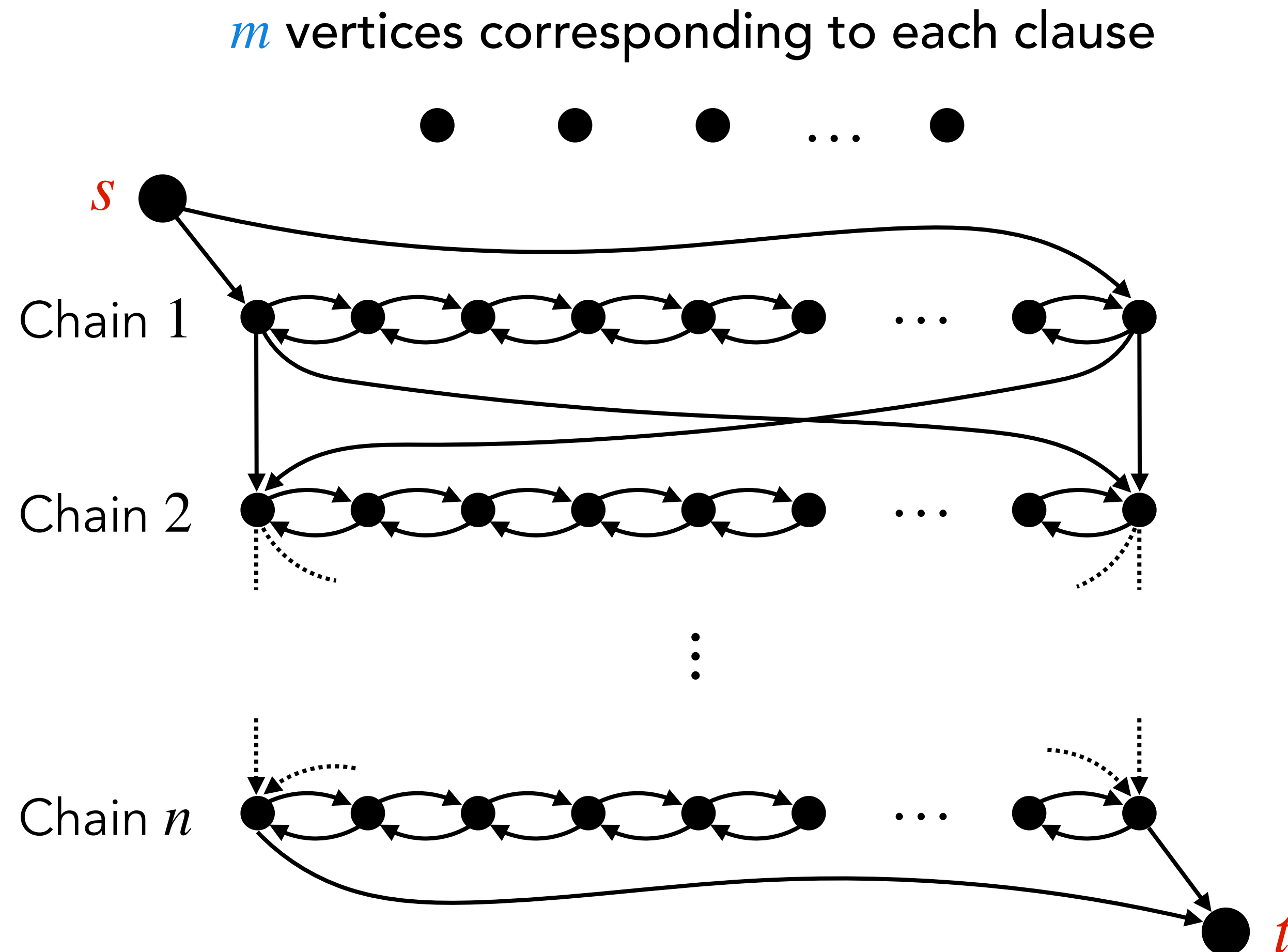
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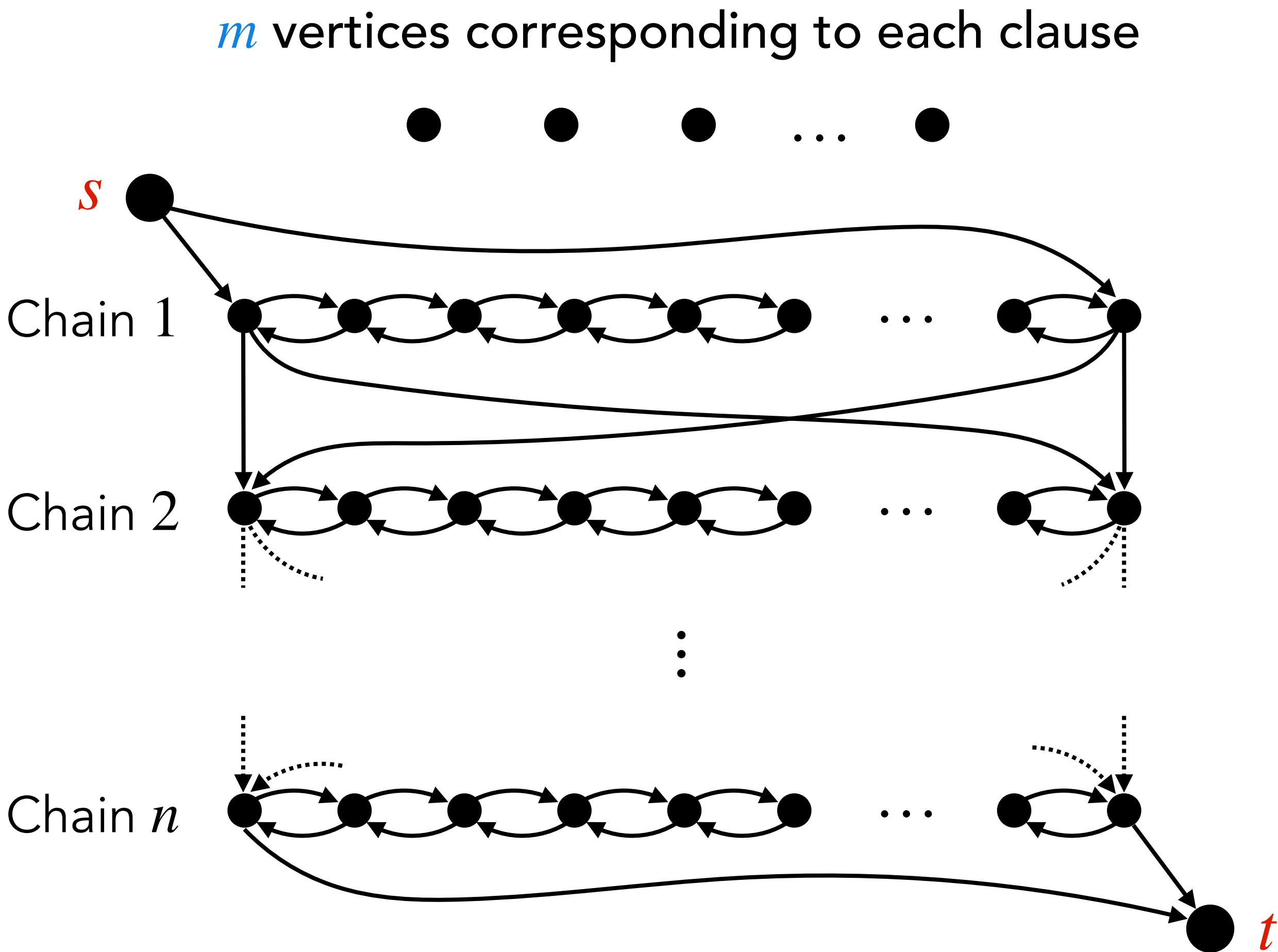
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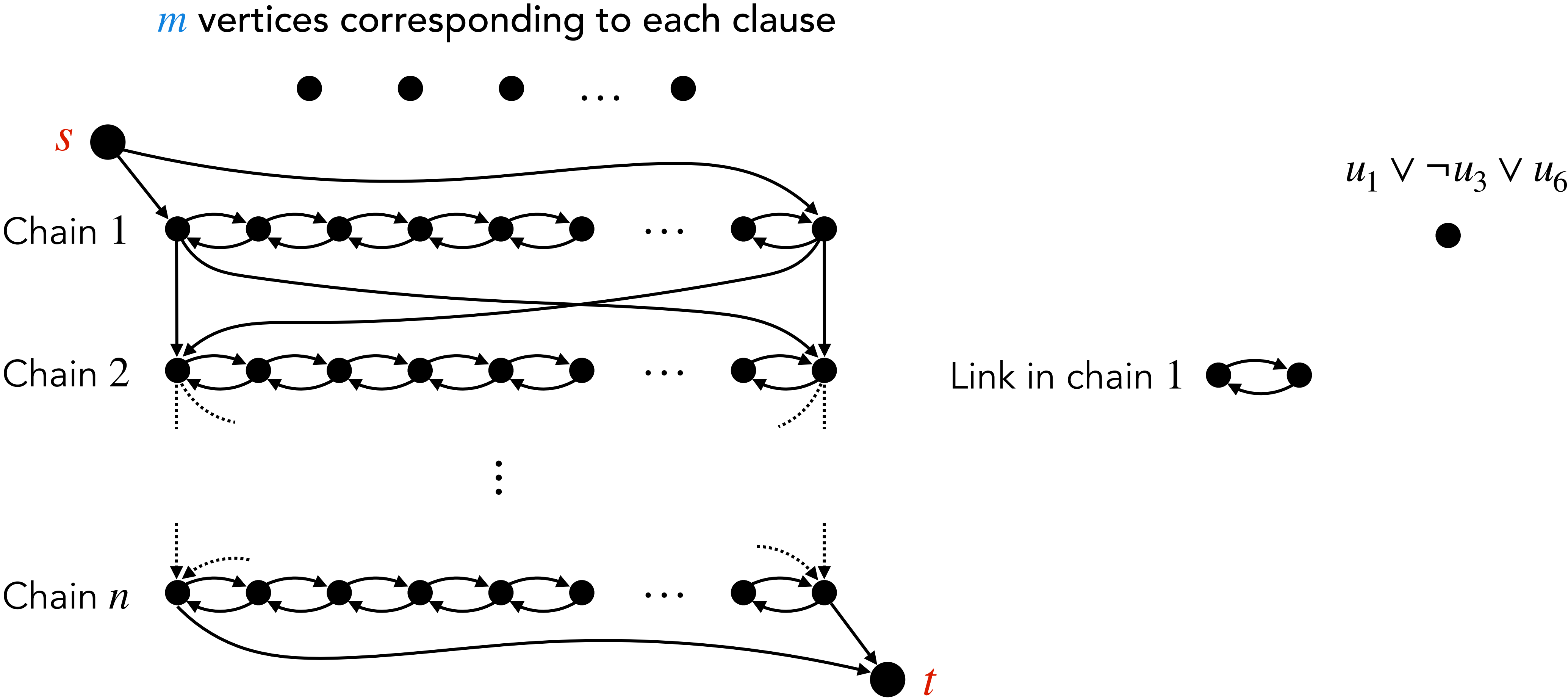


$$u_1 \vee \neg u_3 \vee u_6$$



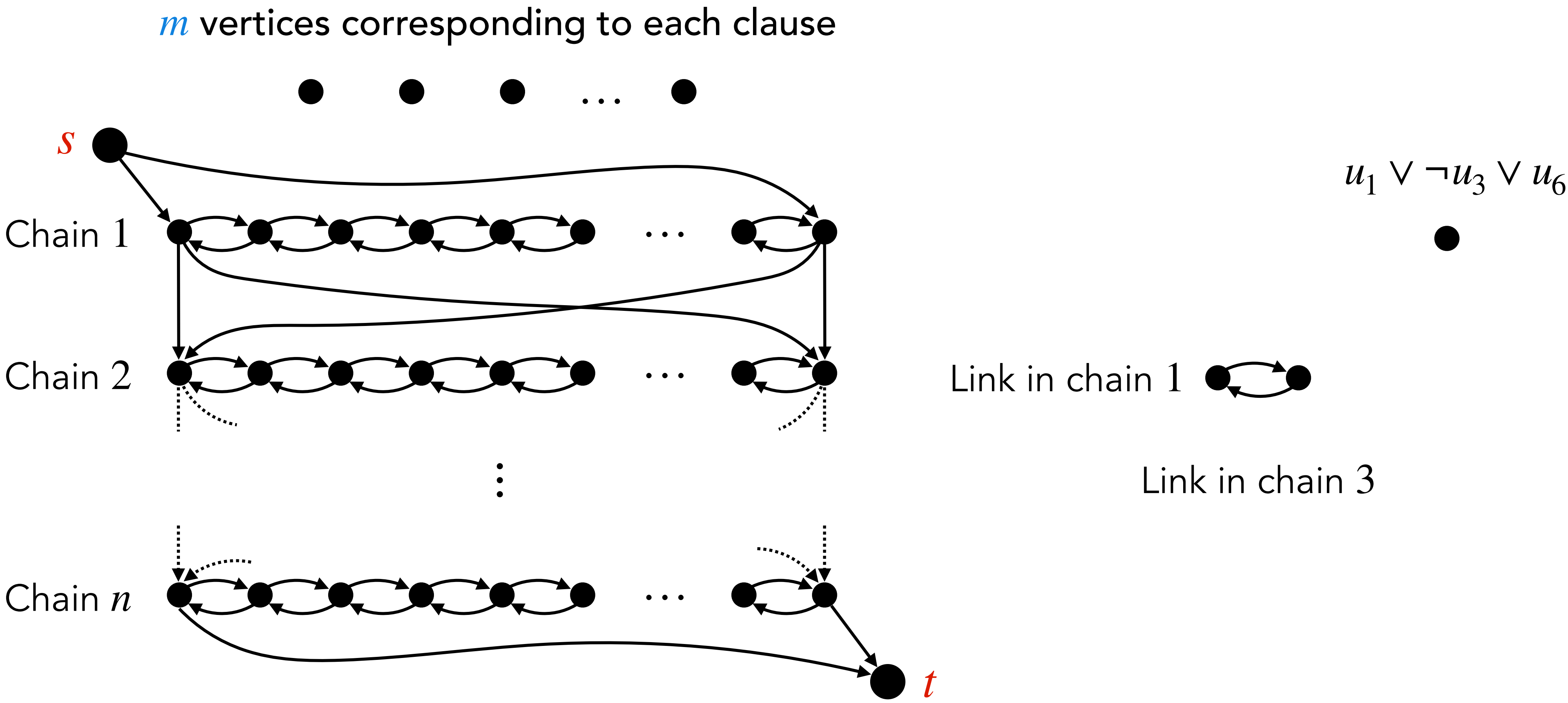
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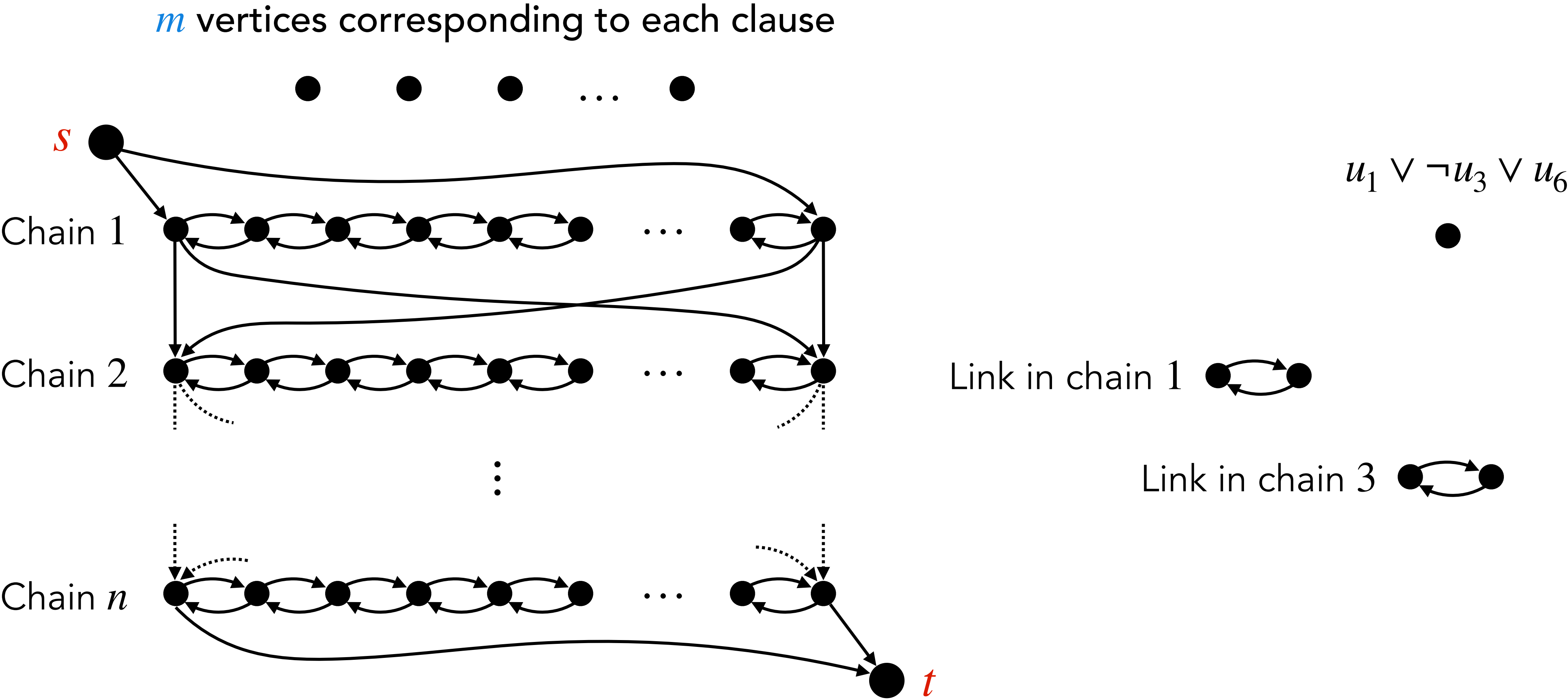
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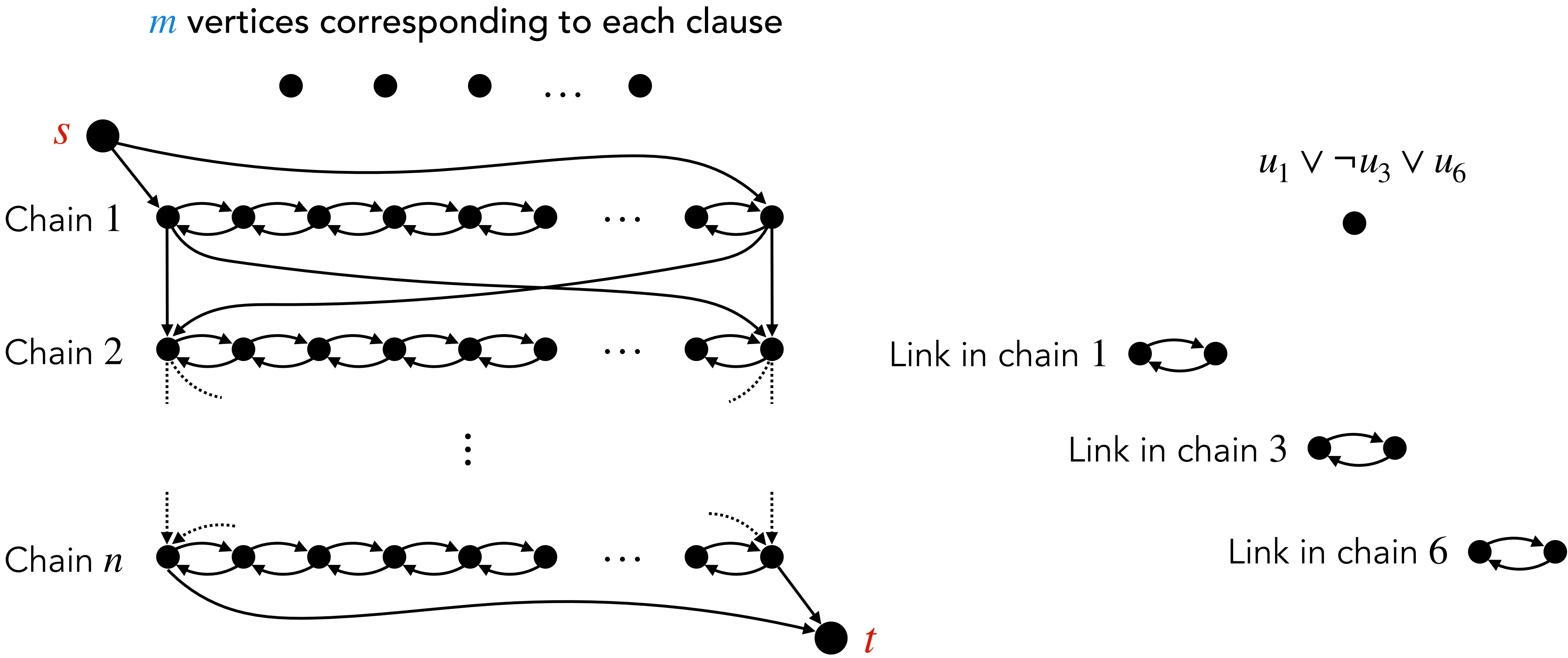
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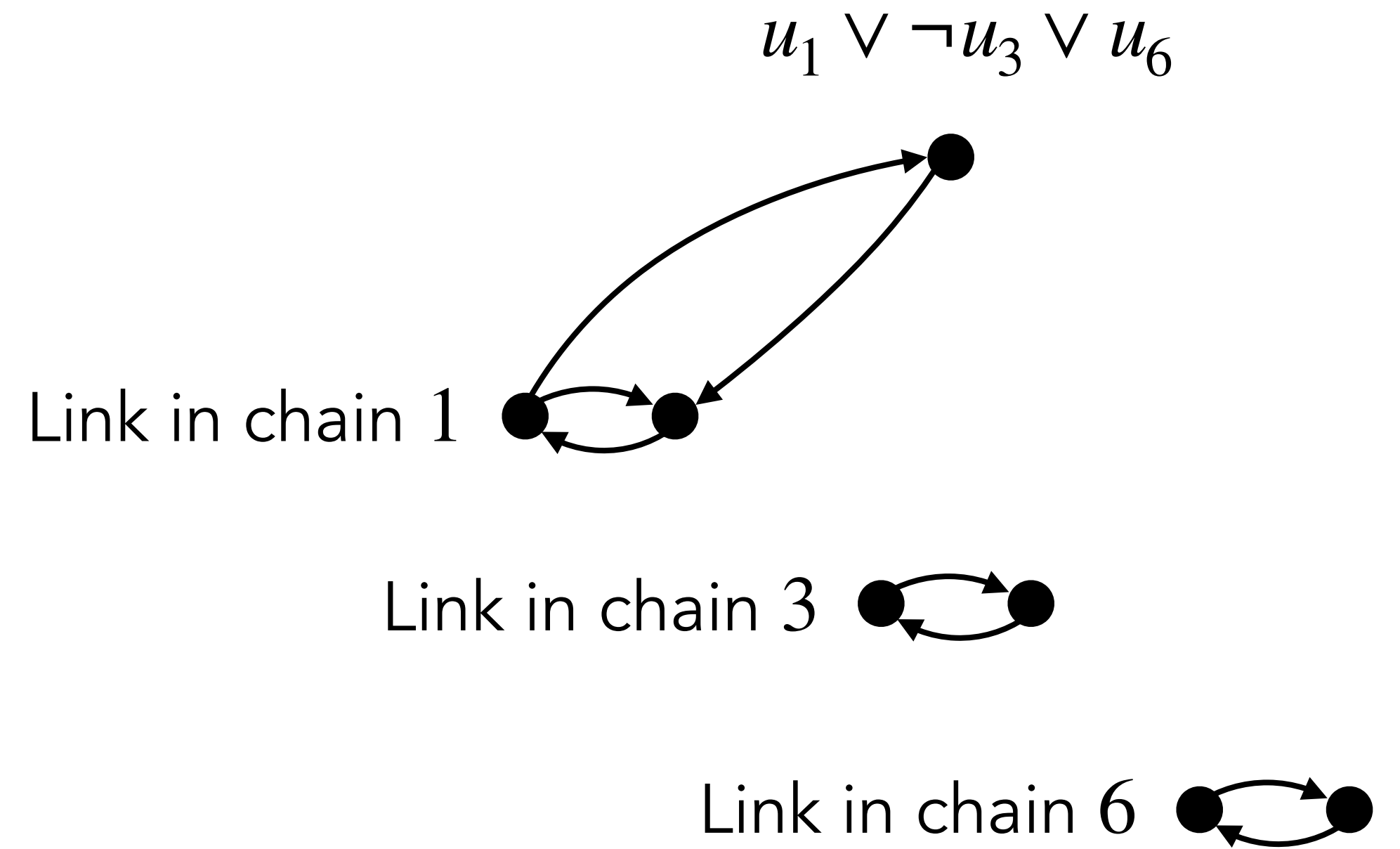
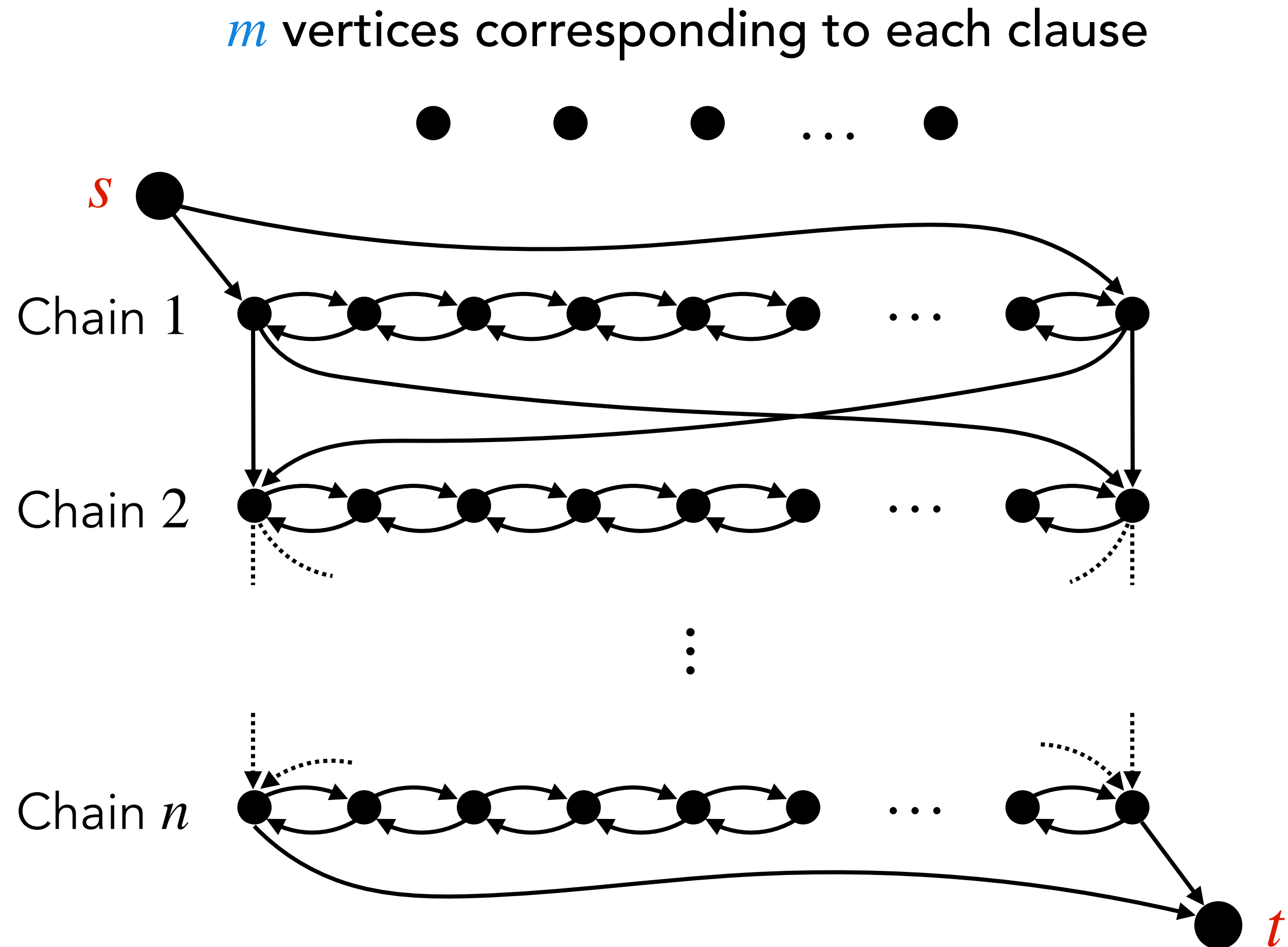
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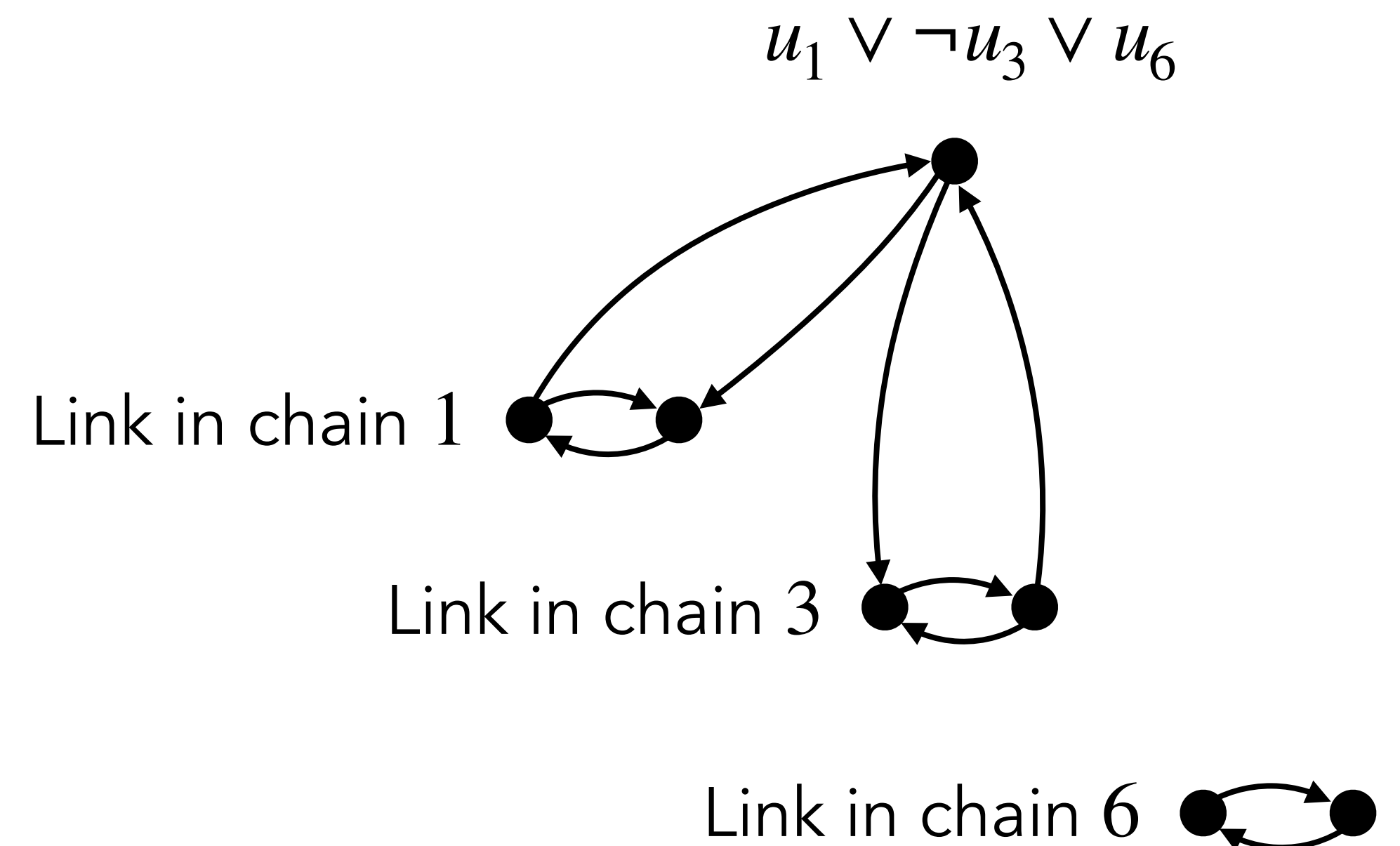
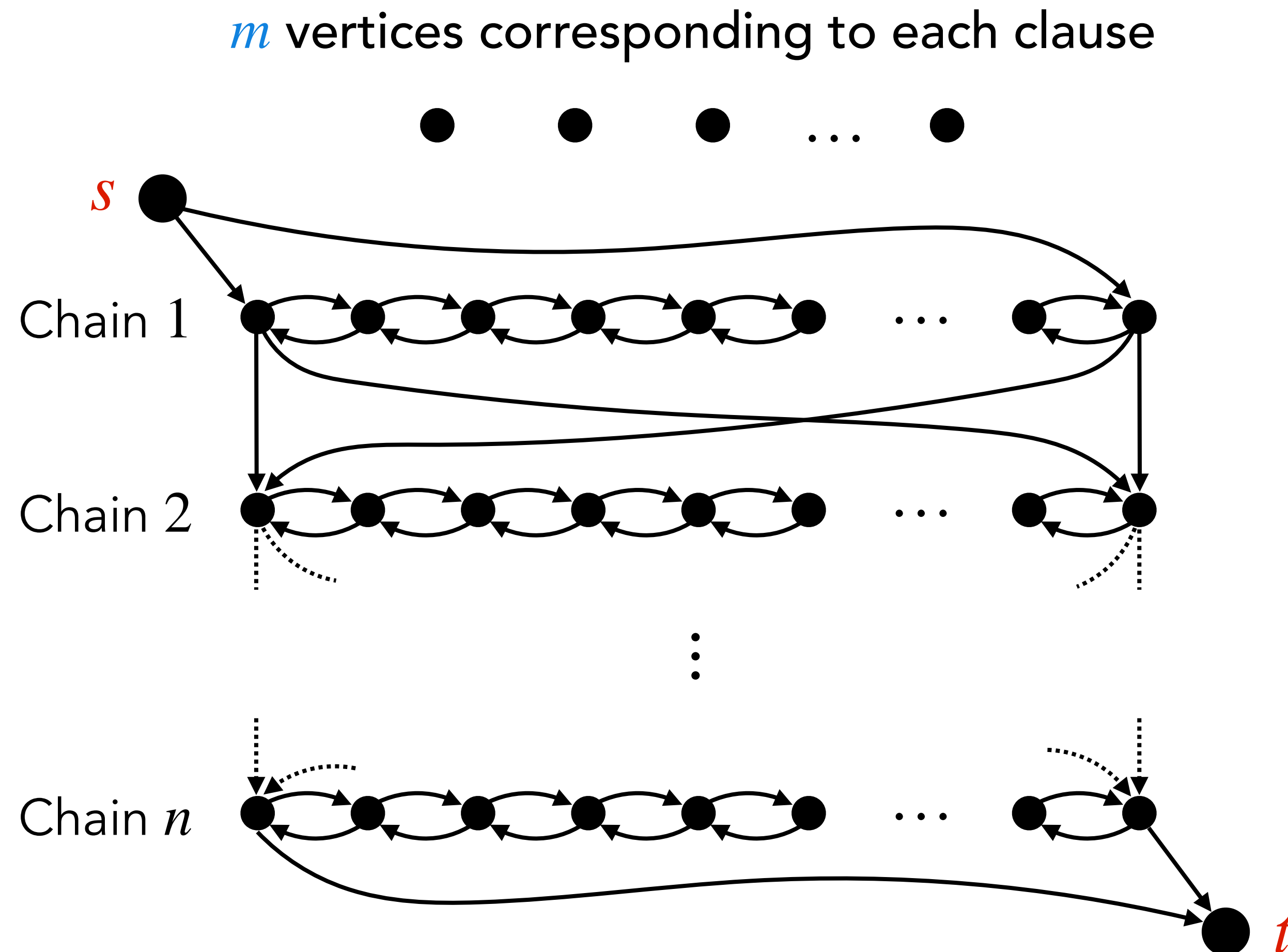
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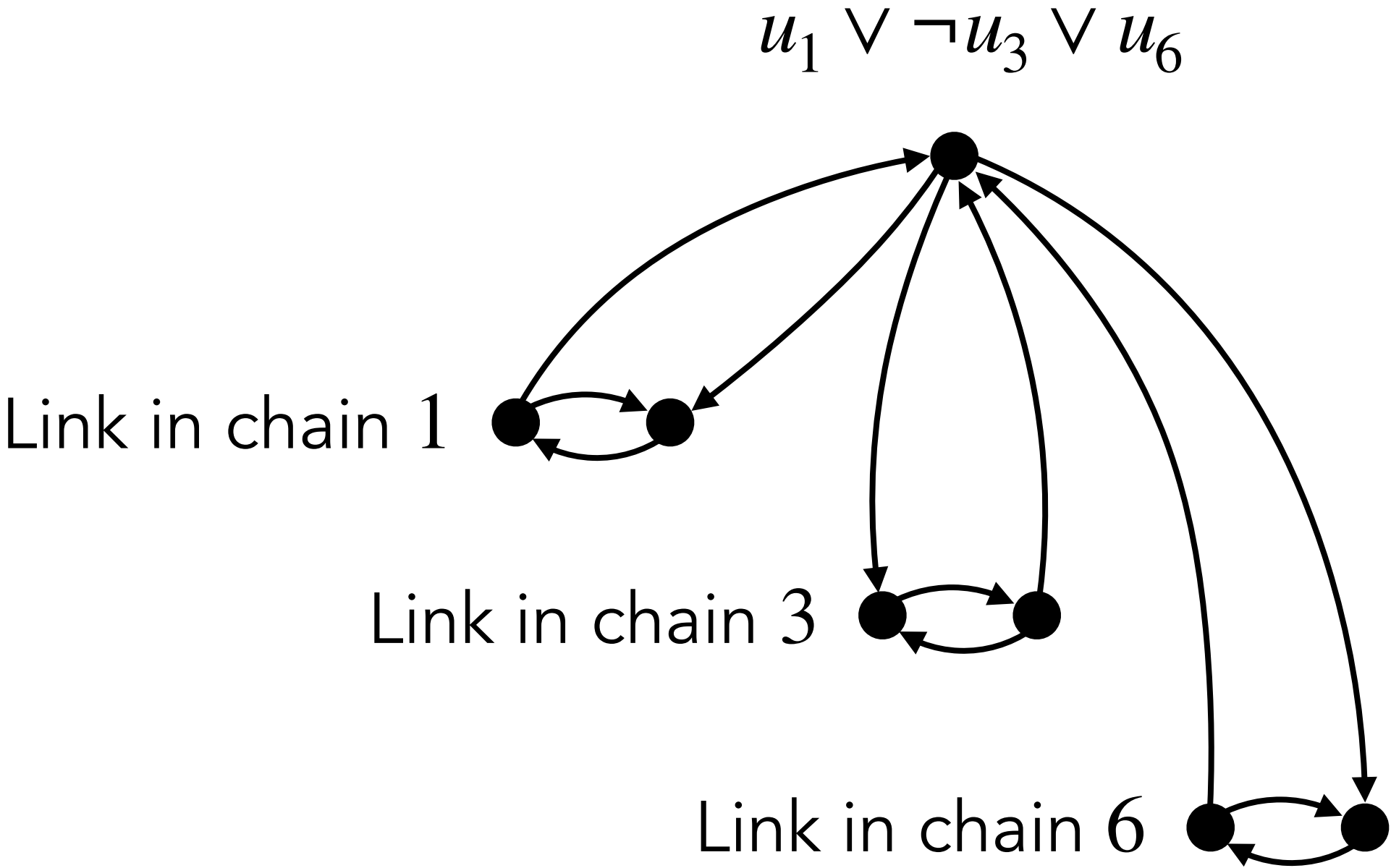
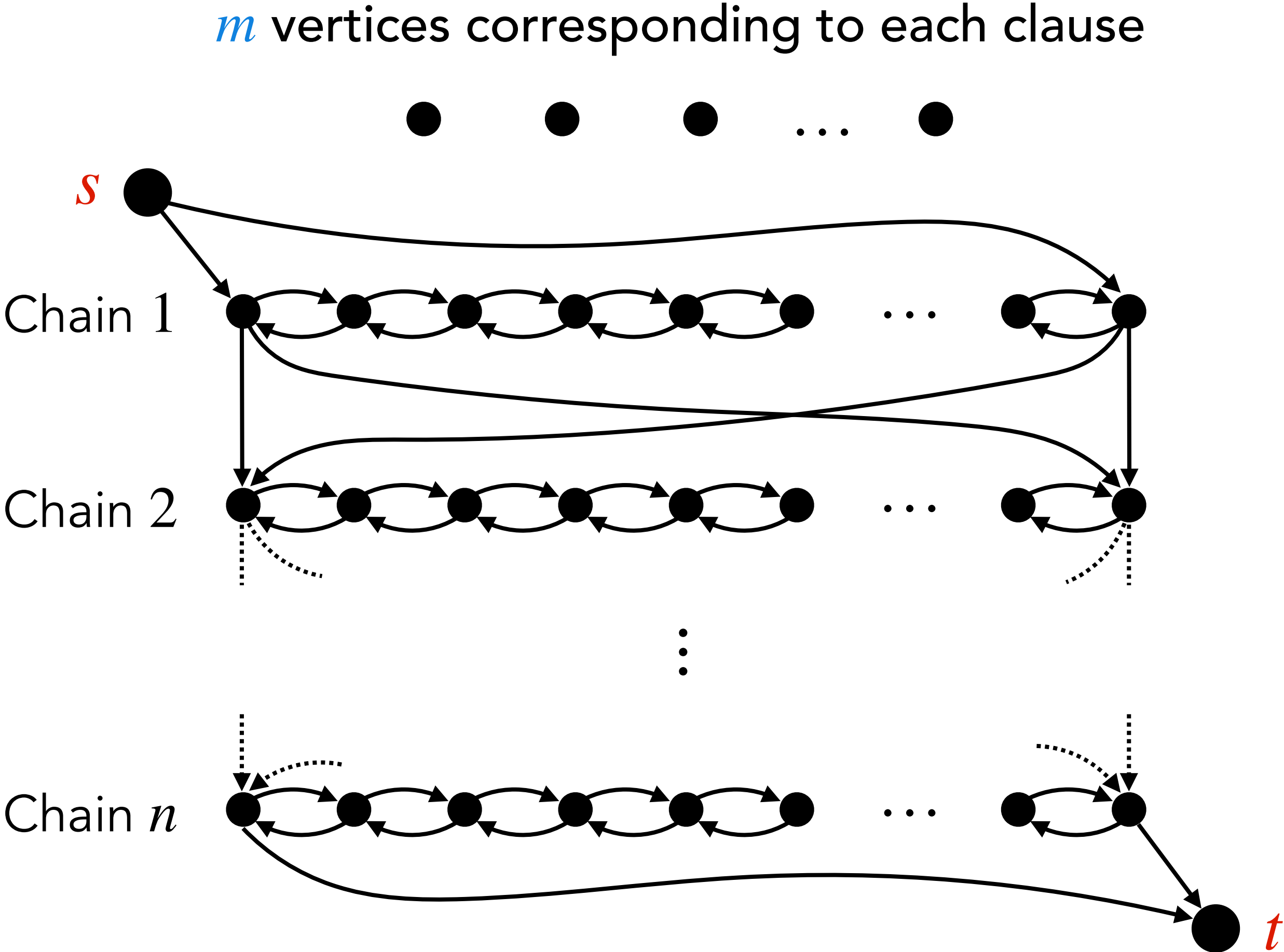
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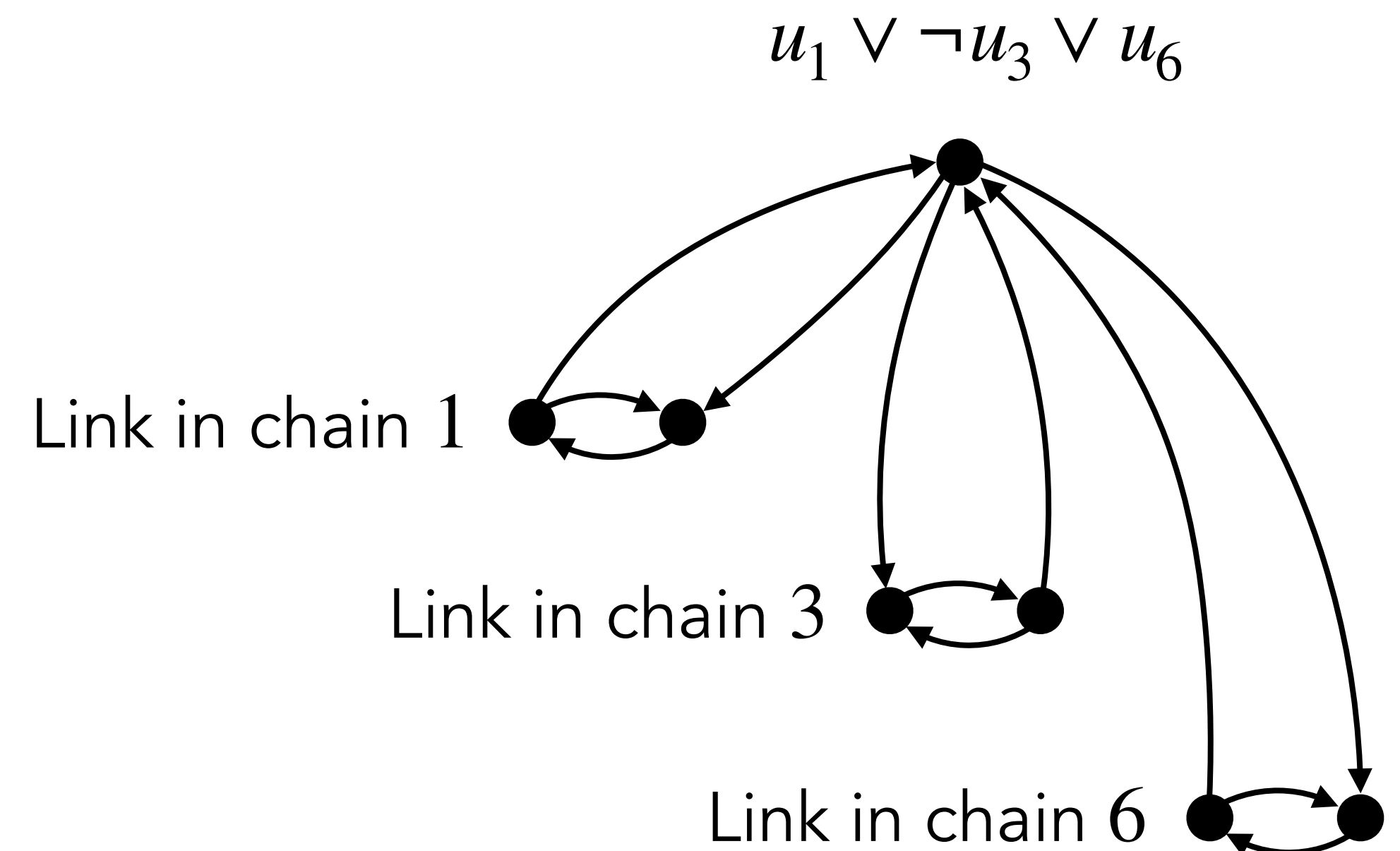
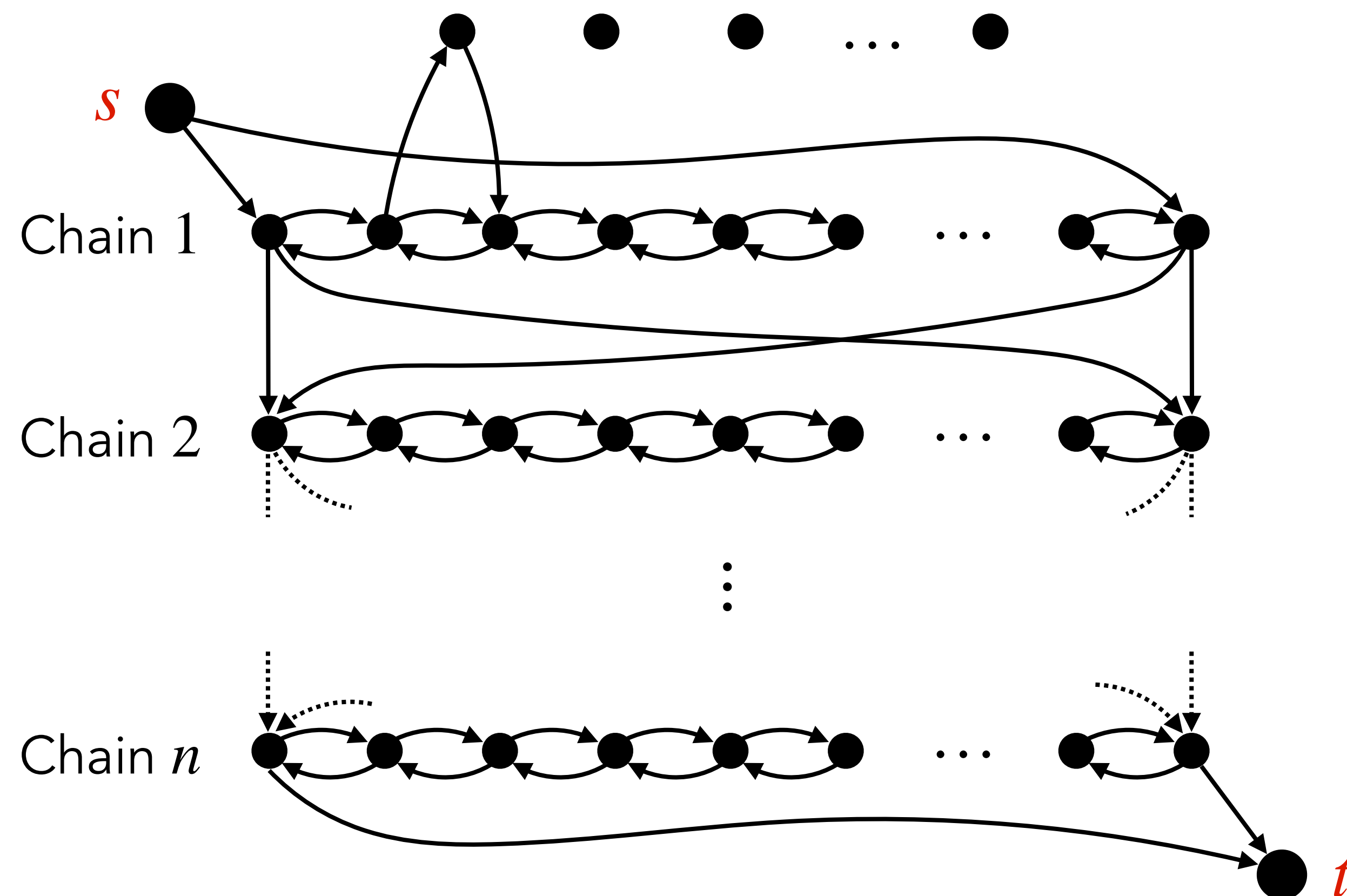
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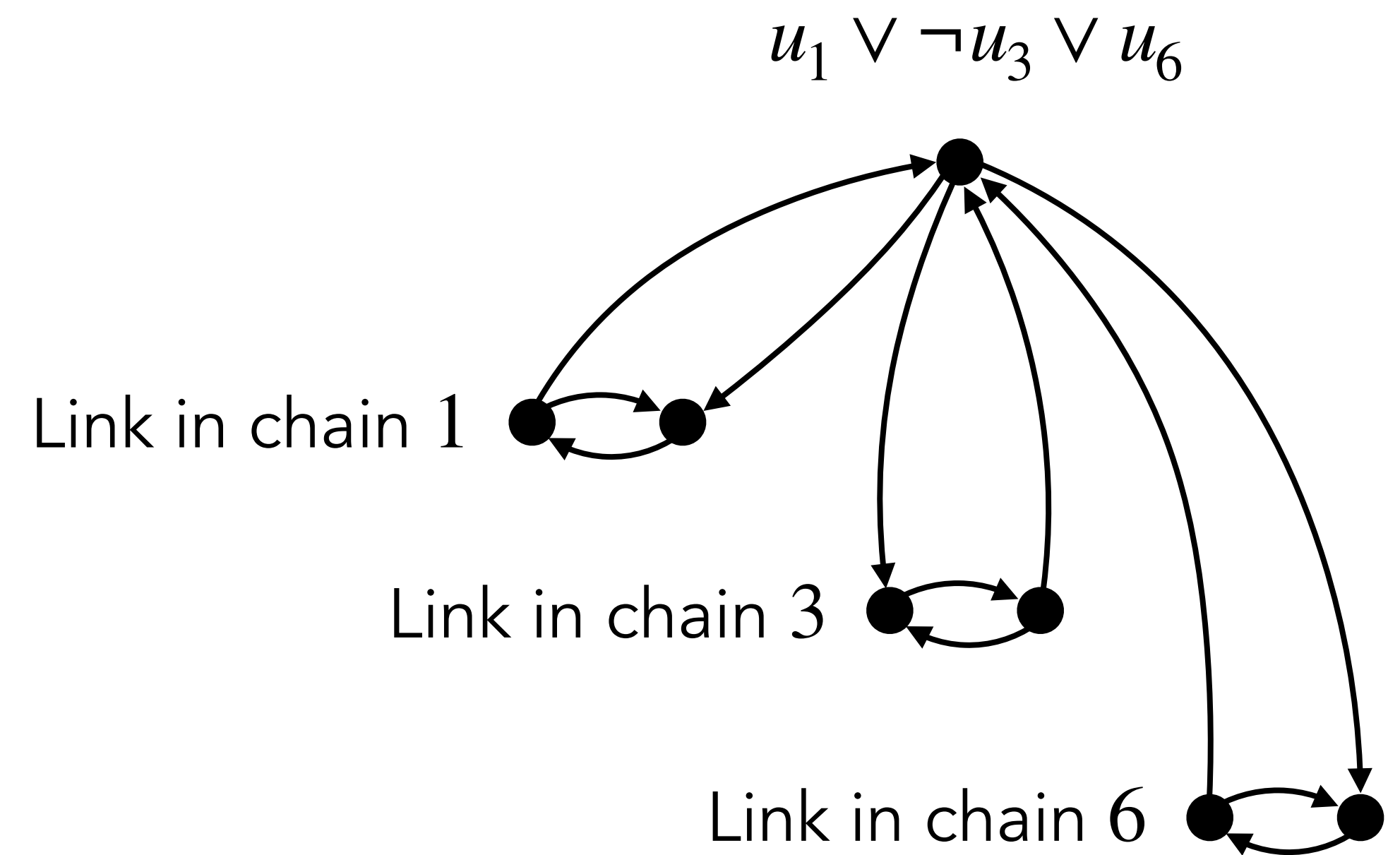
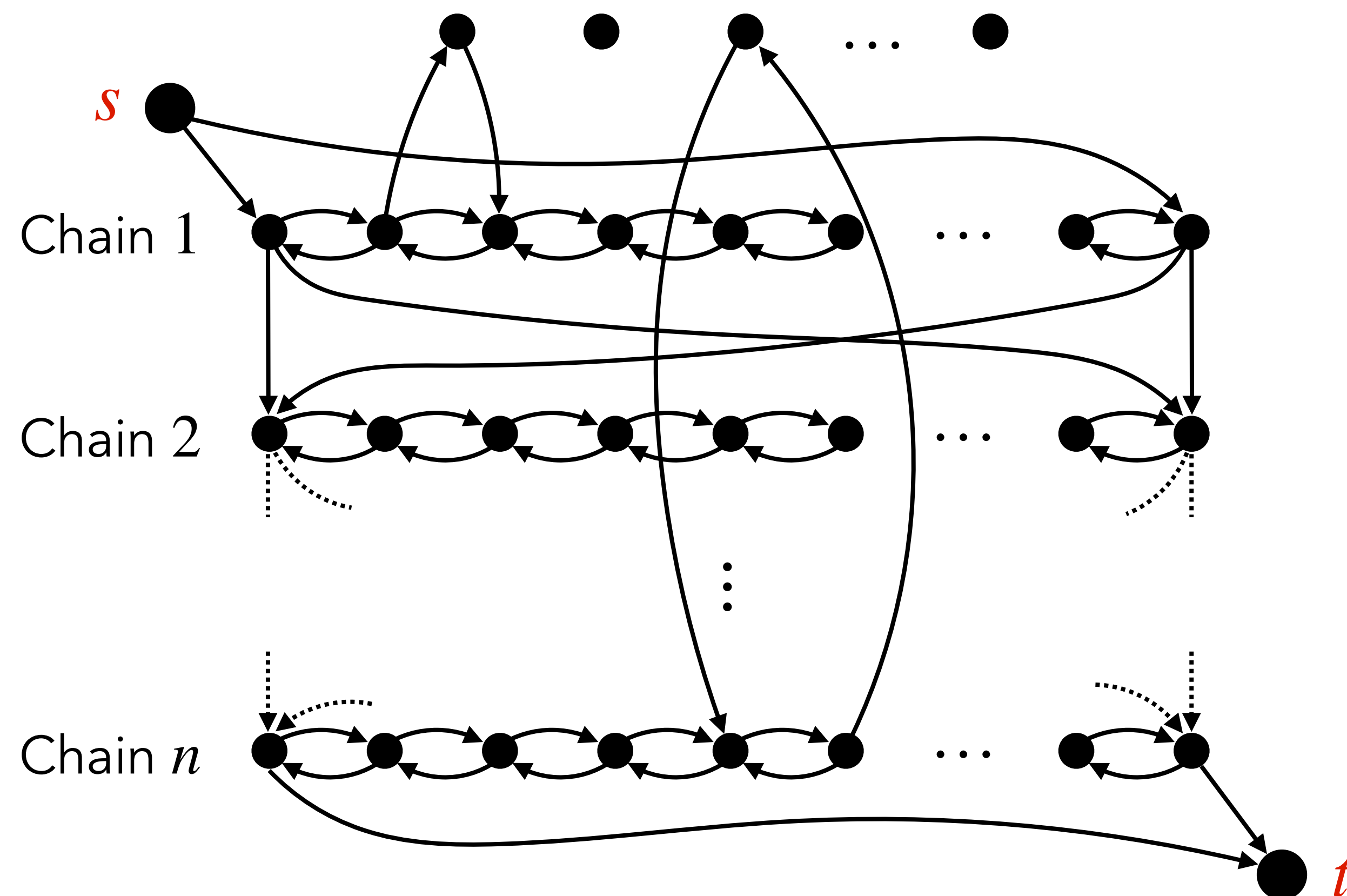
m vertices corresponding to each clause



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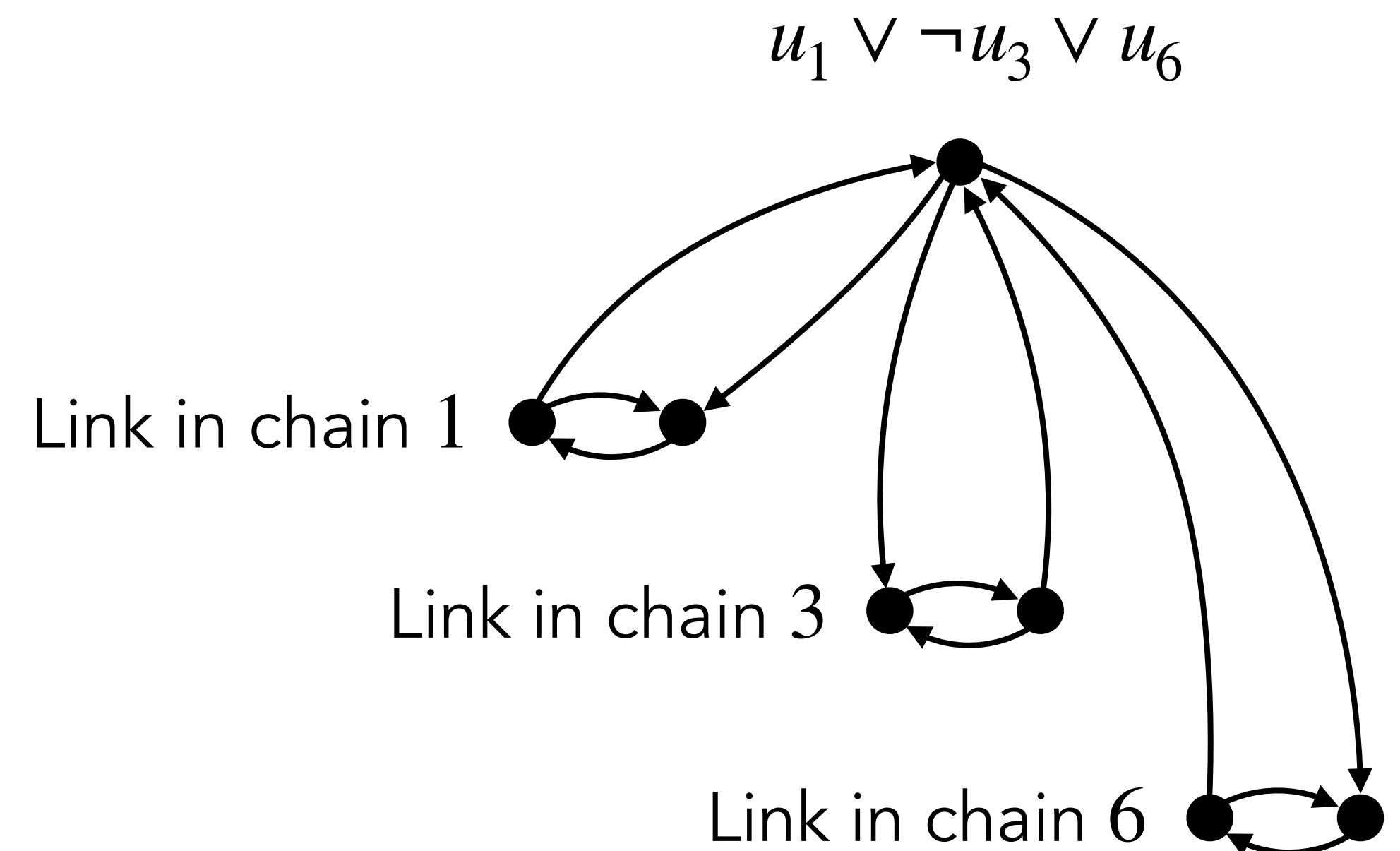
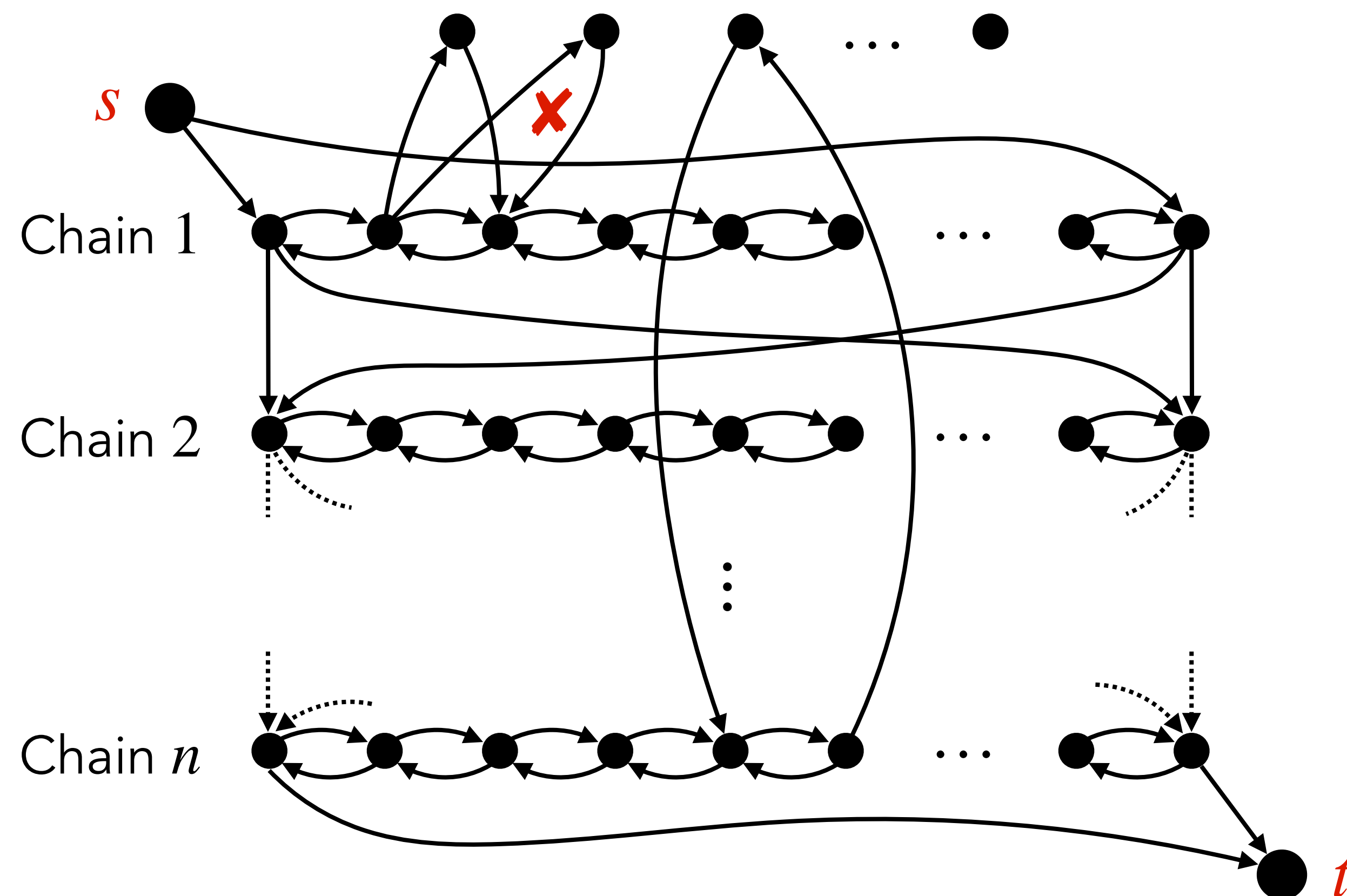
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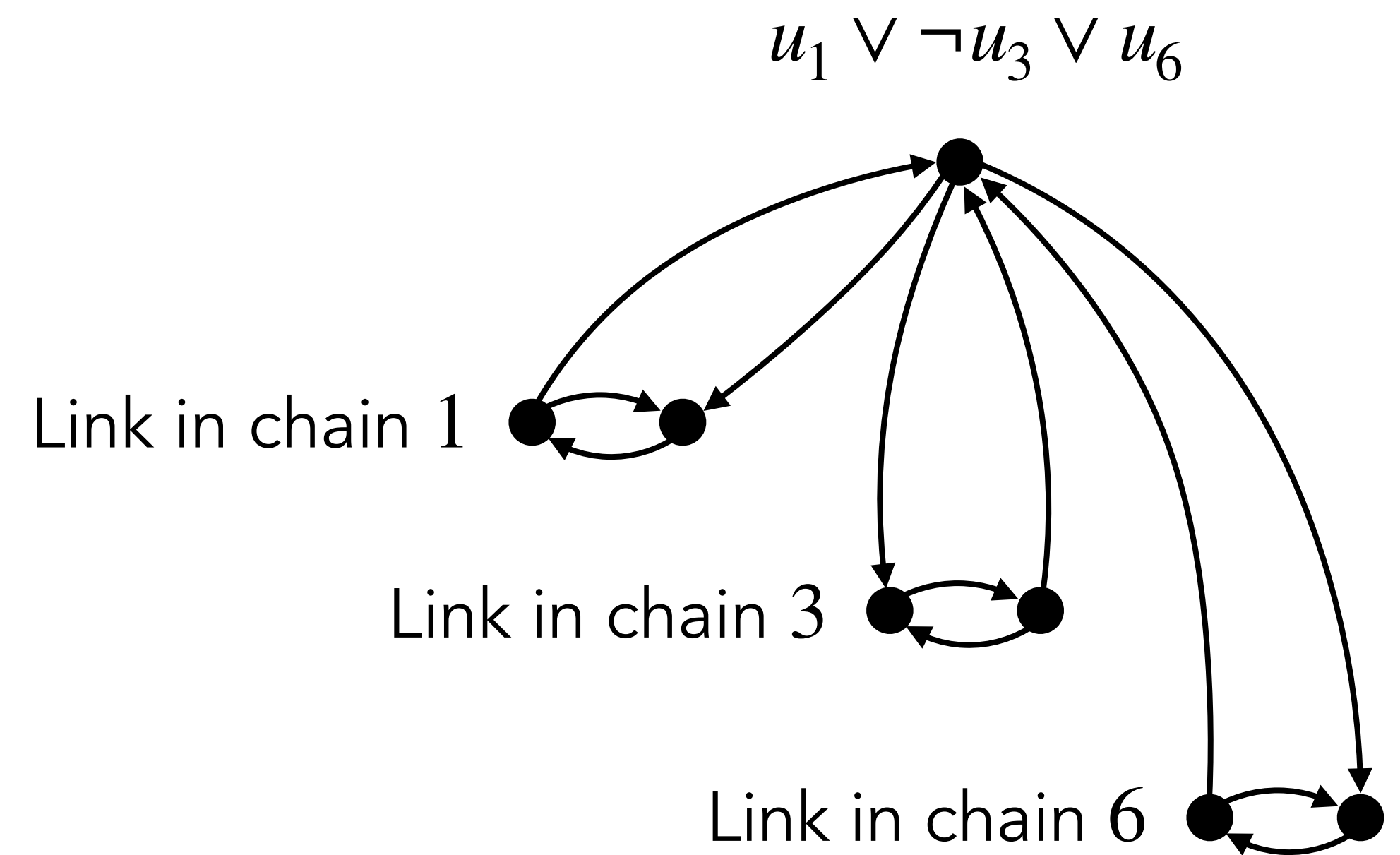
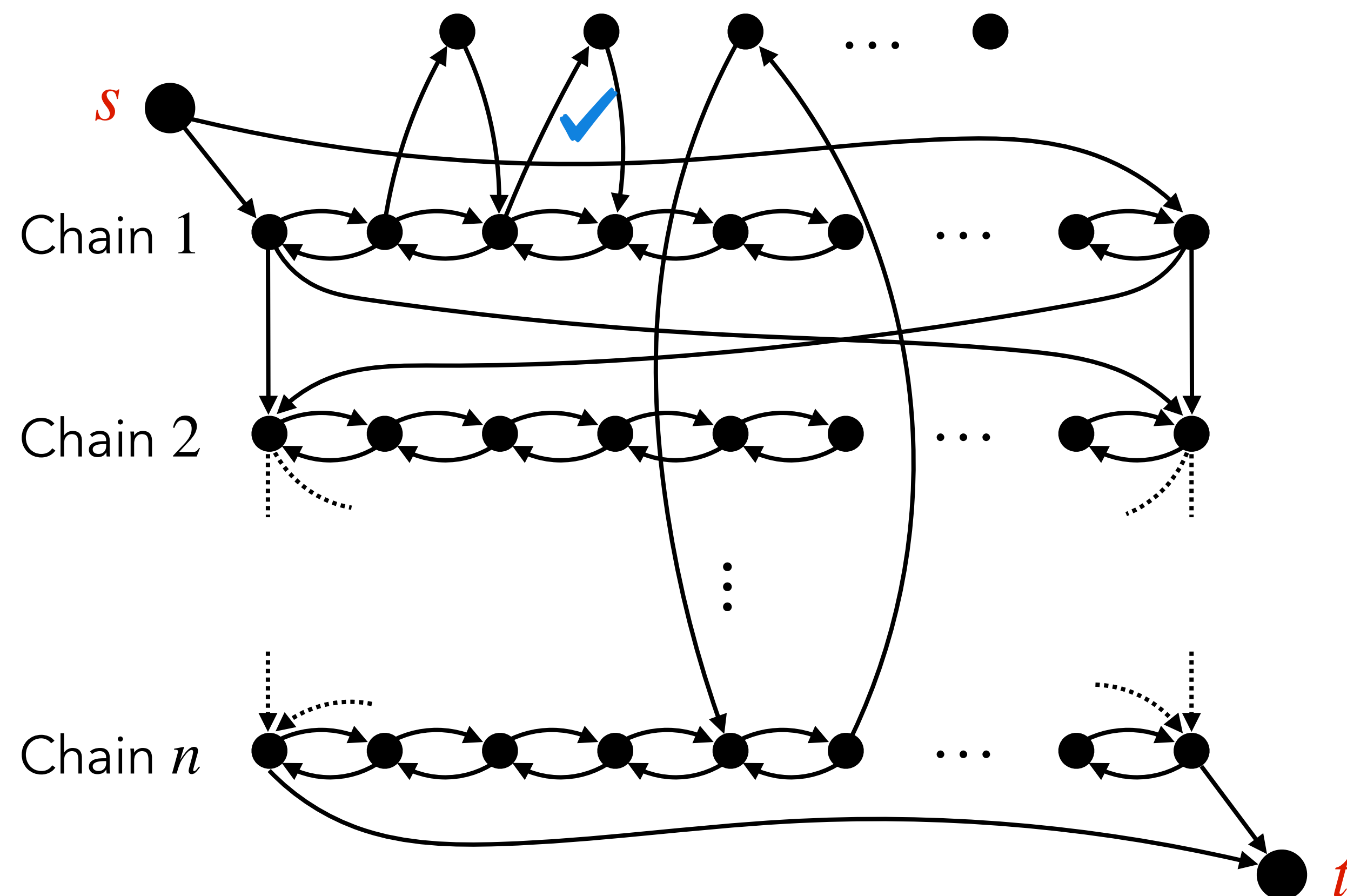
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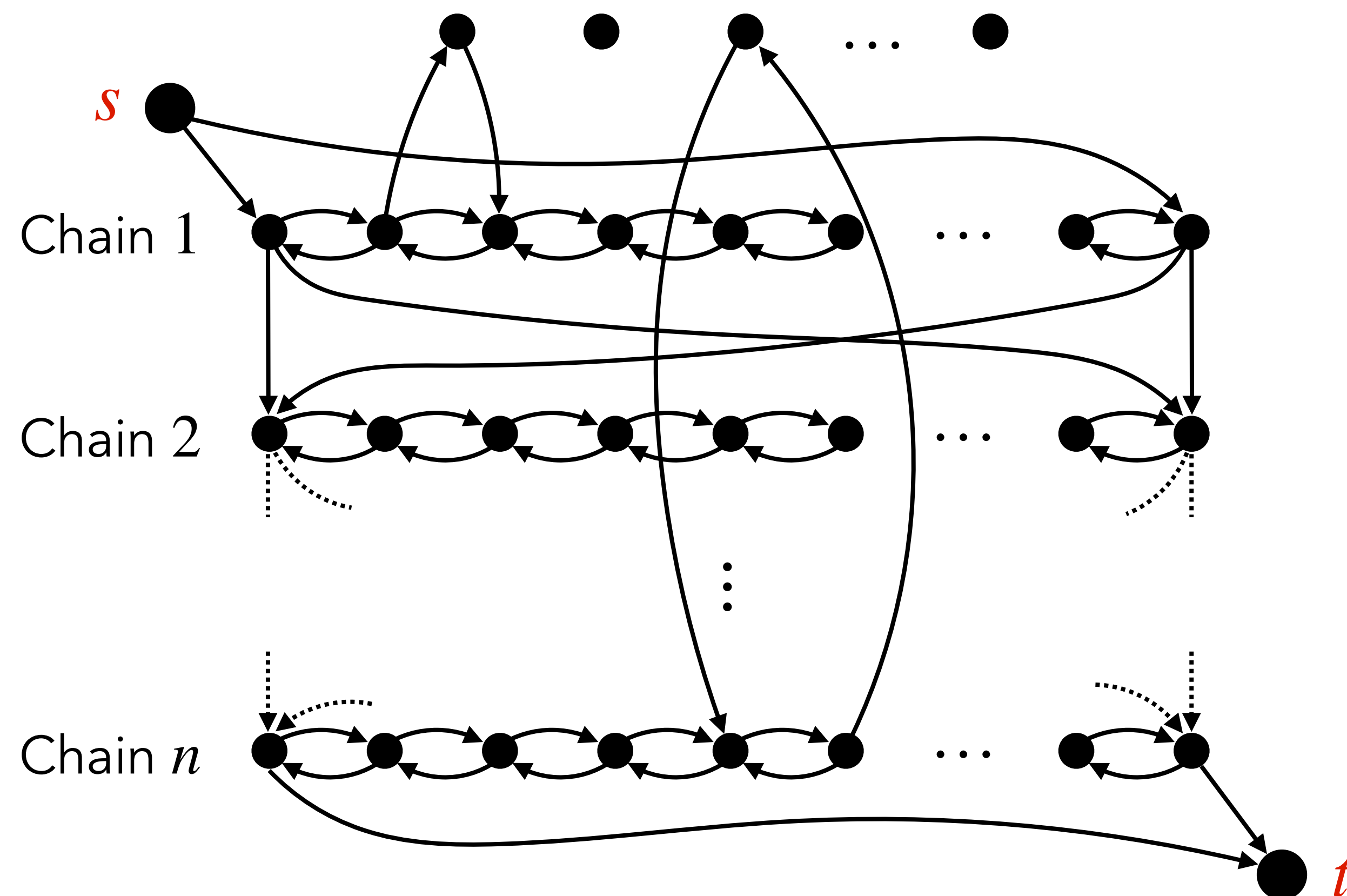
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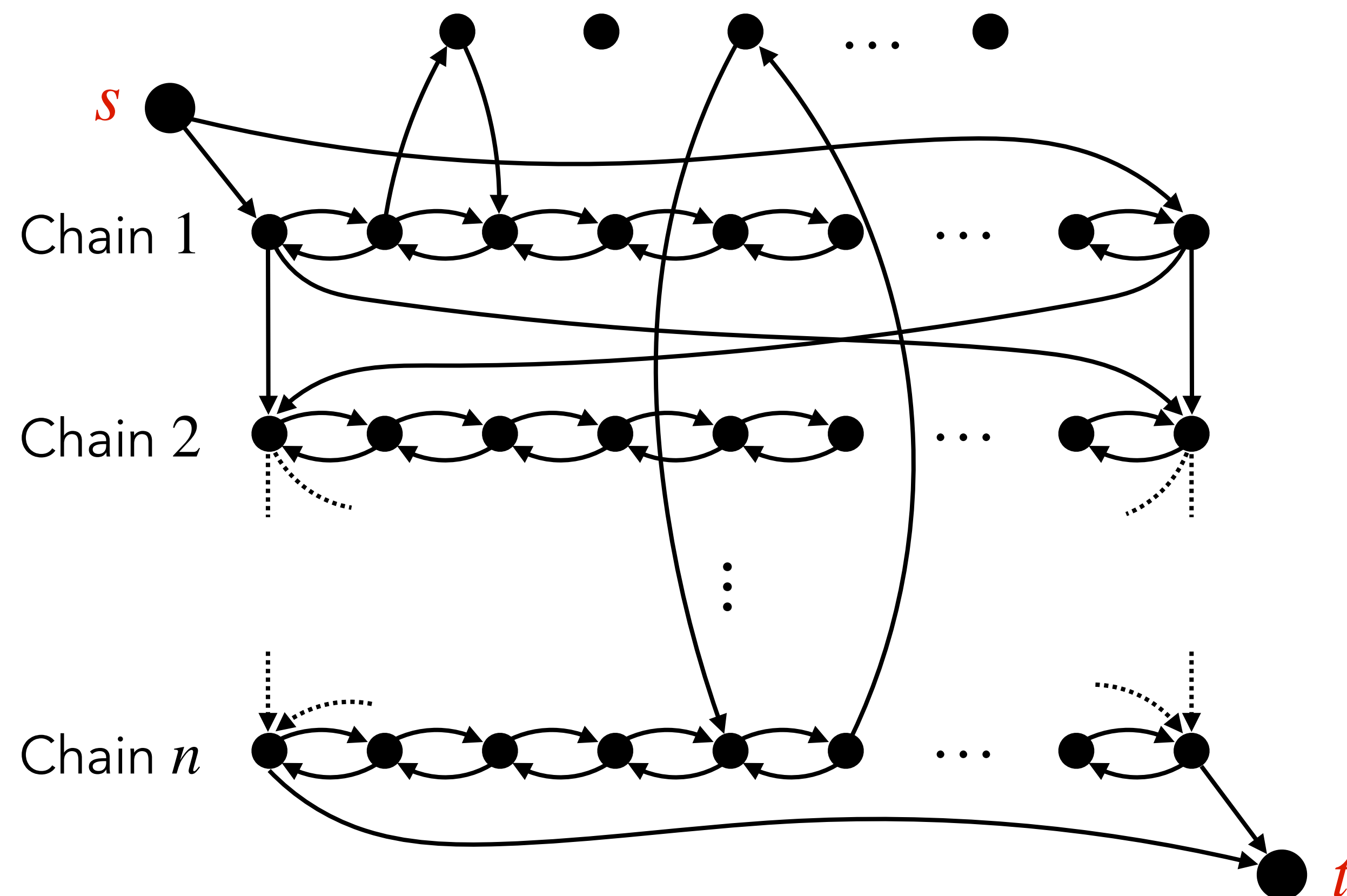
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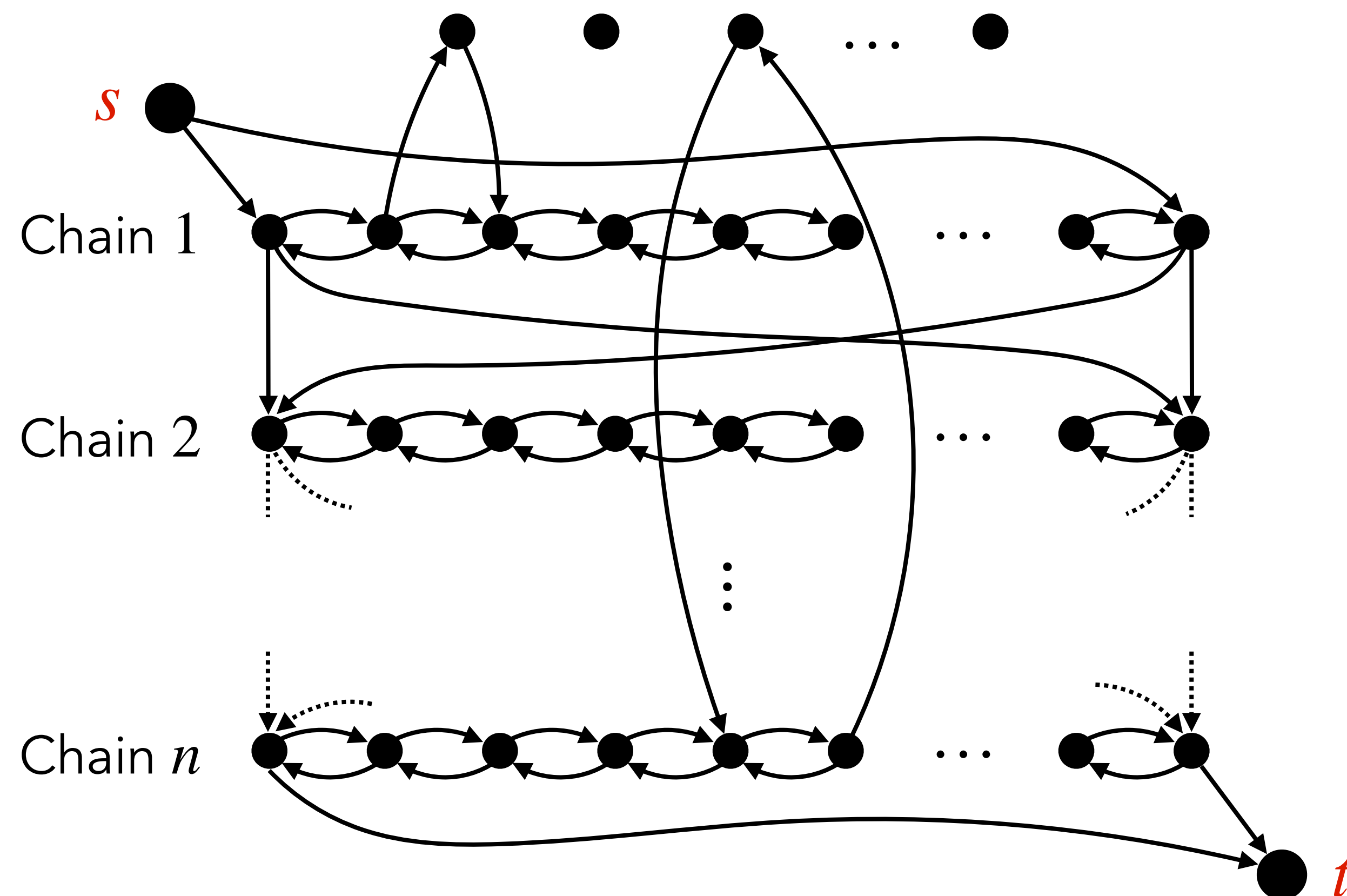
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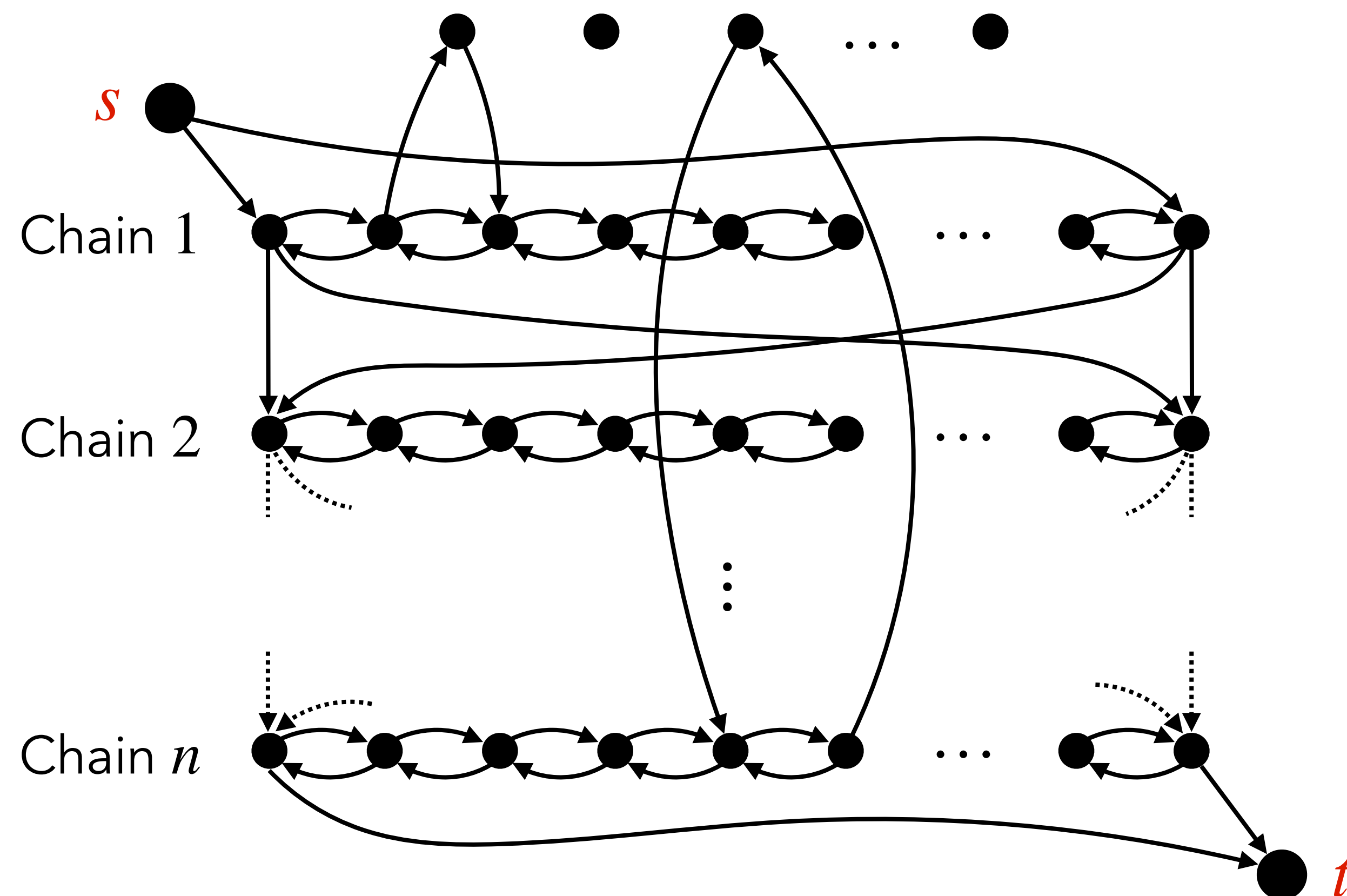
Vertices of G :

- Two special vertices s and t .

$3SAT \leq_p DirHampath$

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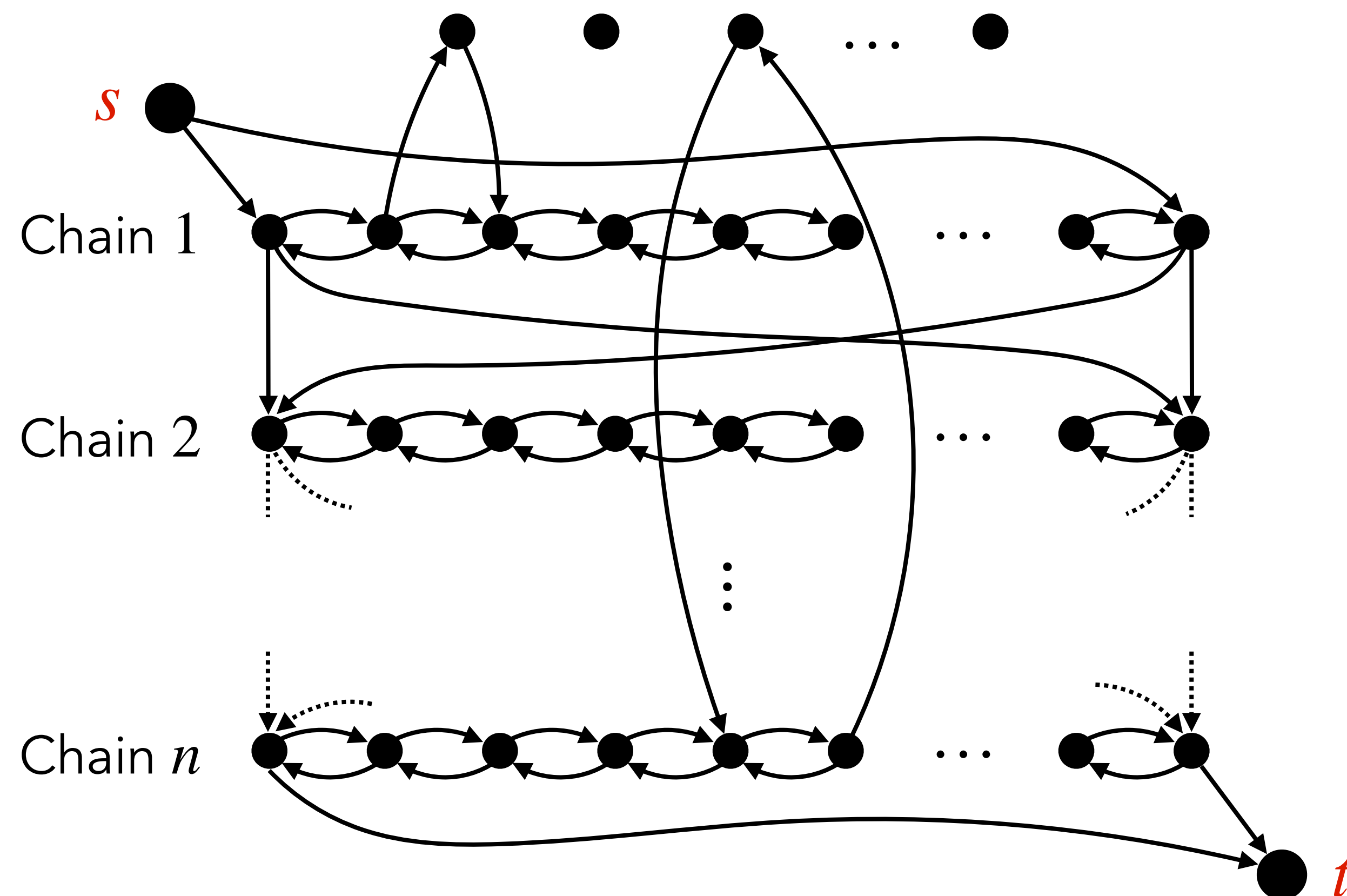
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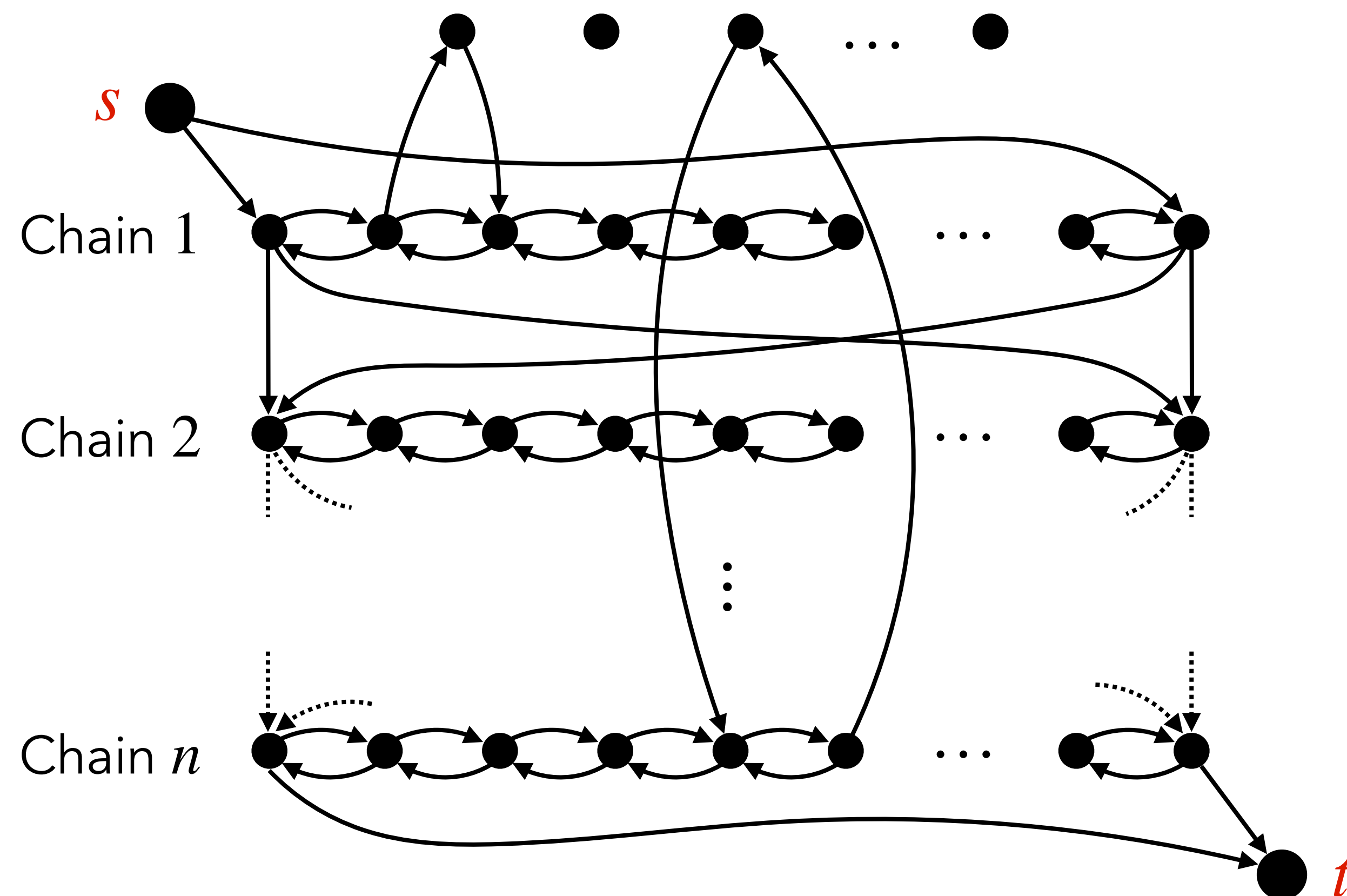
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- A bidirectional chain of $3m$ vertices for every variable.
- A vertex for every clause.

$3SAT \leq_p DirHampath$

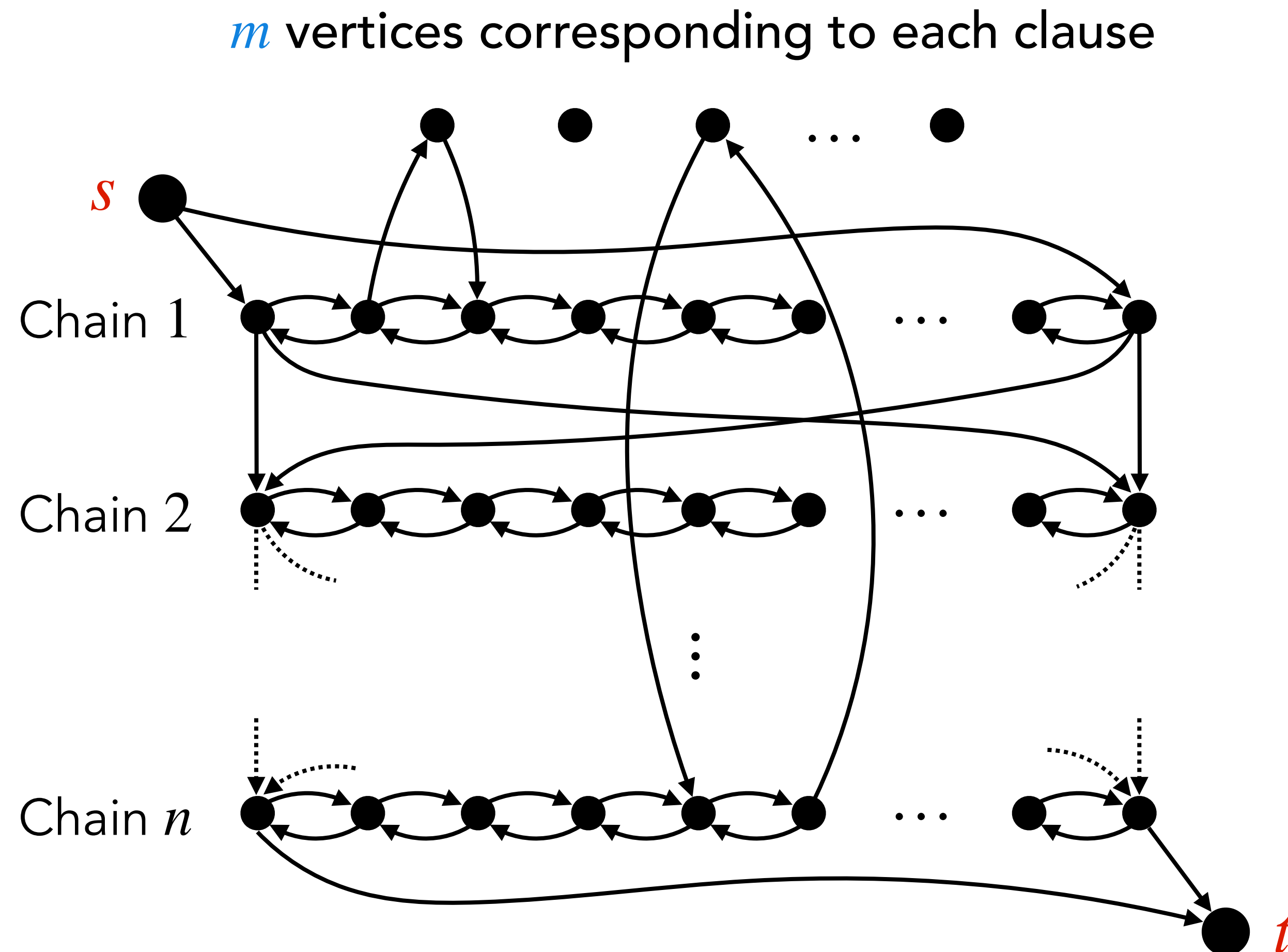
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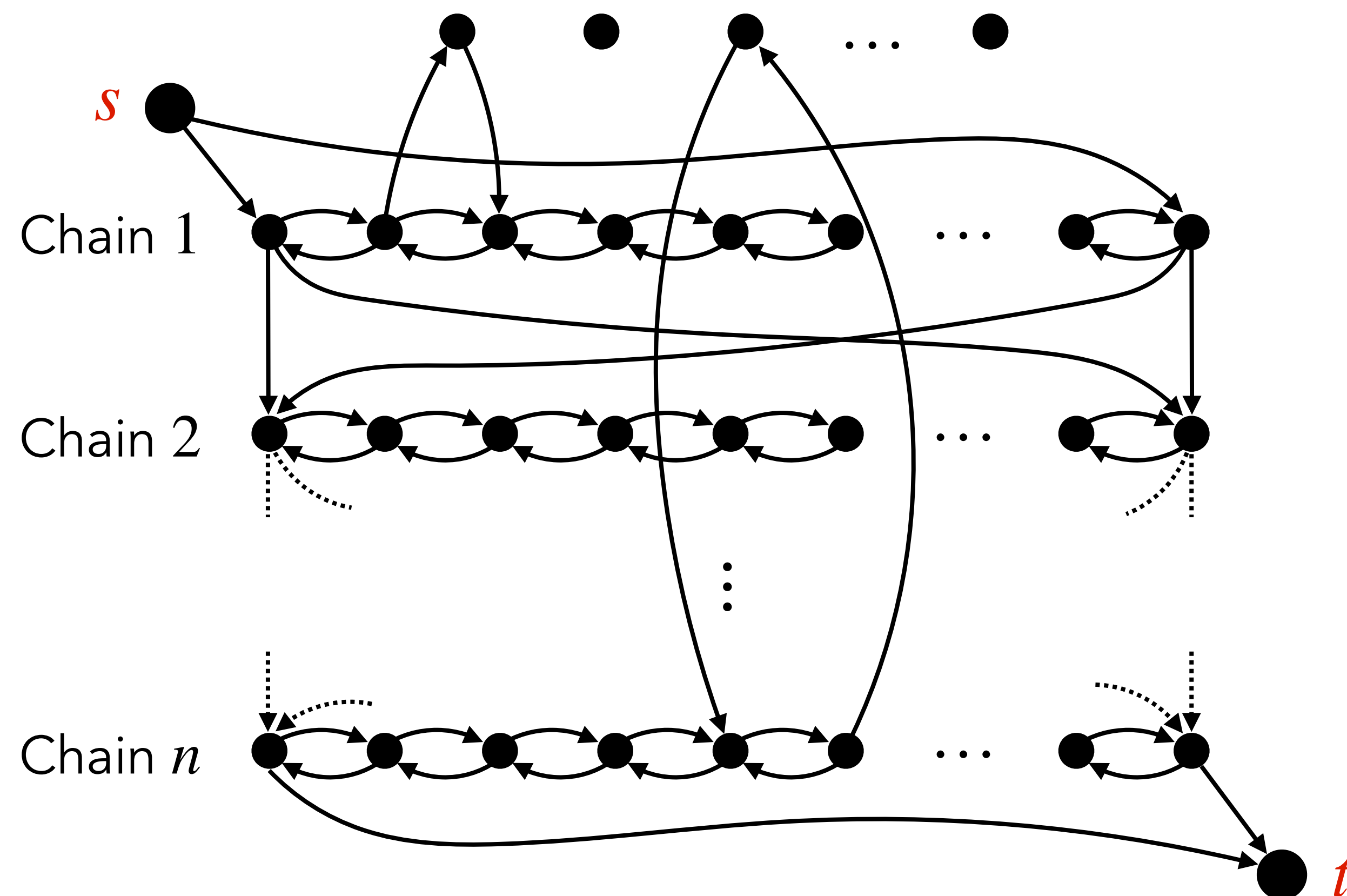


Edges of G :

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m vertices corresponding to each clause



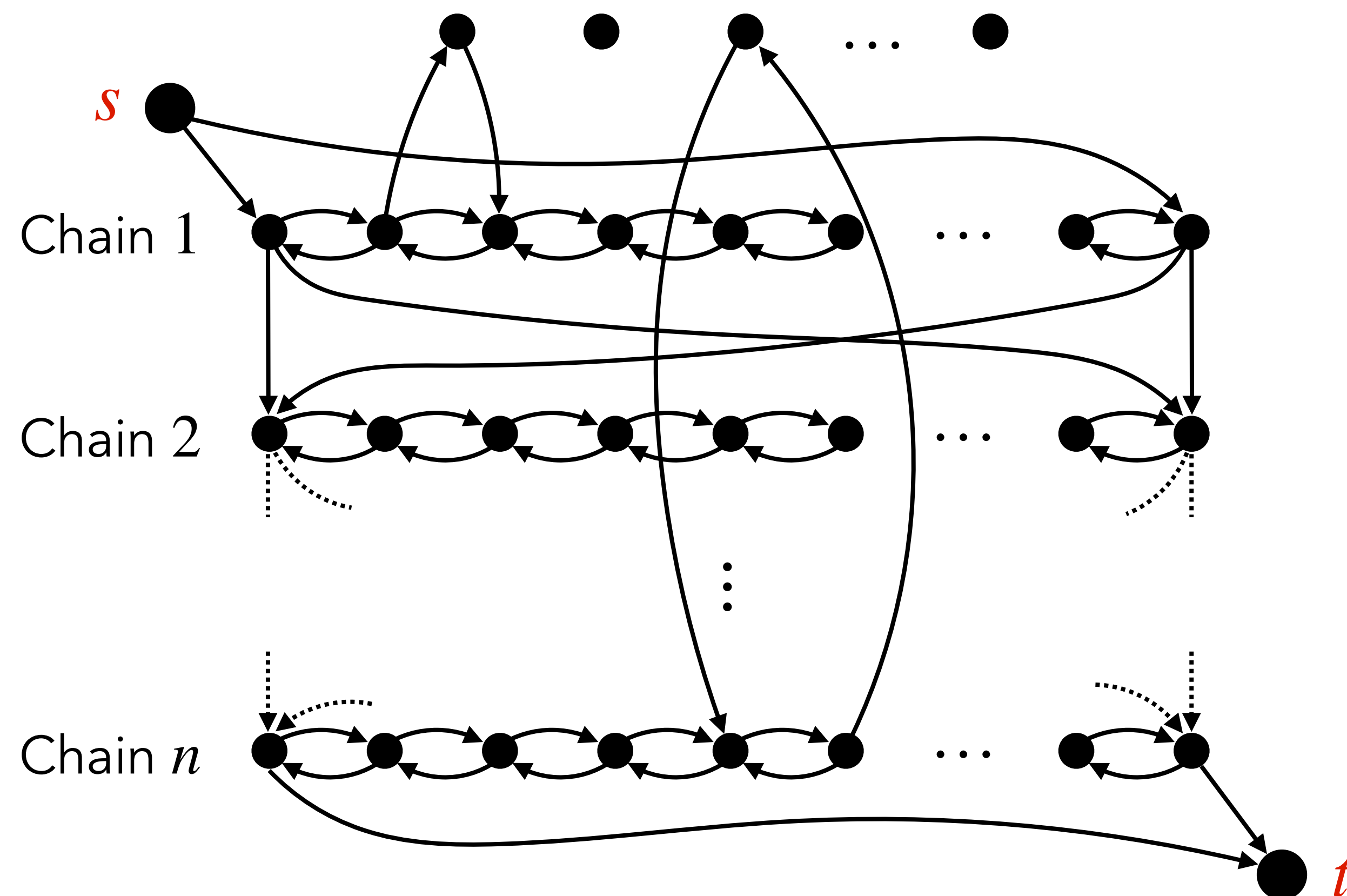
Edges of G :

- Two edges from s to the ends of chain 1.

$$3SAT \leq_p DirHampath$$

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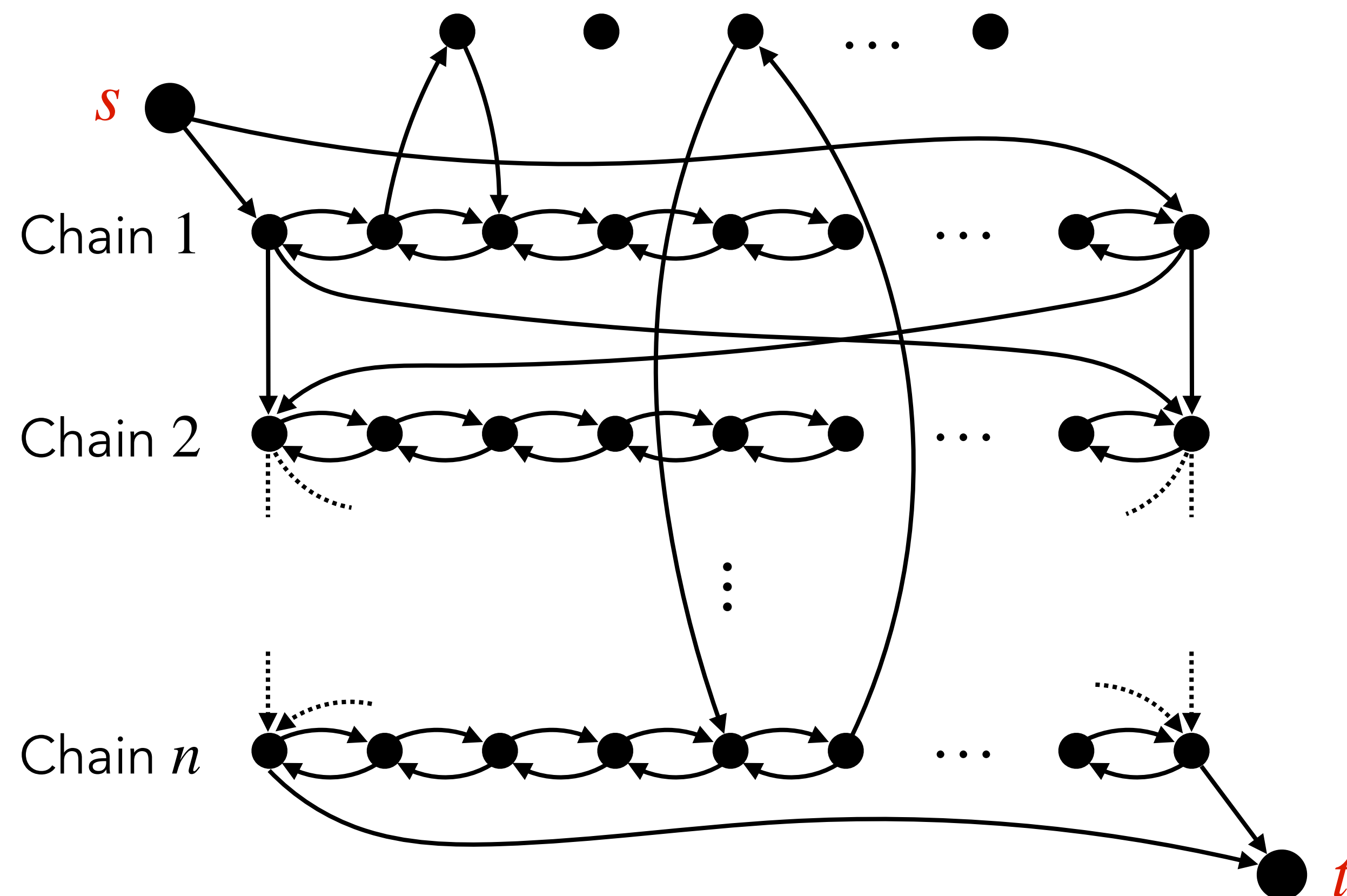
Edges of G :

- Two edges from s to the ends of chain 1.
- Two edges from the ends of chain n to t .

$3SAT \leq_p DirHampath$

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m vertices corresponding to each clause



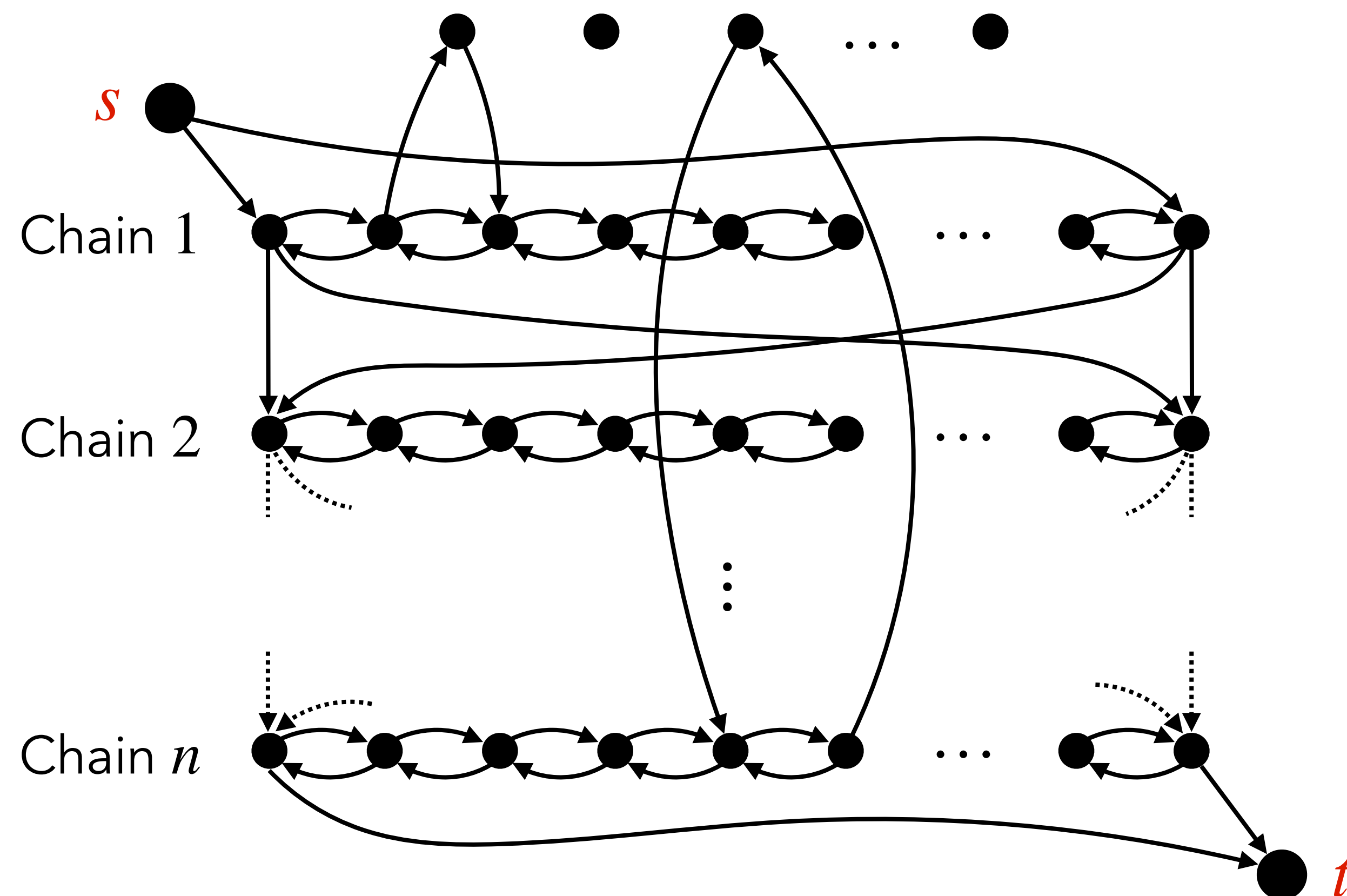
Edges of G :

- Two edges from s to the ends of chain 1.
- Two edges from the ends of chain n to t .
- Edges in the bidirectional chains.

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m vertices corresponding to each clause



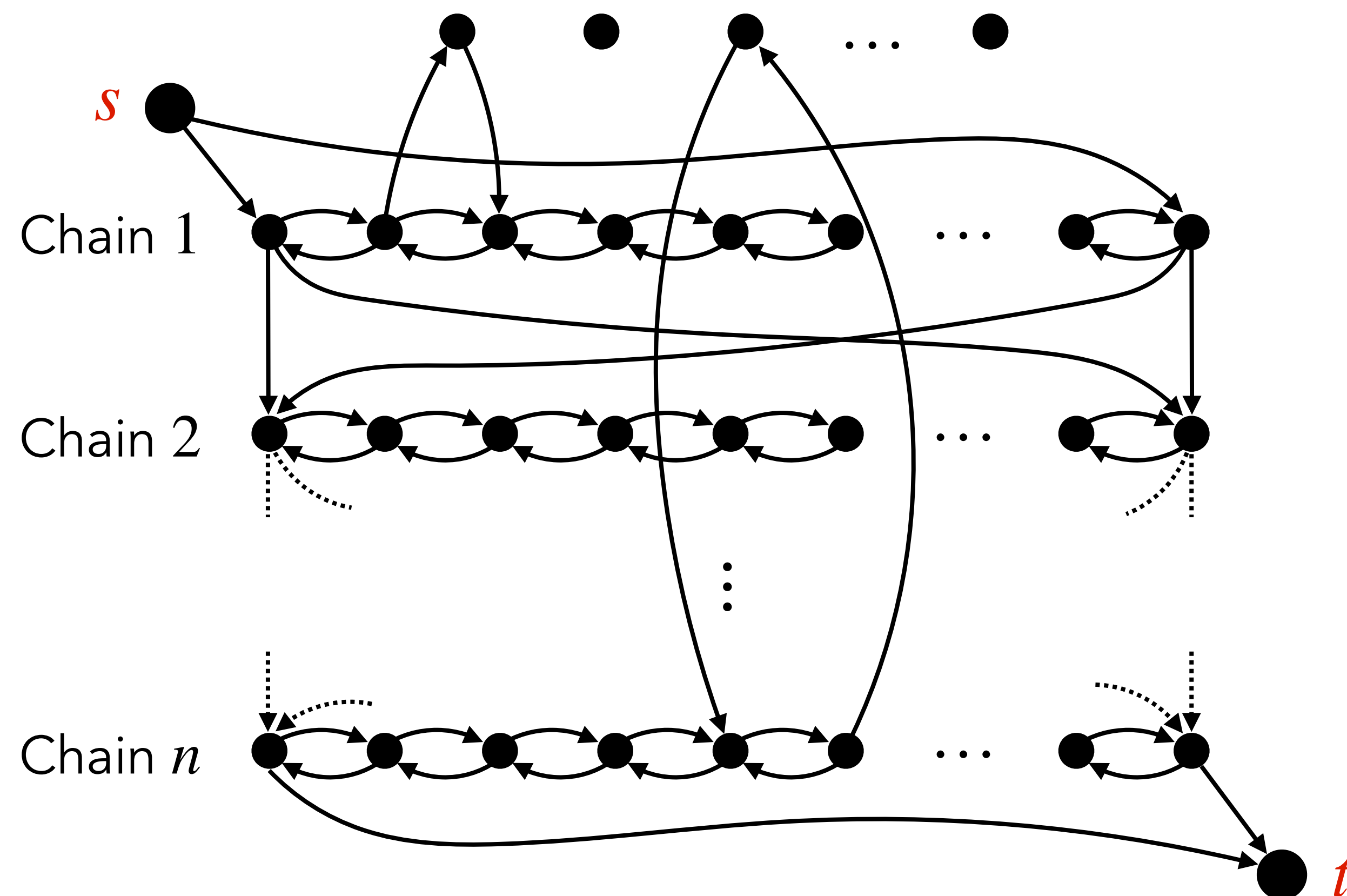
Edges of G :

- Two edges from s to the ends of **chain 1**.
- Two edges from the ends of **chain n** to t .
- Edges in the bidirectional chains.
- Edges from the ends of the i th chain to the ends of the $(i + 1)$ th chain.

$3SAT \leq_p DirHampath$

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m vertices corresponding to each clause

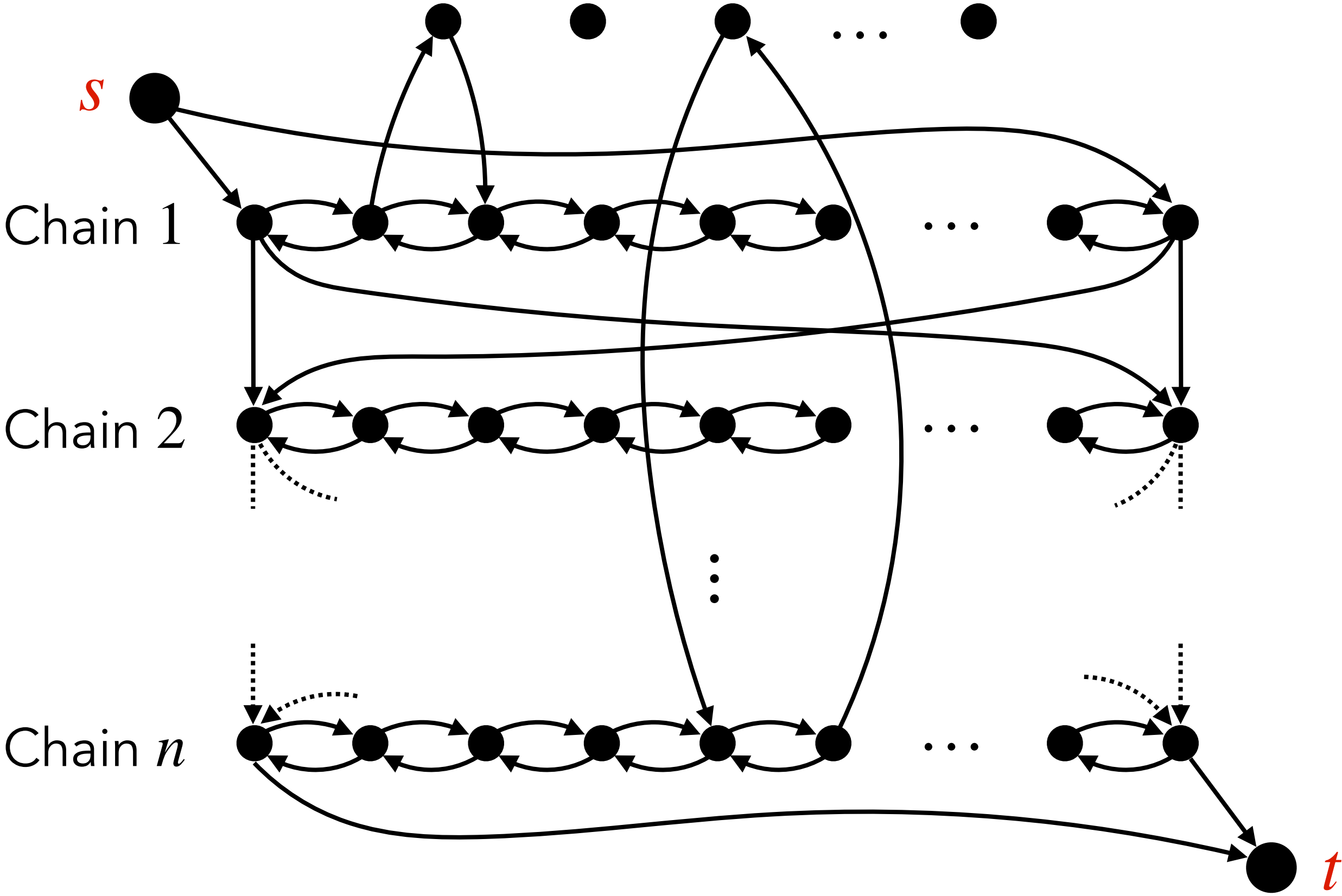


Edges of G :

- Two edges from s to the ends of **chain 1**.
- Two edges from the ends of **chain n** to t .
- Edges in the bidirectional chains.
- Edges from the ends of the i th chain to the ends of the $(i + 1)$ th chain.
- Appropriate edges between the vertices corresponding to clauses & chains without reusing a link.

$$3SAT \leq_p DirHampath$$

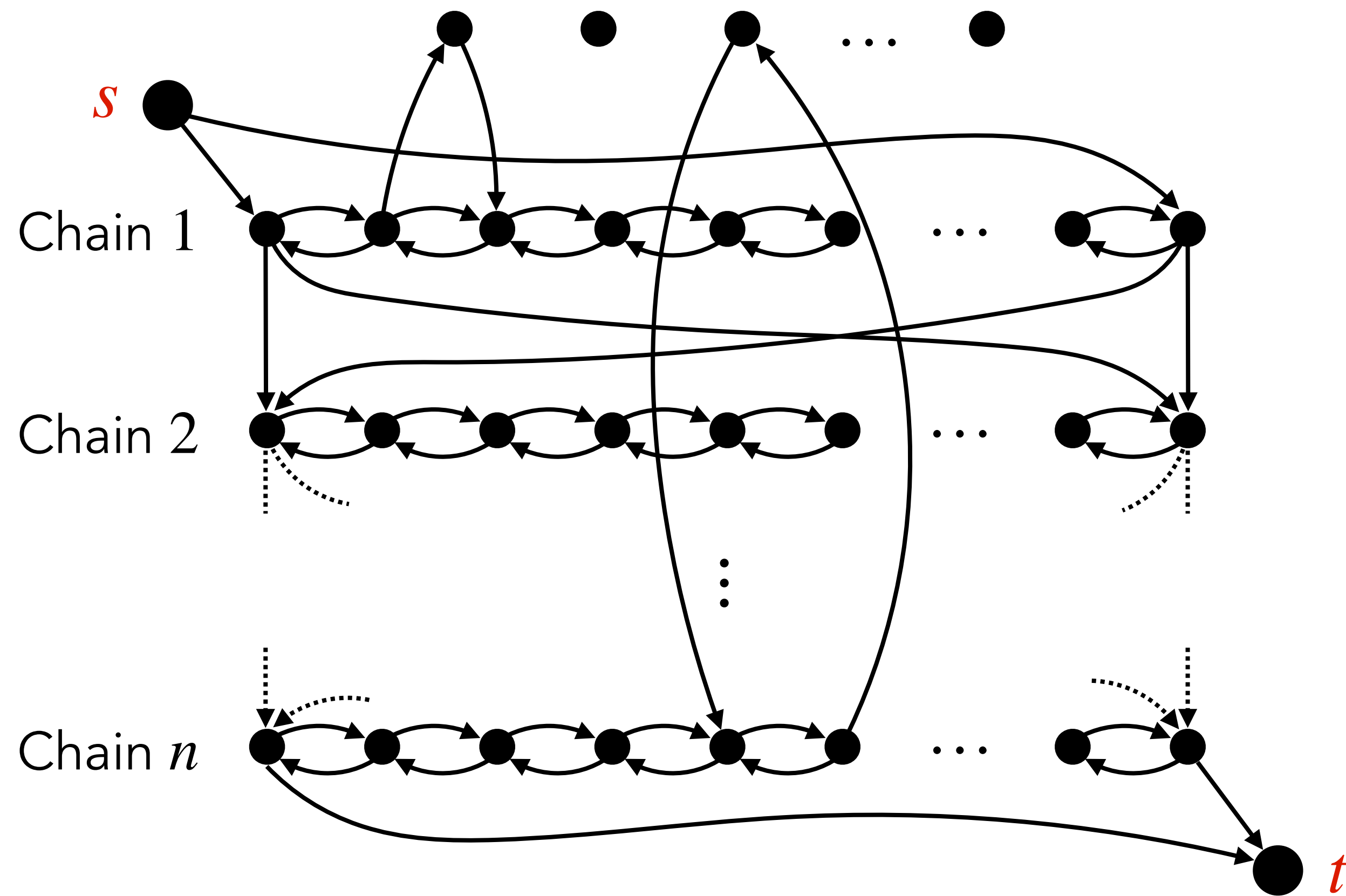
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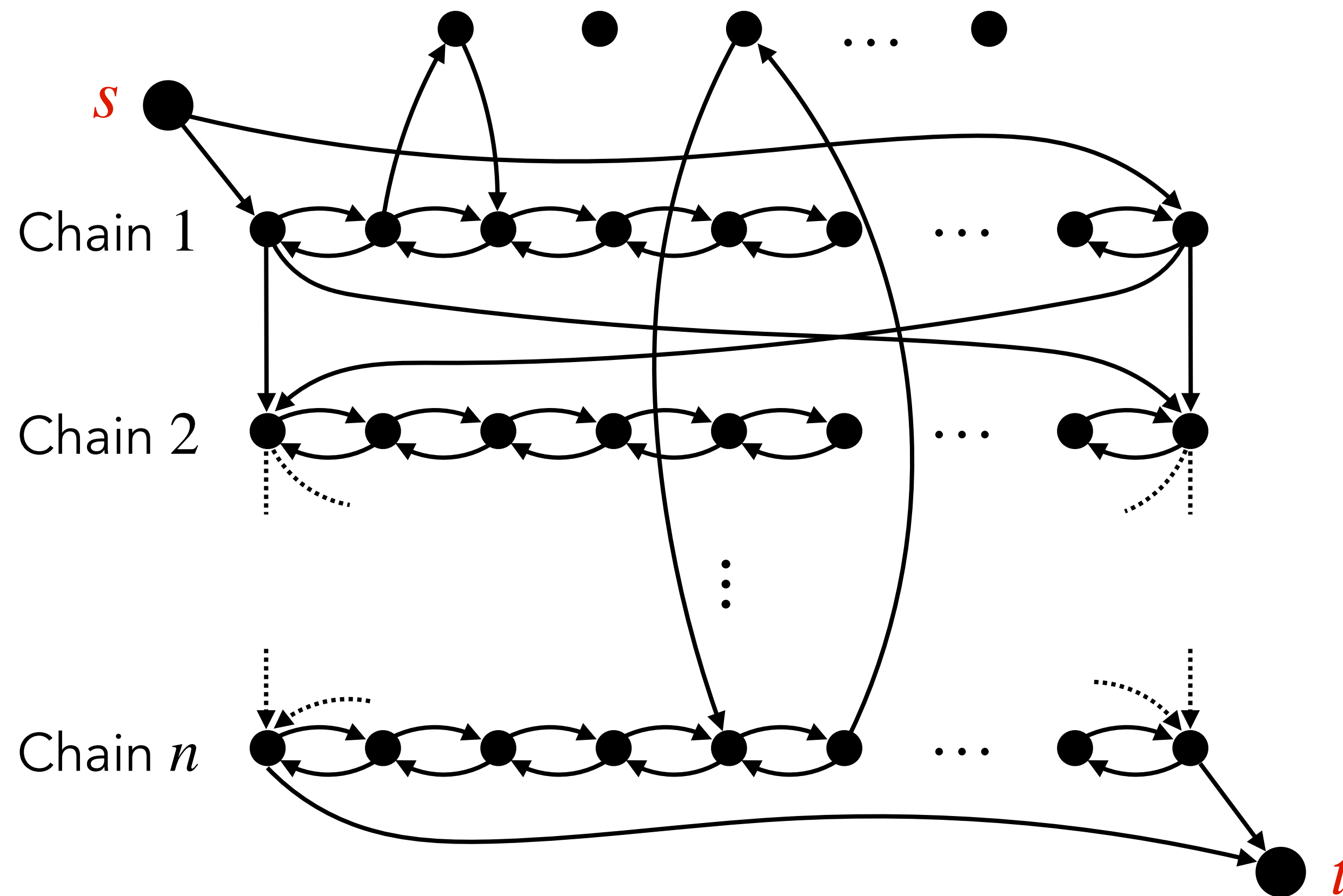
Correctness of Reduction ($\implies$):

m vertices corresponding to each clause



$$3SAT \leq_p DirHampath$$

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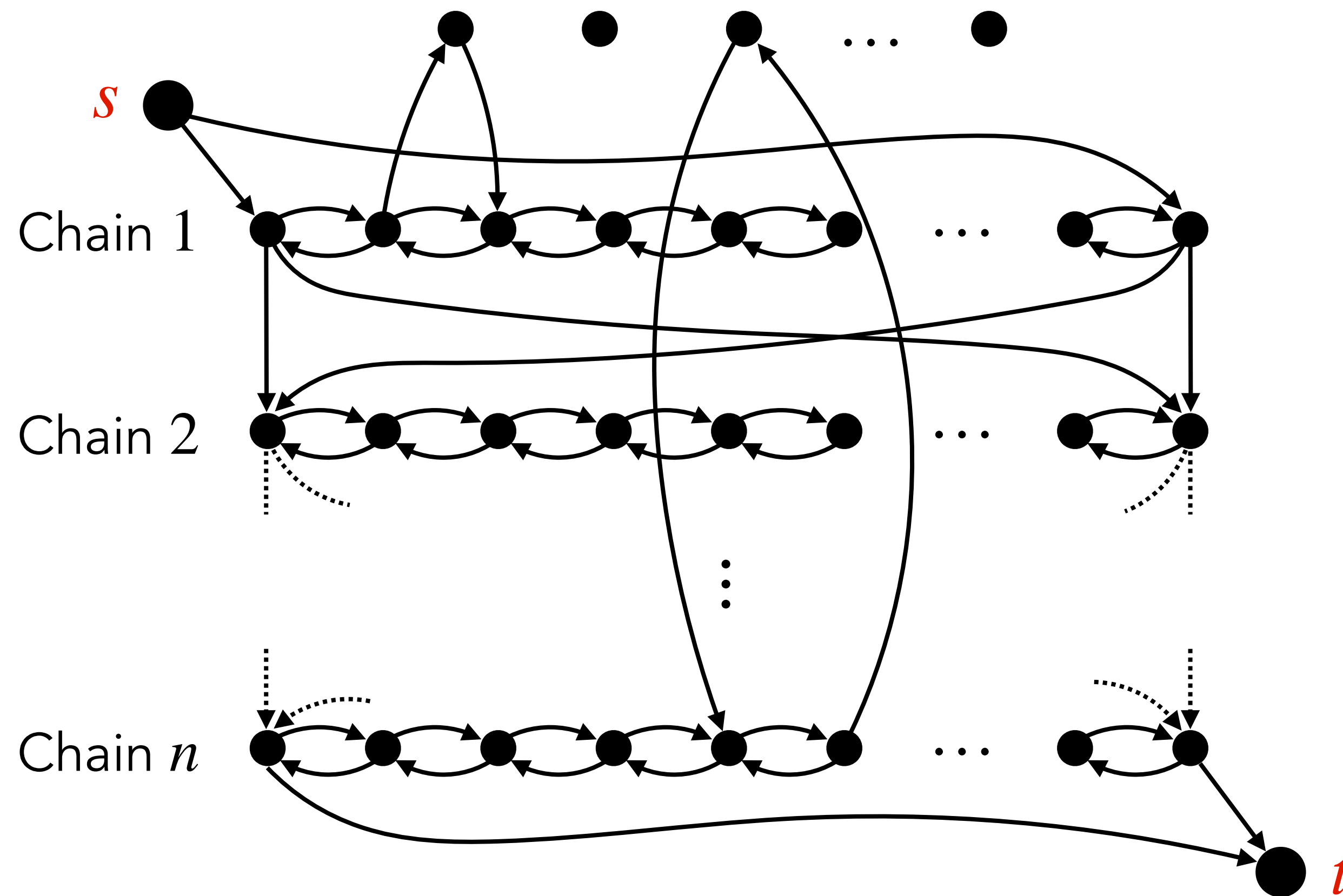


Correctness of Reduction ($\implies$):

- Suppose there exists a satisfying assign. of ϕ .

$$3SAT \leq_p DirHampath$$

m vertices corresponding to each clause

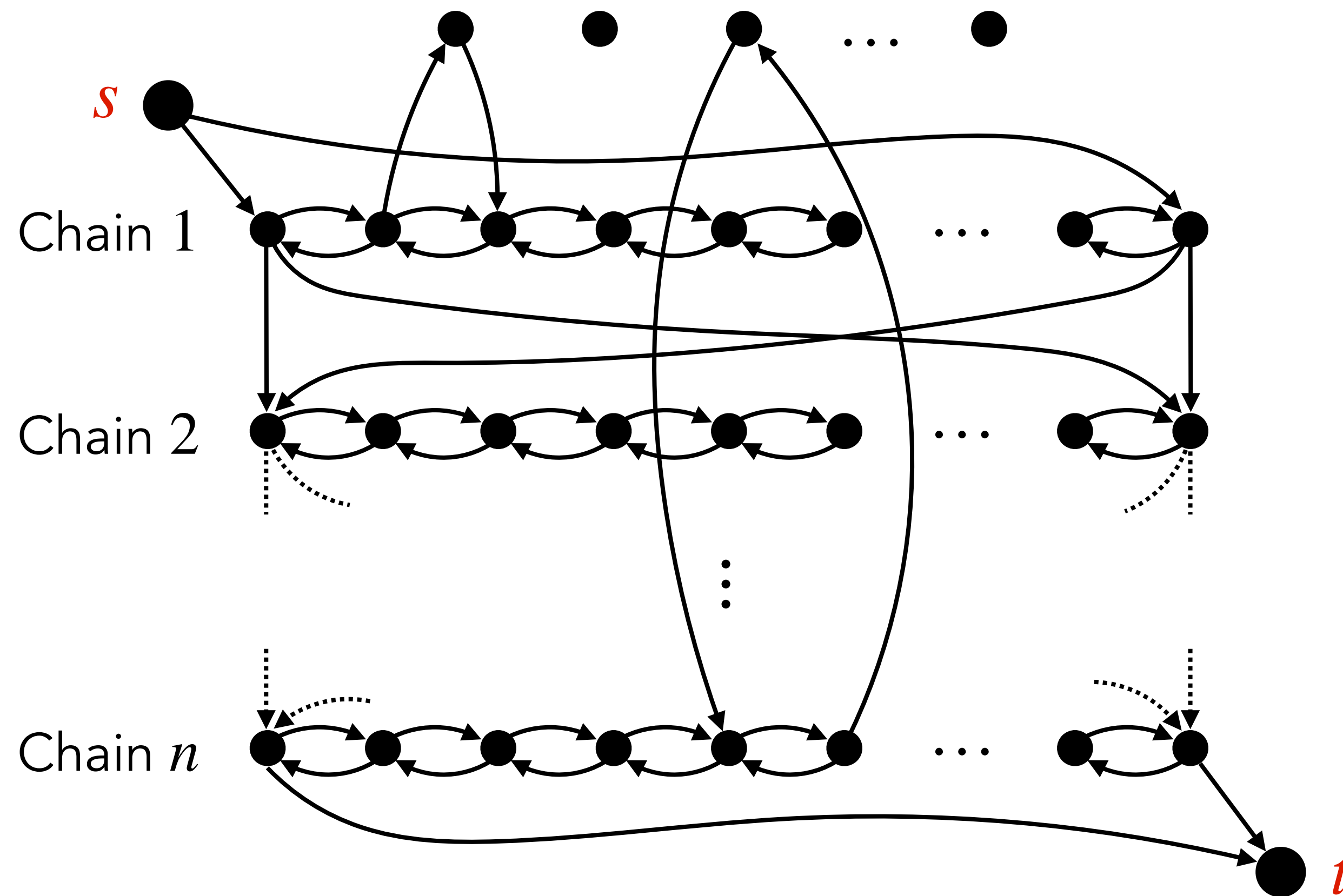


Correctness of Reduction ($\implies$):

- Suppose there exists a satisfying assign. of ϕ .
- Consider the st -path traversing via all the chains such that:

$$3SAT \leq_p DirHampath$$

m vertices corresponding to each clause

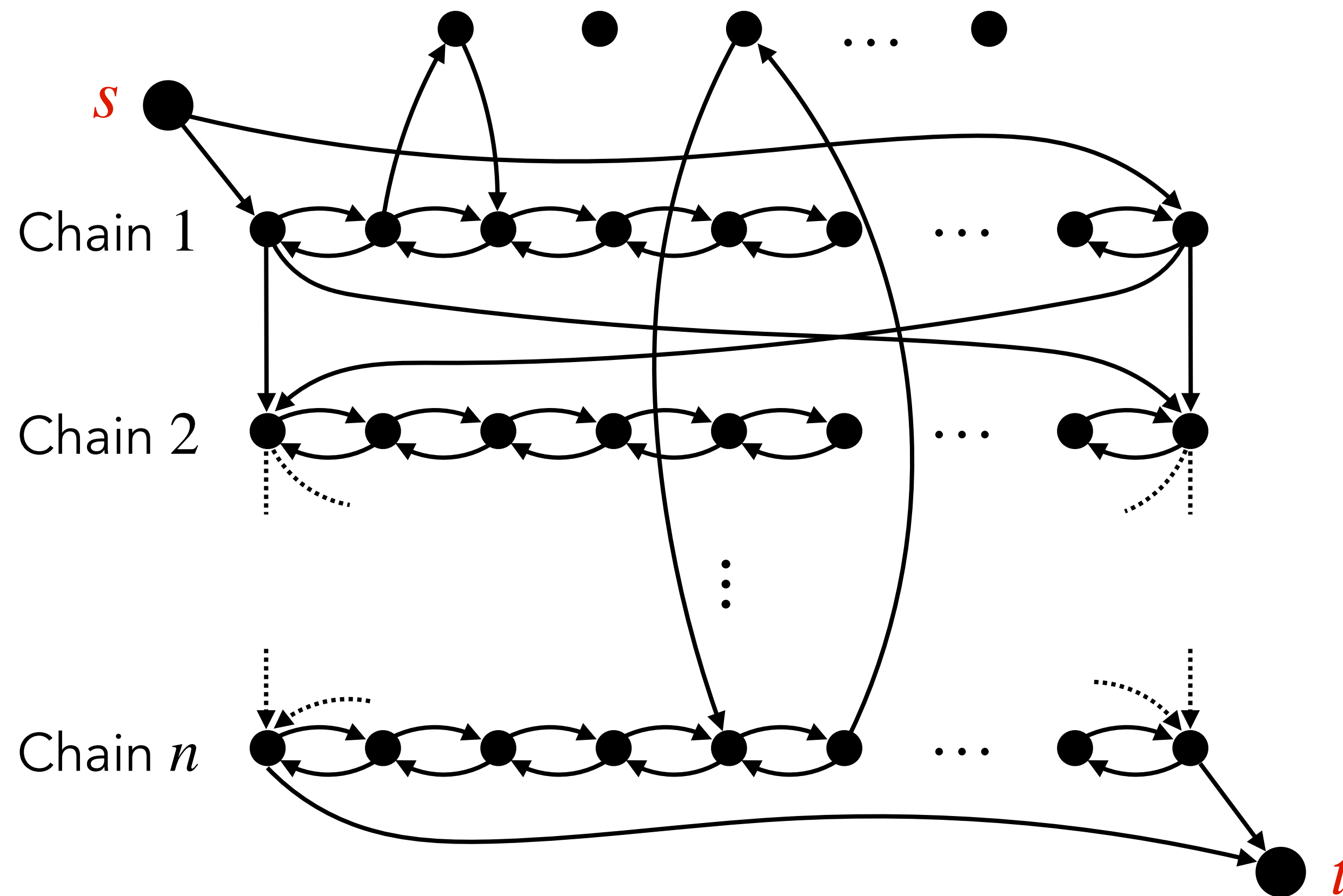


Correctness of Reduction ($\implies$):

- Suppose there exists a satisfying assign. of ϕ .
- Consider the st -path traversing via all the chains such that:
 - If $u_i = 0$, traverse i th chain right-to-left.

$$3SAT \leq_p DirHampath$$

m vertices corresponding to each clause

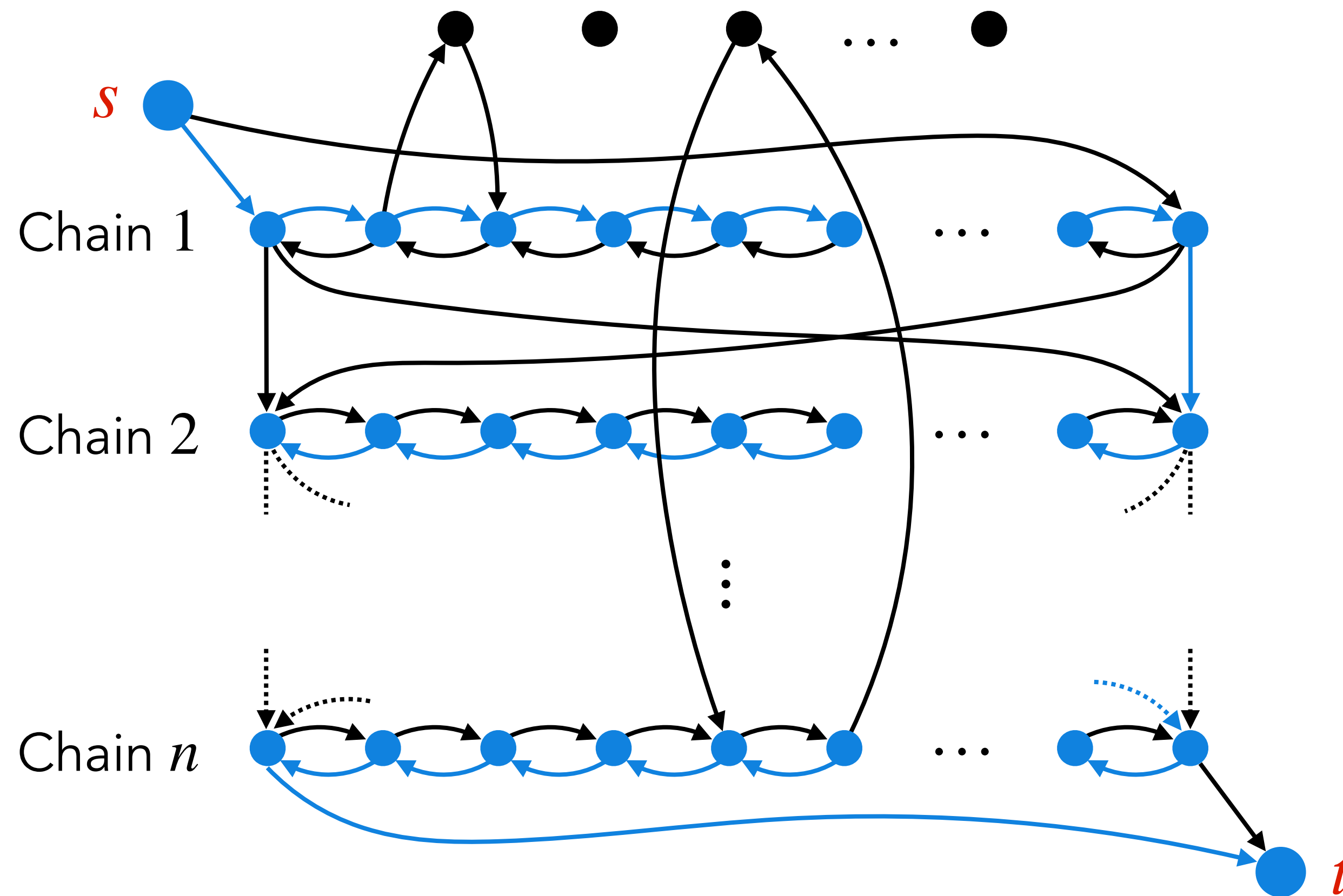


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m vertices corresponding to each clause

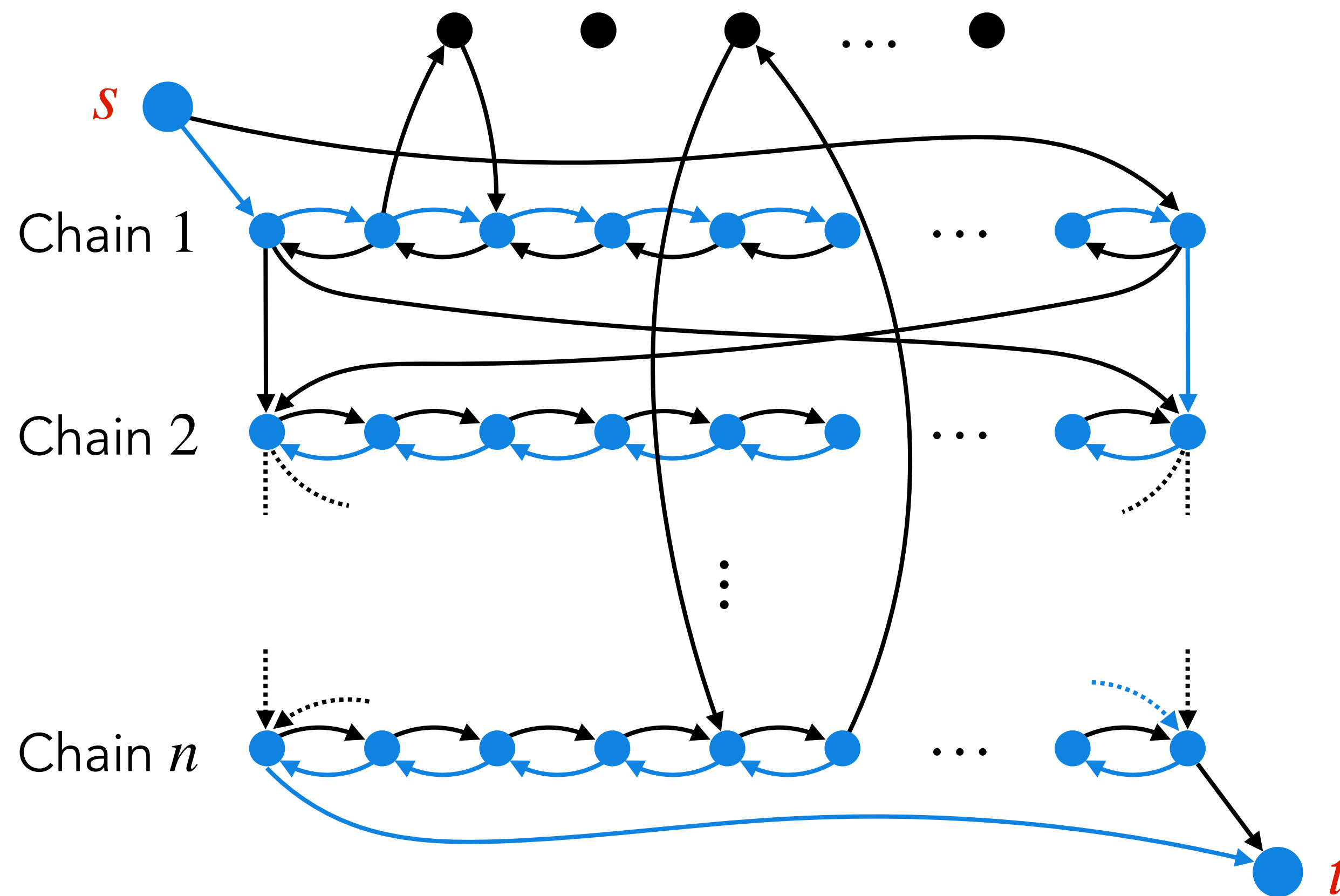


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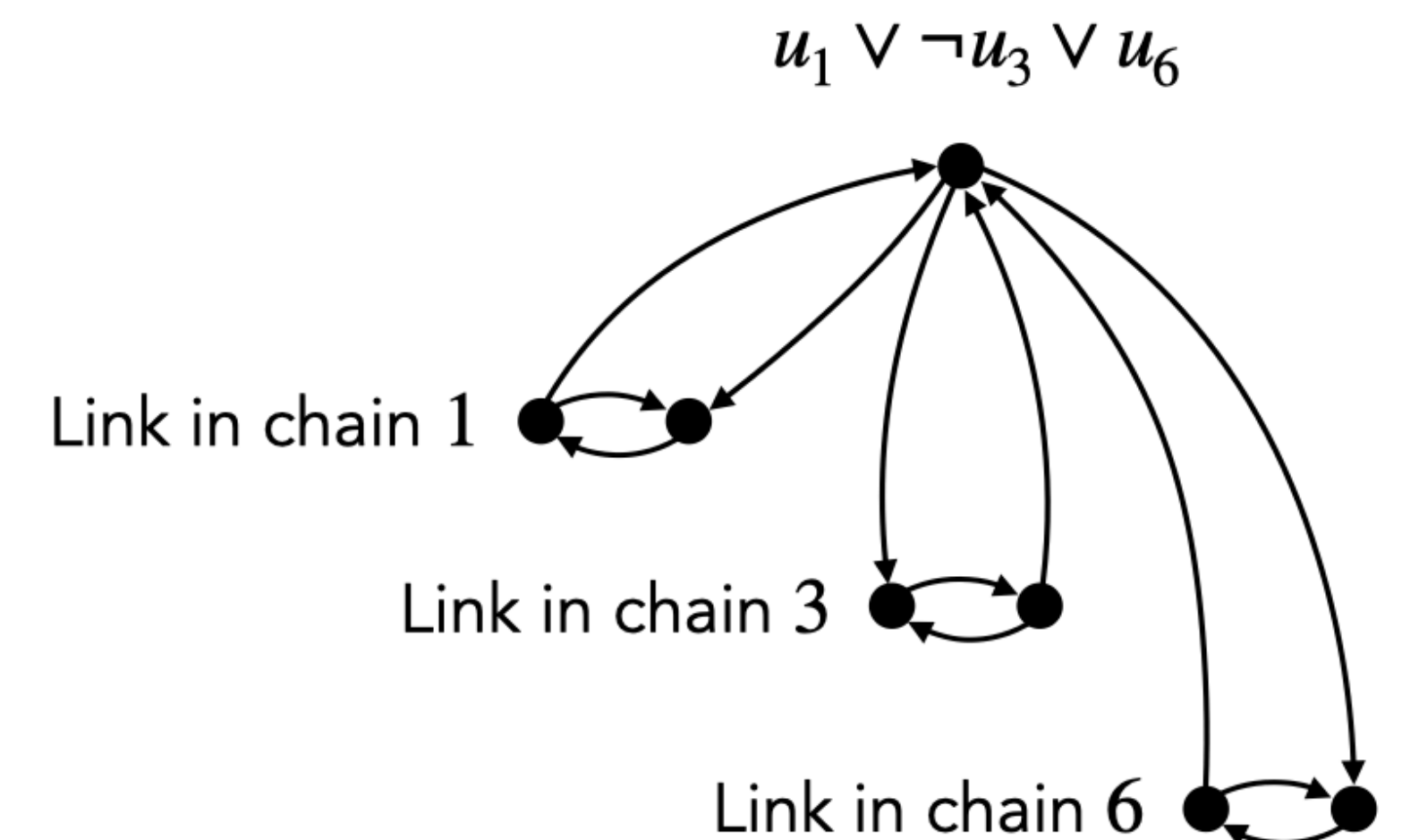
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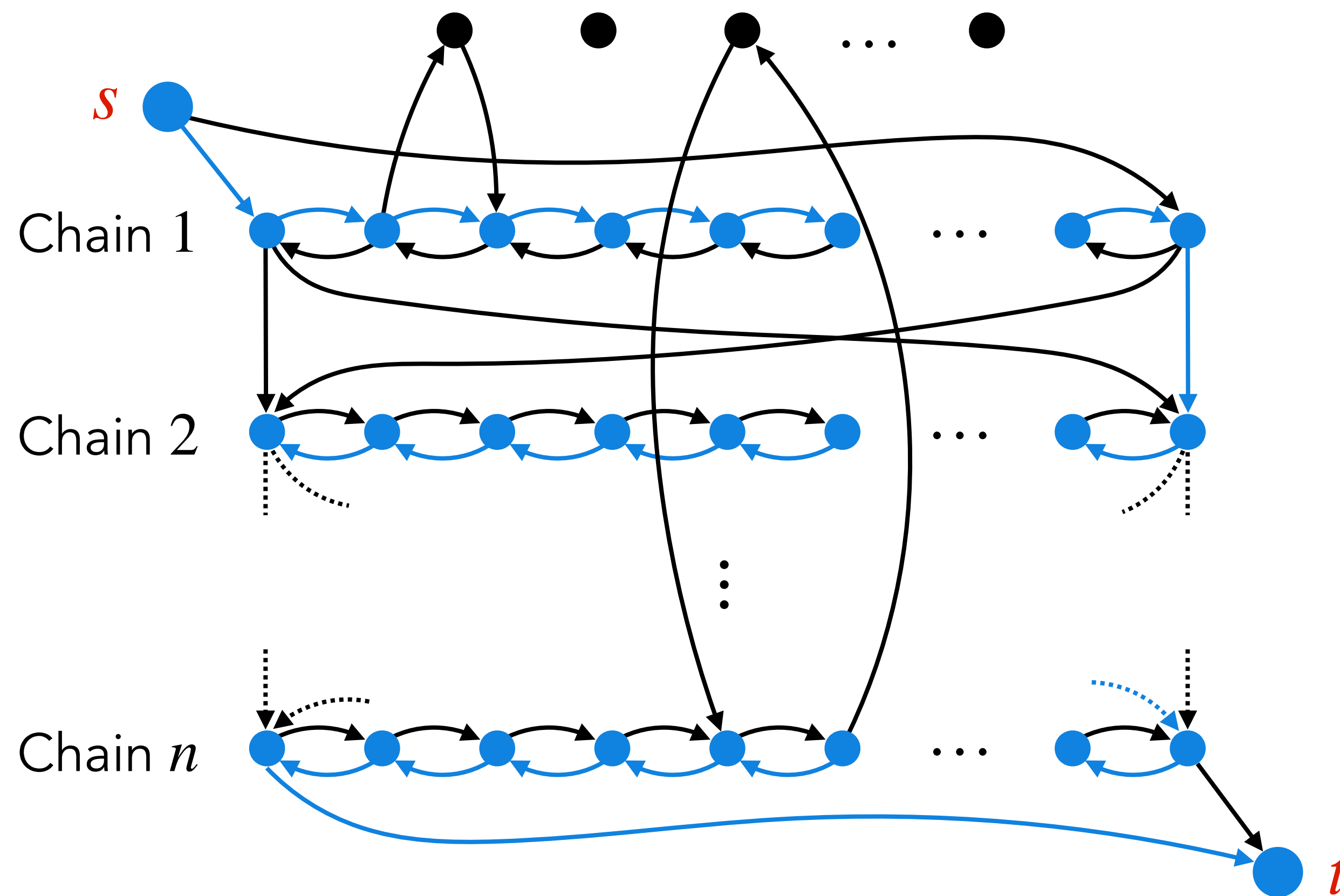
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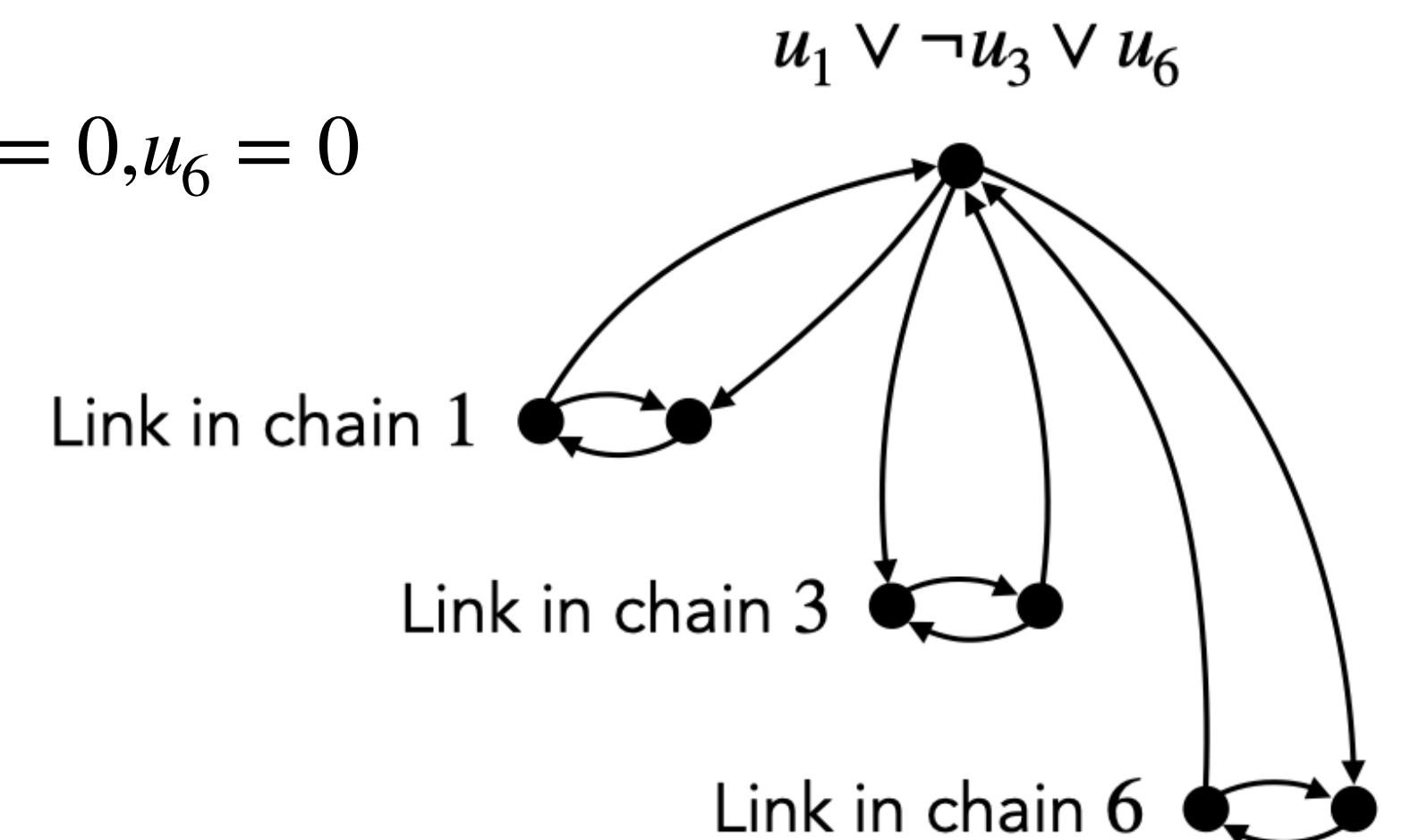
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Correctness of Reduction ($\implies$):

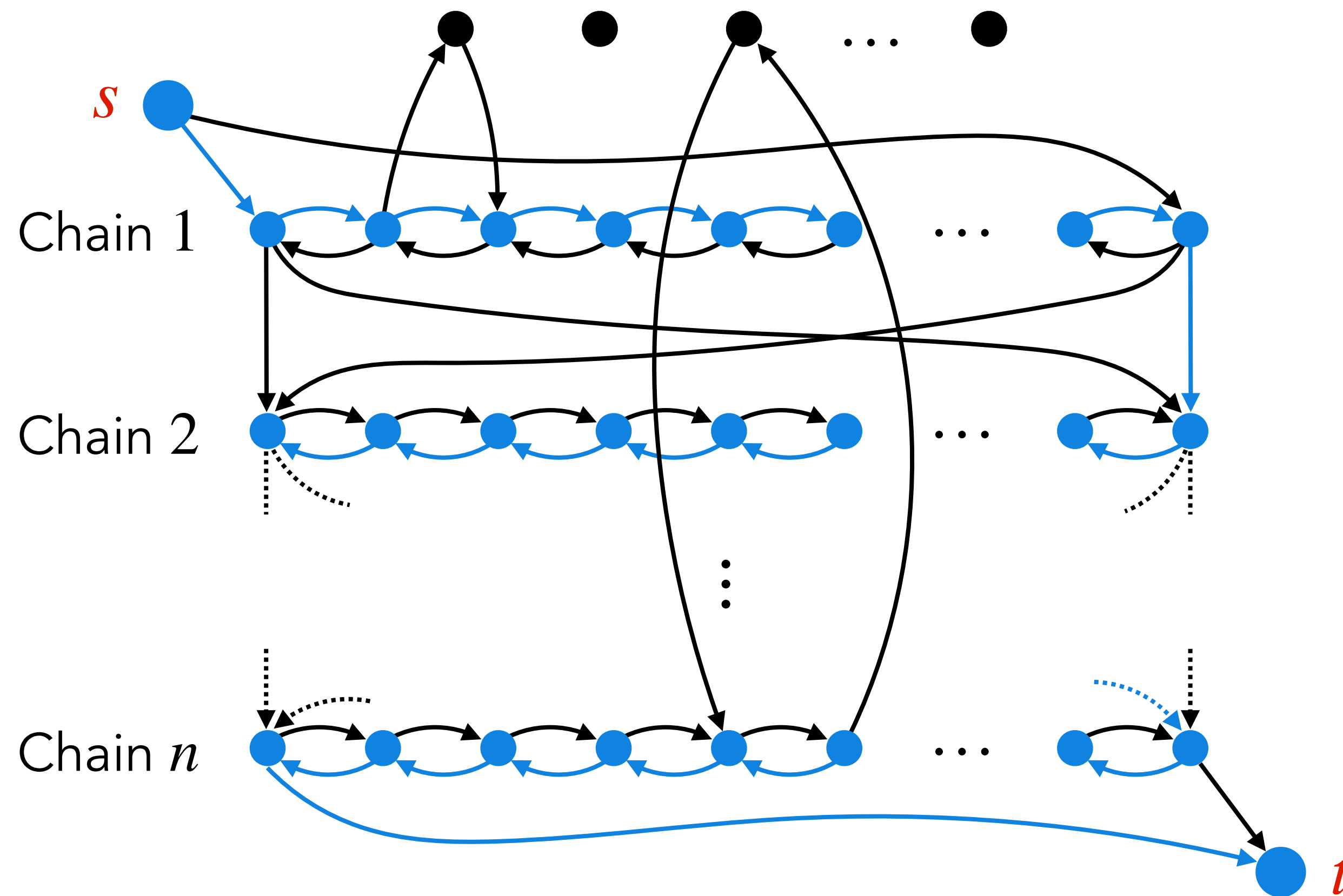
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$$u_1 = 0, u_3 = 0, u_6 = 0$$



$3SAT \leq_p DirHampath$

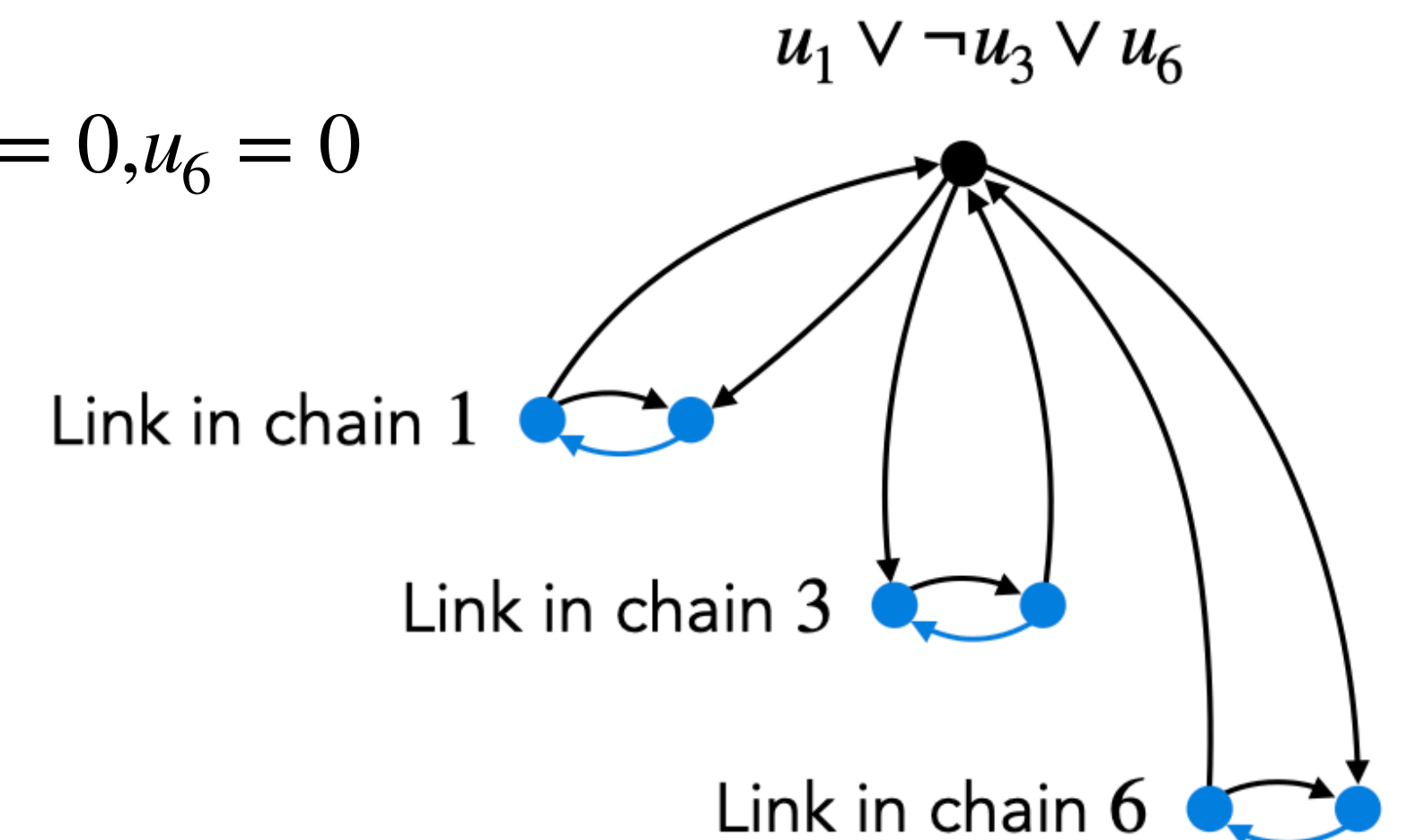
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Correctness of Reduction ($\implies$):

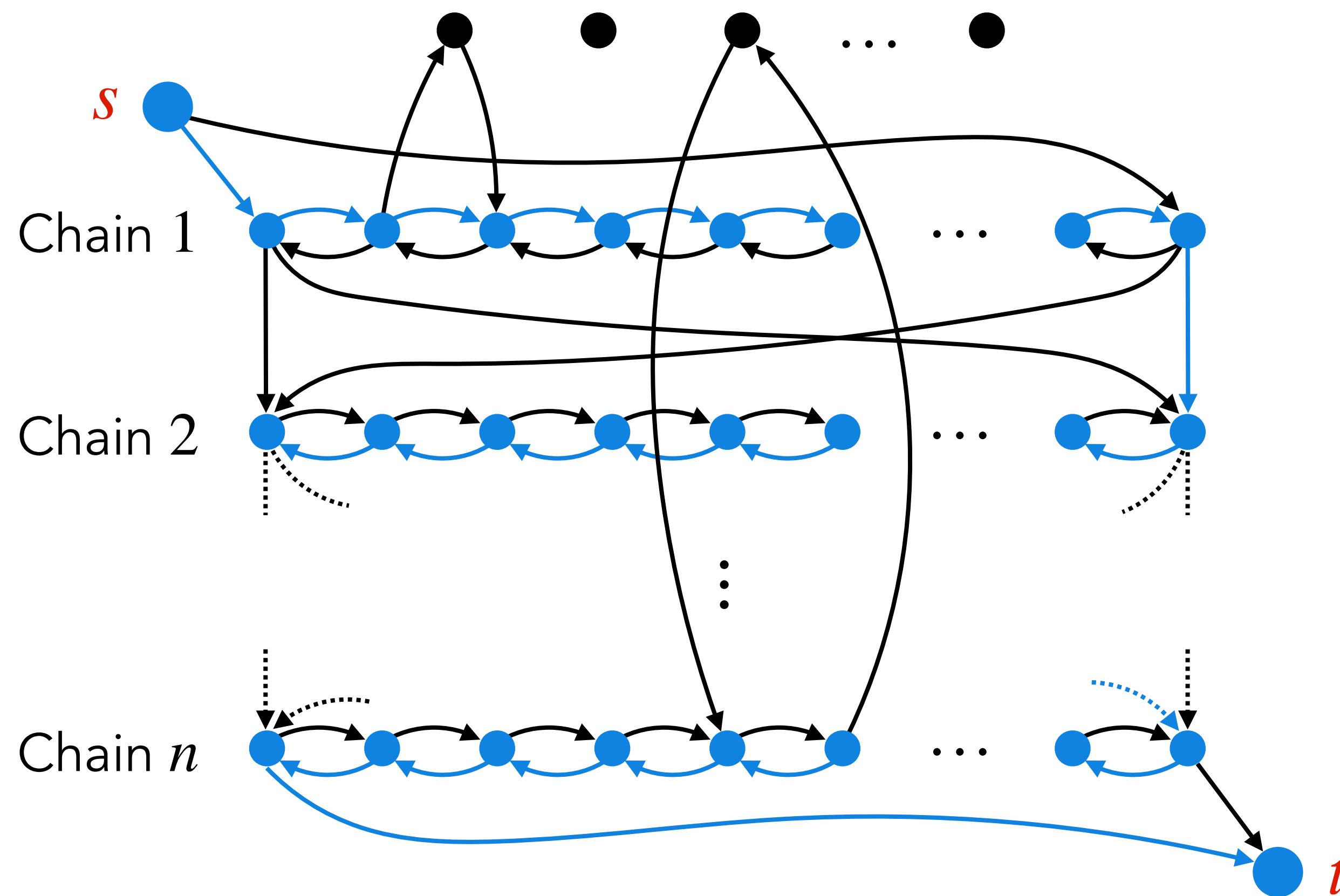
- Suppose there exists a satisfying assign. of ϕ .
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$3SAT \leq_p DirHampath$

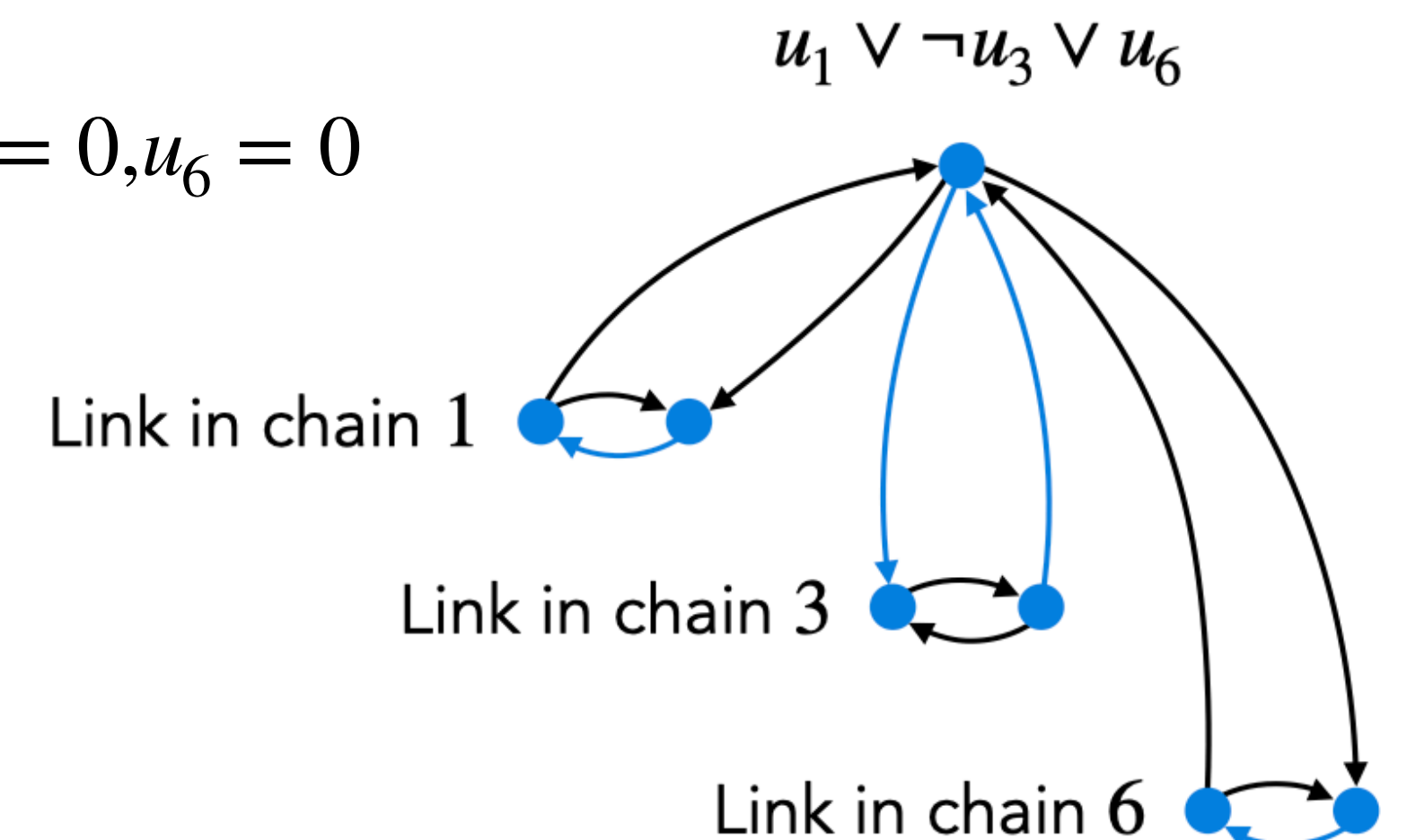
m vertices corresponding to each clause



Correctness of Reduction ($\implies$):

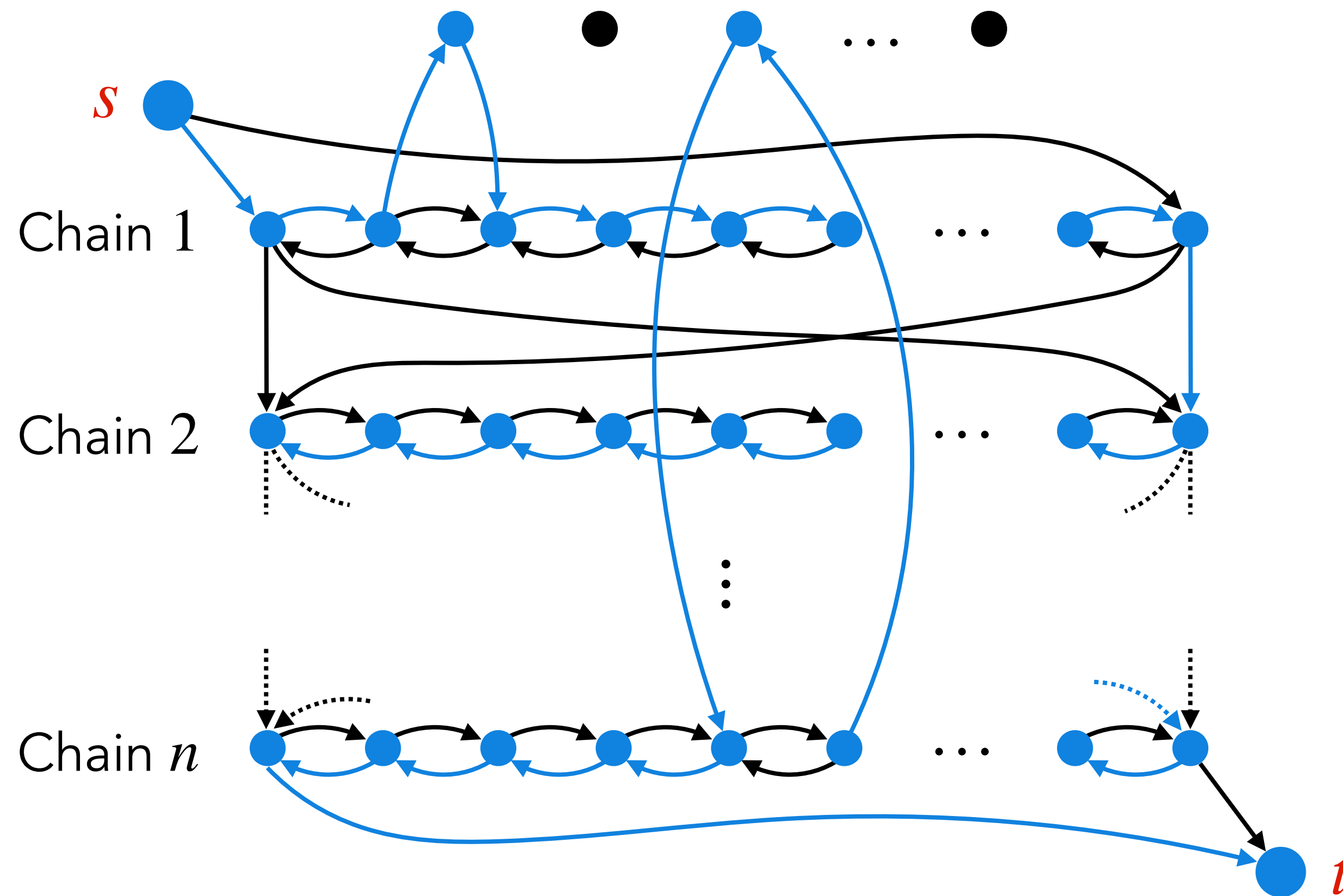
- Suppose there exists a satisfying assign. of ϕ .
- Consider the st -path traversing via all the chains such that:
 - If $u_i = 0$, traverse i th chain right-to-left.
 - If $u_i = 1$, traverse i th chain left-to-right.

$$u_1 = 0, u_3 = 0, u_6 = 0$$



$$3SAT \leq_p DirHampath$$

m vertices corresponding to each clause

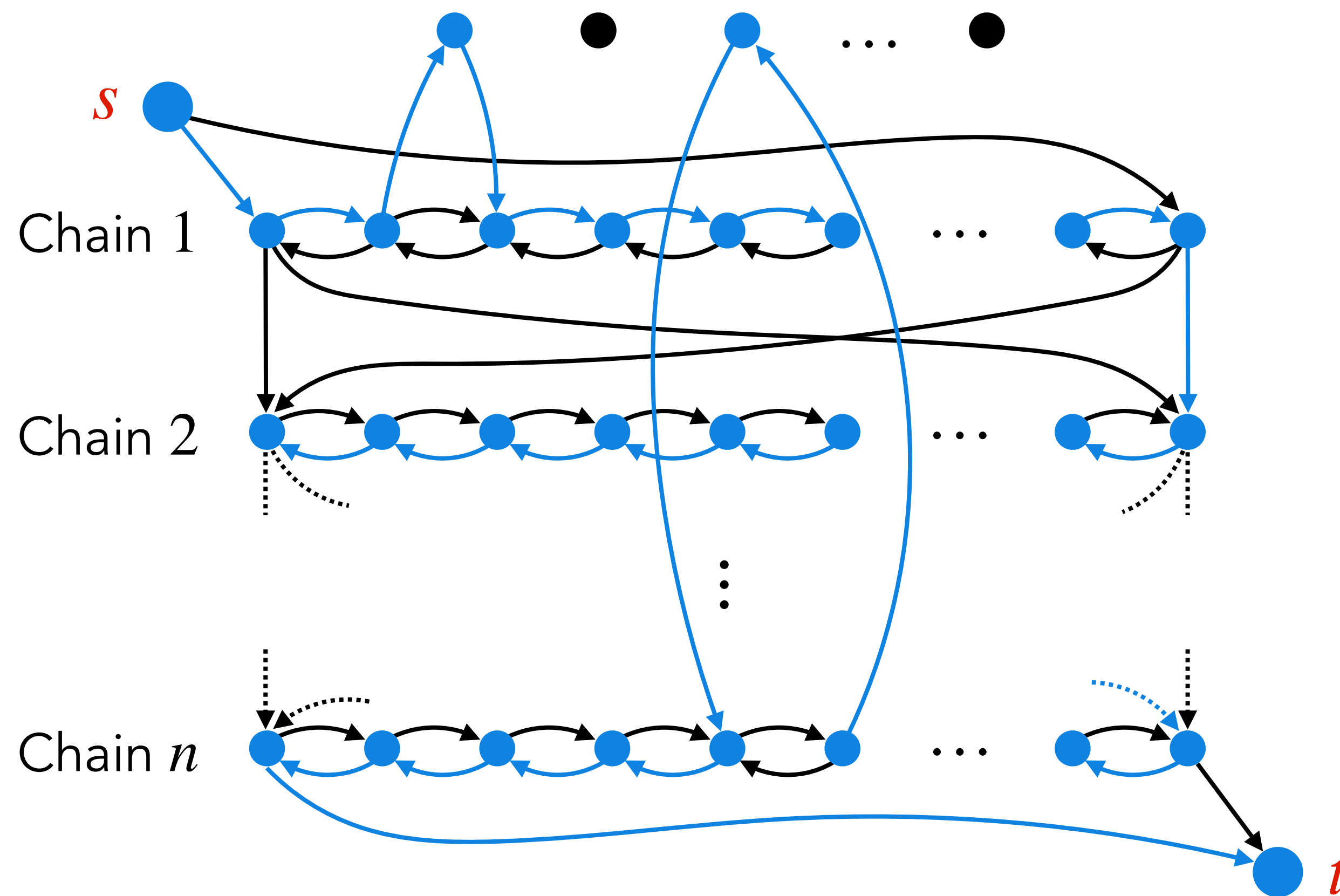


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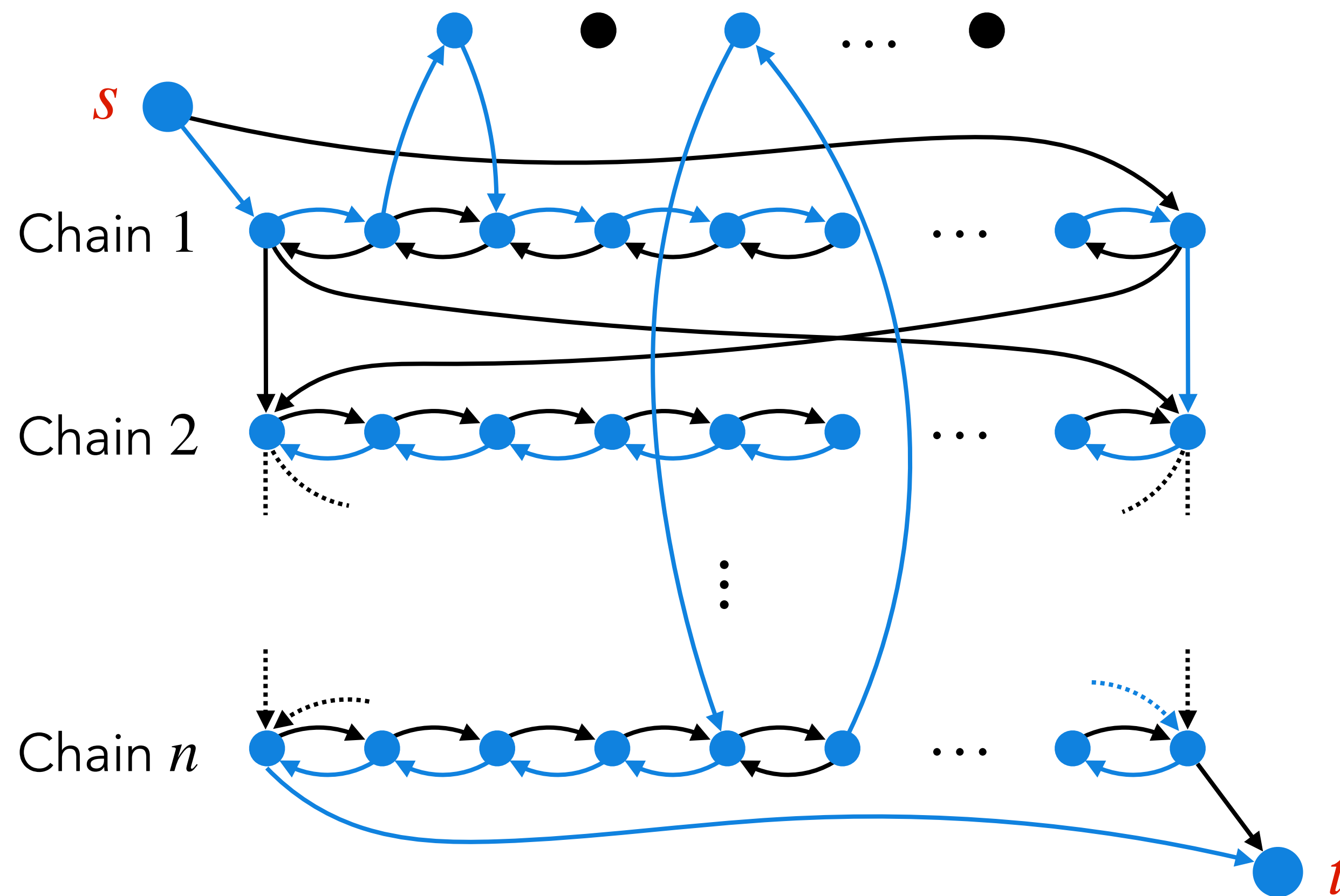


Correctness of Reduction ($\implies$):

- Suppose there exists a satisfying assign. of ϕ .
- Consider the st -path traversing via all the chains such that:
 - If $u_i = 0$, traverse i th chain right-to-left.
 - If $u_i = 1$, traverse i th chain left-to-right.
- If clause C is satisfied because some u_i in it is 1,

$$3SAT \leq_p DirHampath$$

m vertices corresponding to each clause

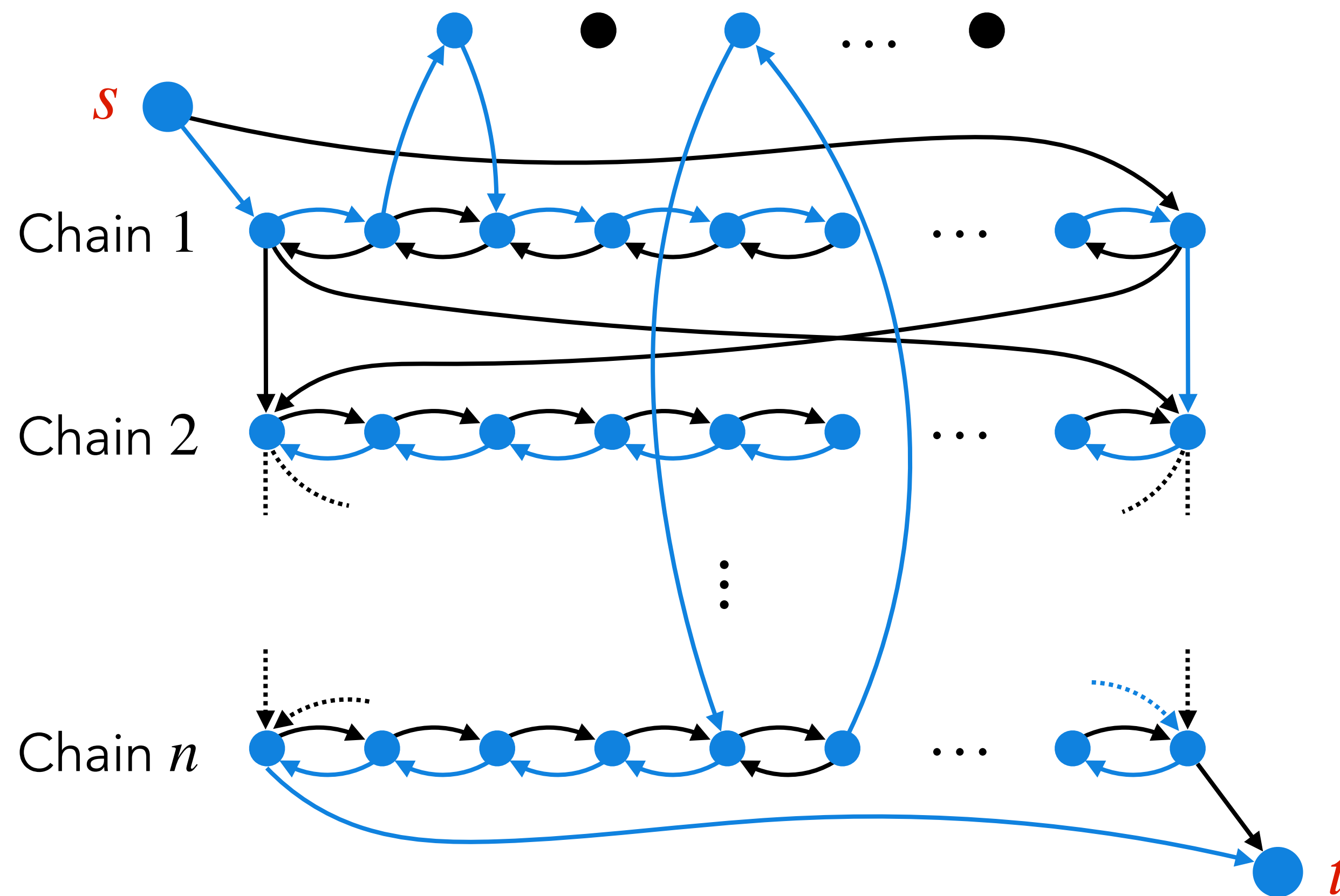


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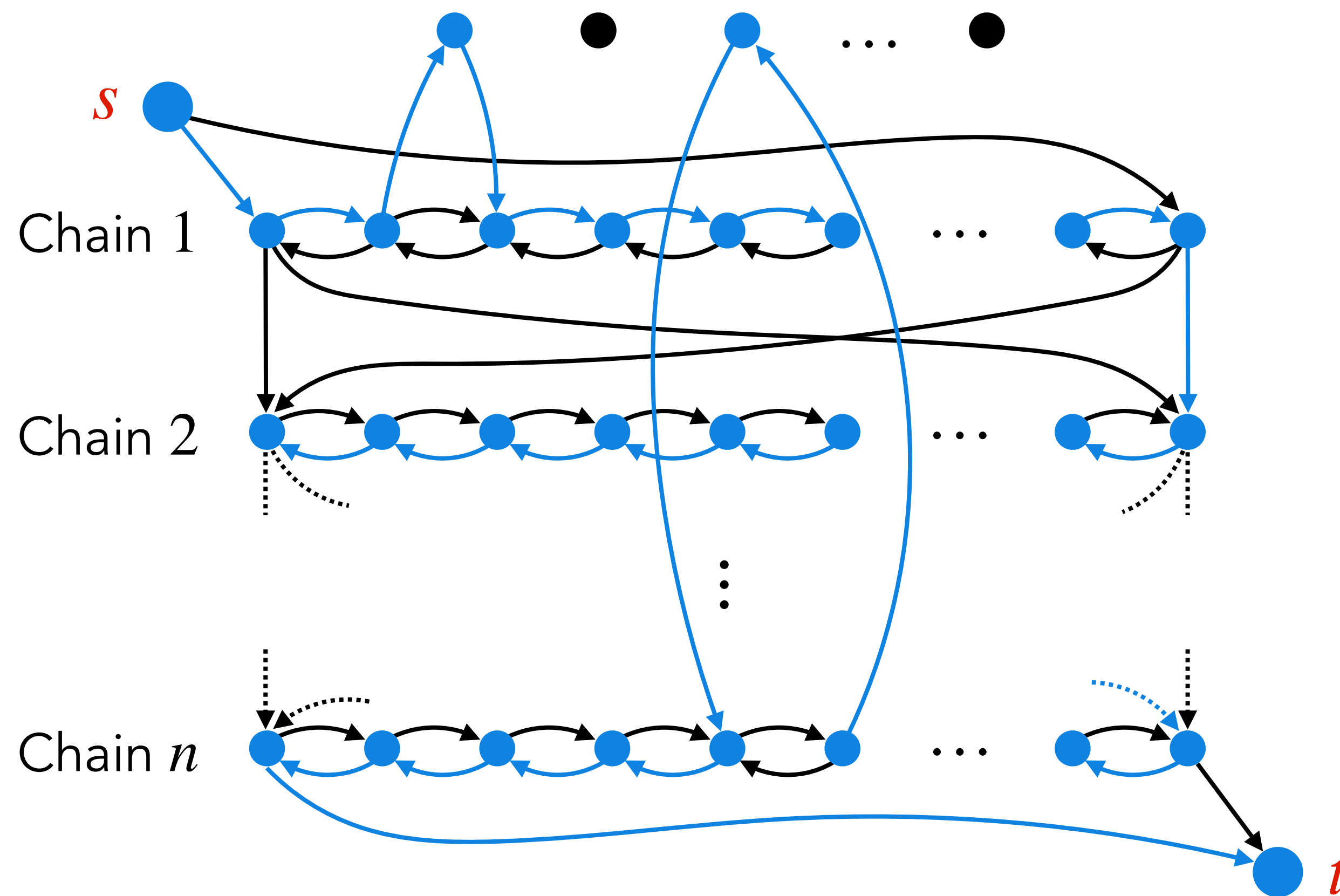


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- If clause C is satisfied because some u_i in it is 1, then visit the vertex corresponding to it by **skipping** the link while going left-to-right

$3SAT \leq_p DirHampath$

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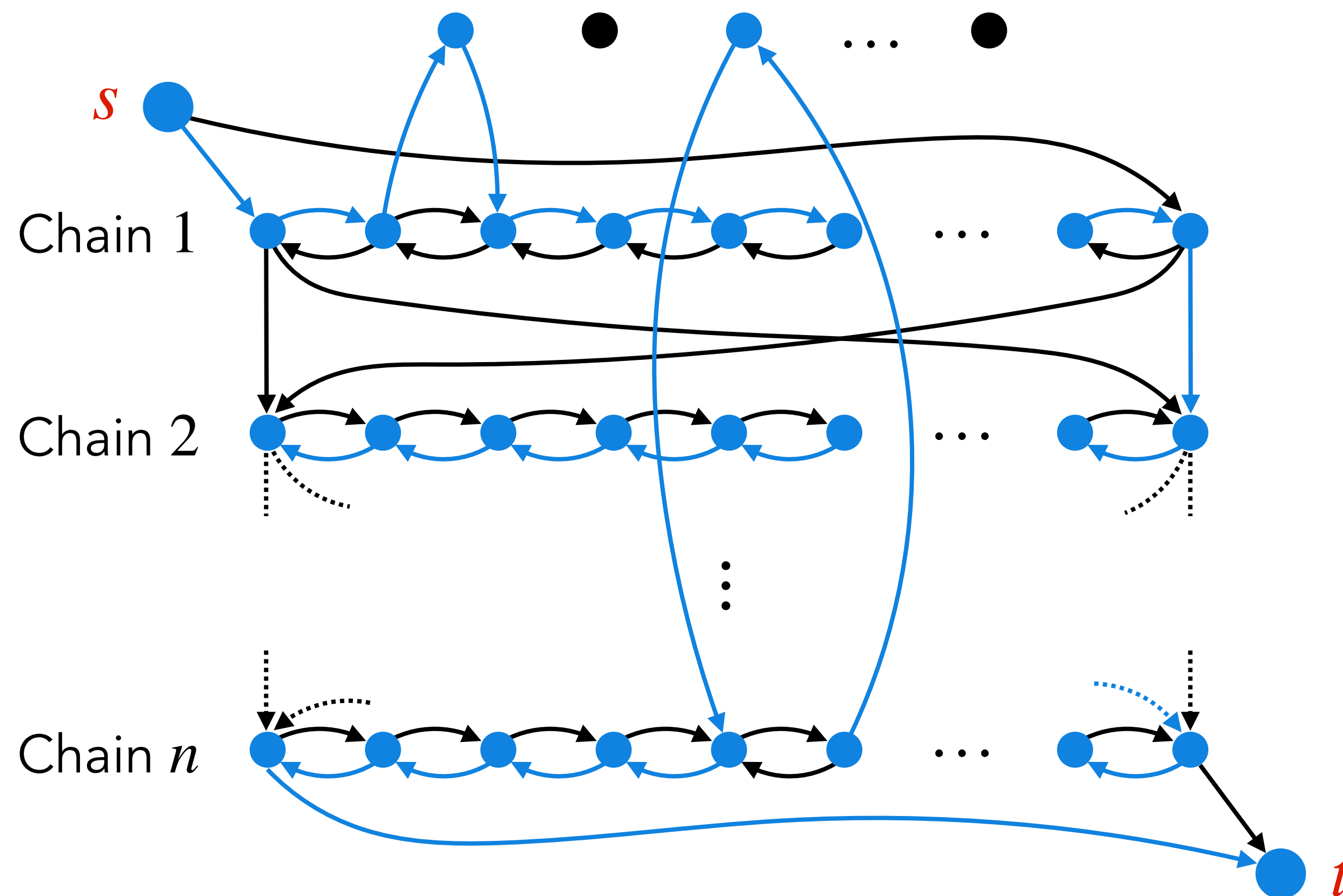


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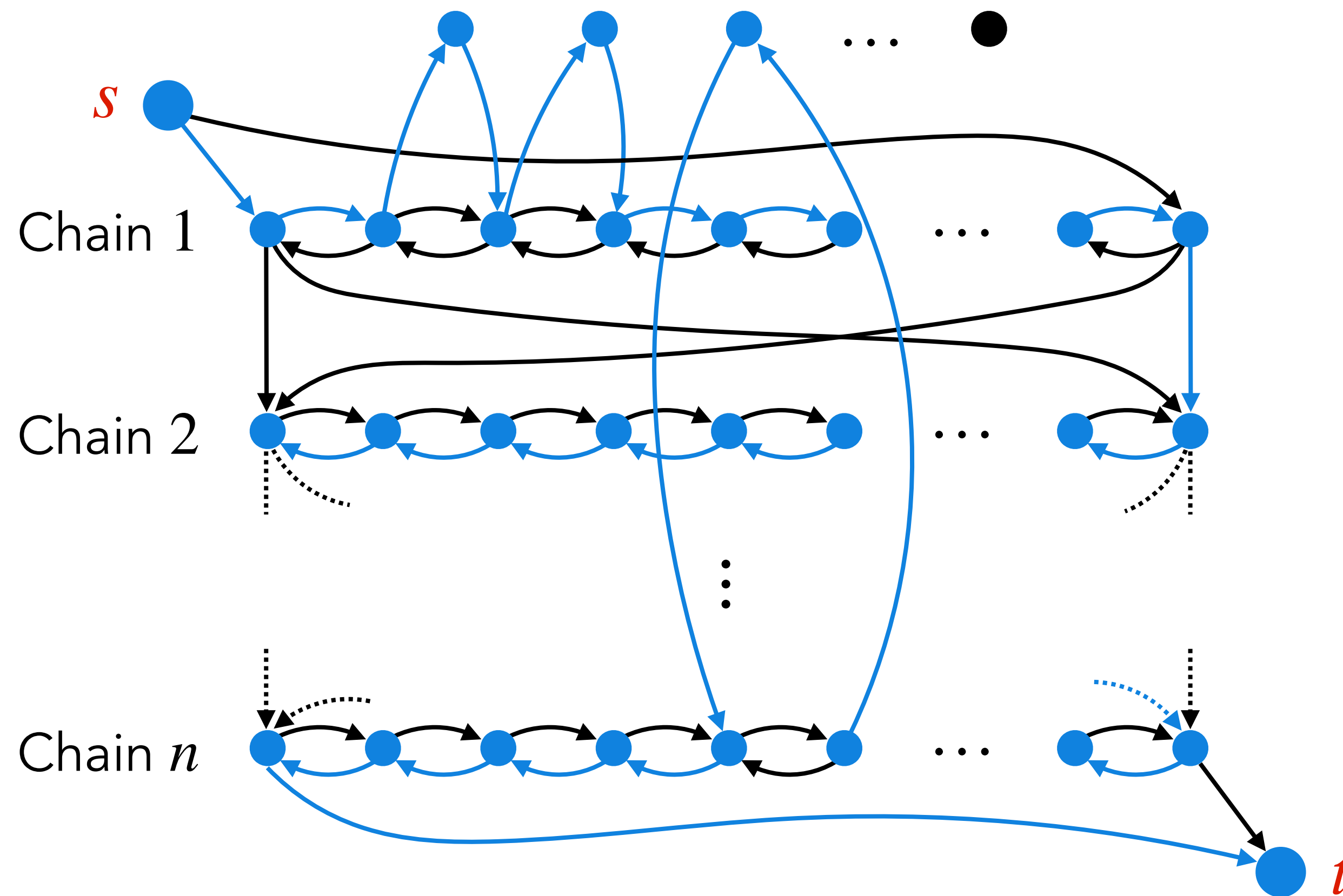


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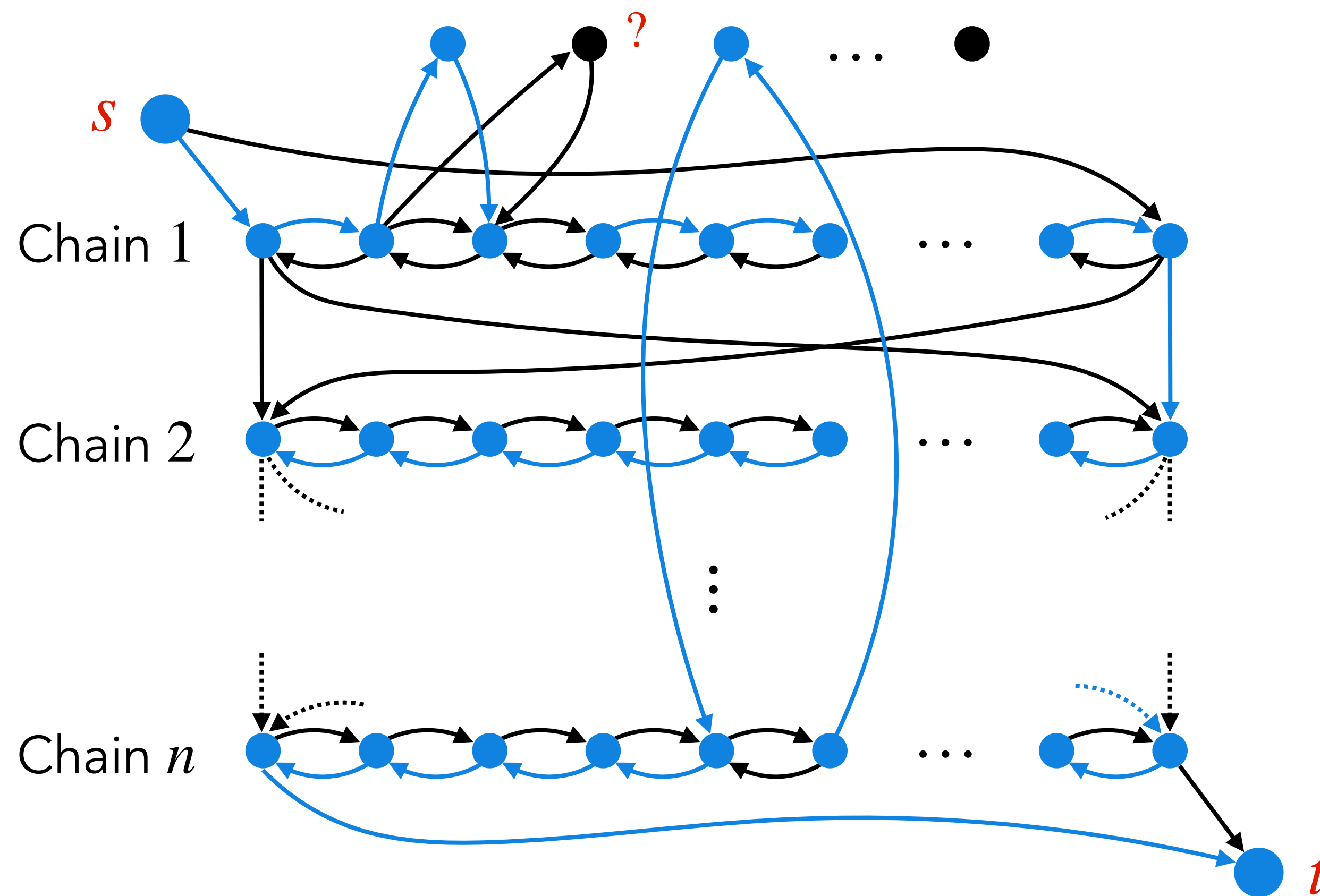


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